4 Air Quality Impact

4.1 Legislation, Standards and Guidelines

4.1.1 General

4.1.1.1 The relevant legislations, standards and guidelines applicable to the present study for the assessment of air quality impacts include:

- Air Pollution Control Ordinance (APCO) (Cap. 311);
- Air Pollution Control (Construction Dust) Regulation;
- Hong Kong Planning Standards and Guidelines (HKPSG); and
- Environmental Impact Assessment Ordinance (EIAO) (Cap. 499), Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM), Section 1 of Annex 4, and Annex 12.

4.1.2 Air Quality Objectives

4.1.2.1 The principal legislation for controlling air pollutants is the Air Pollution Control Ordinance (APCO) (Cap 311) and its subsidiary regulations, which defines statutory Air Quality Objectives (AQOs).

4.1.2.2 The APCO (Cap.311) provides the power for controlling air pollutants from a variety of stationary and mobile sources and encompasses a number of AQOs. In addition to the APCO, the following overall policy objectives are laid down in Chapter 9 of the HKPSG as follows:

- Limit the contamination of the air in Hong Kong, through land use planning and through the enforcement of the APCO to safeguard the health and well-being of the community; and
- Ensure that the AQOs for 7 common air pollutants are met as soon as possible.

4.1.2.3 The principal legislation for controlling air pollutants is the APCO (Cap 311) which provides a statutory framework for establishing the AQOs and stipulating the anti-pollution requirements for air pollution sources The AQOs are listed in the **Table 4.1**.

Table 4.1 Hong Kong Air Quality Objectives (HKAQOs)

Pollutant	**Limits on Concentration, µg/m^3 [1] (Number of Exceedance per calendar year allowed is shown in brackets)**				
	10-min	**1-hour**	**8-hour**	**24-hour [2]**	**Annual [2]**
Sulphur Dioxide (SO_2)	500 (3)			125 (3)	
Respirable Suspended Particulates (RSP) [3]				100 (9)	50

Pollutant	Limits on Concentration, µg/m³ [1] (Number of Exceedance per calendar year allowed is shown in brackets)				
	10-min	1-hour	8-hour	24-hour [2]	Annual [2]
Fine Suspended Particulates (FSP) [4]				75 (9)	35
Carbon Monoxide (CO)		30,000 (0)	10,000 (0)		
Nitrogen Dioxide (NO_2)		200 (18)			40
Ozone (O_3)			160 (9)		
Lead (Pb)					0.5

Notes:

[1] All measurements of the concentration of gaseous air pollutants, i.e., sulphur dioxide, nitrogen dioxide, ozone and carbon monoxide, are to be adjusted to a reference temperature of 293Kelvin and a reference pressure of 101.325 kilopascal.

[2] Arithmetic mean.

[3] Respirable suspended particulates (RSP) means suspended particles in air with a nominal aerodynamic diameter of 10 µm or less.

[4] Fine suspended particulates (FSP) means suspended particles in air with a nominal aerodynamic diameter of 2.5 µm or less.

4.1.3 Air Pollution Control (Construction Dust) Regulation

4.1.3.1 The Air Pollution Control (Construction Dust) Regulation specifies processes that require special dust control. The Contractors are required to inform the EPD and adopt proper dust suppression measures while carrying out "Notifiable Works" (which requires prior notification by the regulation) and "Regulatory Works" to meet the requirements as defined under the regulation.

4.1.4 Total Suspended Particulate Criterion

4.1.4.1 There is no criterion on TSP under the AQOs. In accordance with Annex 4 of EIAO-TM, a limit of 500µg/m^3 for 1-hour TSP concentration at any sensitive receivers should be adopted for construction dust impact assessment.

4.1.5 Odour Criterion

4.1.5.1 In accordance with the Annex 4 of the EIAO-TM, odour level at an air sensitive receiver shall meet 5 odour units based on an averaging time of 5 seconds.

4.2 Baseline Conditions

4.2.1 Existing Air Quality

4.2.1.1 The latest air quality monitoring data (available up to 2016) of the various air pollutants monitored at the nearest Yuen Long air quality monitoring station operated by EPD are shown in **Table 4.2** and have been compared with the AQOs for information.

Table 4.2 Air quality monitoring data (Yuen Long Station, 2012 – 2016)

Pollutant	Parameter	Concentrations (µg/m^3)						AQOs (µg/m^3)
		2012	2013	2014	2015	2016	5-year mean	
SO_2	4th highest 10-minutes	N/M	N/M	92	51	58	67 [13%]	500 (3)
	4th highest 24-hour	29	33	27	17	17	25 [20%]	125 (3)
NO_2	19th highest 1-hour	147	183	165	162	149	161 [81%]	200 (18)
	Annual	**49**	**54**	**52**	**45**	**46**	49 [123%]	40
CO	Max. 1-hour	2200	2690	2560	2460	2080	2398 [8%]	30,000
	Max. 8-hour	1950	1950	2110	2140	1470	1924 [19%]	10,000
O_3	10th highest 8-hour	**185**	**163**	**177**	**161**	143	166 [104%]	160 (9)
RSP	10th highest 24-hour	100	**142**	**124**	**102**	86	111 [111%]	100 (9)
	Annual	44	**56**	50	44	37	46 [92%]	50
FSP	10th highest 24-hour	65	**106**	**86**	**78**	63	80 [107%]	75 (9)
	Annual	29	**37**	35	30	23	31 [89%]	35

Note:

[1] N/M - Not Measured.

[2] Number of exceedance allowed under the AQO is shown in ( ), % of the AQO is shown in [ ]. The 5-year mean is the average of the yearly maximum.

[3] Monitoring results exceeding the AQO are bolded.

[4] Monitoring data for 10-min SO_2 for Years 2012 and 2013 are not available.

4.2.1.2 It can be seen from **Table 4.2** that there was no obvious trend for the 19th highest 1-hour NO_2 concentration and the range was from 147µg/m^3 in 2012 to 183µg/m^3 in 2013, all complying with the AQOs of 200µg/m^3. No trend of annual NO_2 concentration was also observed from 2012 to 2016. It ranged from 46µg/m^3 in Year 2016 to 54µg/m^3 in Year 2013, all exceeding the AQOs of 40µg/m^3.

4.2.1.3 Decreasing trend of RSP concentration was observed from Years 2013 to 2016. The 10th highest daily RSP concentration records exceeded the AQO during the 5 years period except Years 2012 and 2016. The annual concentrations ranged from 37μg/m^3 in Year 2016 to 56μg/m^3 in Year 2013. Exceedance was recorded in Year 2013.

4.2.1.4 Similar pattern as RSP was observed for FSP. The 10th highest daily FSP concentrations ranged from 63μg/m^3 to 106μg/m^3, which exceeded the AQO during the 5 years period except Years 2012 and 2016. The annual FSP concentrations were in the range of 23μg/m^3 to 37μg/m^3. Exceedance was also recorded in Year 2013.

4.2.1.5 The 10th highest 8-hour averaged O_3 concentrations exceeded the AQOs during the 5 years period except Year 2016, ranging from 143μg/m^3 in 2016 to 185μg/m^3 in 2012.

4.2.1.6 Monitoring records of SO_2 and CO indicated that these two pollutants were in relatively low level. Both pollutants were well within the AQO.

4.2.2 Future Air Quality

4.2.2.1 It should be noted that the ambient air quality conditions described in above sections are the historical data in the past 5 years. During the 16th Hong Kong-Guangdong Joint Working Group Meeting on Sustainable Development and Environmental Protection (January 2017), the Hong Kong and Guangdong Governments jointly endorsed a Work Plan and will continue to implement the Pearl River Delta (PRD) Regional Air Quality Management Plan up to year 2020. Key emission reduction measures to be implemented by Hong Kong and PRDEZ include:

Hong Kong Government

- tightening of vehicle emission standards;
- phasing out highly polluting commercial diesel vehicles;
- retrofitting Euro II and Euro III franchised buses with selective catalytic reduction devices;
- strengthening inspection and maintenance of petrol and liquefied petroleum gas vehicles;
- requiring ocean-going vessels to switch to using low sulphur fuel while at berth;
- tightening the permissible sulphur content level of locally supplied marine diesel;
- controlling emissions from off-road vehicles/equipment;
- further tightening of emission caps on power plants and increasing use of clean energy for electricity generation; and
- controlling VOC contents of solvents used in printing and construction industry.

Pearl River Delta Economic Zone

- installing desulphurization and denitrification systems at large-scale coal-fired power generating units;
- closing down small-scale power generating units;
- phasing out heavily polluting cement plants as well as iron and steel plants;

- installing vapour recovery systems at petrol filling stations, oil depots and on tanker trucks;
- implementing new pollutant emission standards for boilers as well as specific industries such as cement, furniture manufacturing, printing, shoe-making and surface coating (automobile manufacturing) industries;
- installing denitrification systems at new dry-type cement kilns;
- tightening the emission standards for newly registered petrol vehicles to Guangdong IV standard; and
- progressively supplying diesel at National IV standard and petrol at Guangdong IV standard.

4.2.2.2 In order to predict the future ambient air quality taking into account the measures to improve air quality, PATH-2016 (Pollutants in the Atmosphere and their Transport over Hong Kong), a regional air quality model, has been developed by EPD to simulate air quality over Hong Kong against the PRD as background.

4.2.2.3 The project involves 30 grids in the PATH-2016. The hourly pollutant concentration data predicted by PATH-2016 for year 2020 are provided by EPD and are summarised in the following tables. **Figure 4.1** illustrates the locations of concerned PATH grids for the study area.

Table 4.3 Future ambient air quality for concerned PATH grids (Year 2020)

Pollutant	Parameter	Concentrations in various PATH Grids (μg/m³)															AQOs [2] (μg/m³)
		20_47	20_48	21_46	21_47	21_48	22_45	22_46	22_47	22_48	23_43	23_44	23_45	23_46	23_47	23_48	
SO_2	4th highest 10-minutes [1]	144	170	117	122	132	138	143	142	132	113	112	111	128	124	114	500 (3)
	4th highest 24-hour	33	32	29	30	30	26	26	28	29	25	26	26	26	27	27	125 (3)
NO_2	19th highest 1-hour	95	96	98	95	93	100	99	98	97	86	89	94	97	99	100	200 (18)
	Annual	20	19	21	20	19	23	22	21	20	18	19	20	21	22	21	40
CO	Max. 1-hour	996	1006	996	1025	1050	993	1007	1061	1102	997	995	1002	997	1018	1068	30,000
	Max. 8-hour	838	851	838	854	868	865	861	883	899	871	873	876	880	897	911	10,000
O_3	10th highest 8-hour	156	158	157	159	159	156	157	158	158	159	158	156	159	160	**161**	160 (9)
RSP	10th highest 24-hour	85	84	83	83	83	82	81	81	81	87	86	83	81	81	80	100 (9)
	Annual	36	36	36	35	35	36	35	34	34	37	37	36	35	34	34	50
FSP	10th highest 24-hour [3]	64	63	62	62	62	62	61	61	61	65	65	62	61	61	60	75 (9)
	Annual [3]	26	25	25	25	25	25	25	24	24	27	26	25	25	24	24	35

Pollutant	Parameter	Concentrations in various PATH Grids (μg/m³)															AQOs [2] (μg/m³)
		24_43	24_44	24_45	24_46	24_47	25_43	25_44	25_45	25_46	25_47	26_45	26_46	26_47	27_46	27_47	
SO_2	4th highest 10-minutes [1]	111	111	110	114	111	111	111	111	112	112	112	111	112	111	112	500 (3)
	4th highest 24-hour	26	26	26	26	26	26	25	26	26	26	25	26	26	25	26	125 (3)
NO_2	19th highest 1-hour	95	90	95	101	106	103	101	100	112	114	103	101	104	91	95	200 (18)
	Annual	19	18	20	24	25	19	19	19	26	28	17	19	20	16	17	40
CO	Max. 1-hour	1002	1001	1014	1024	1010	1000	999	1013	1064	1004	1005	1002	1001	1000	1002	30,000
	Max. 8-hour	885	887	893	901	898	894	899	908	922	917	899	906	907	896	905	10,000
O_3	10th highest 8-hour	158	158	156	158	158	156	155	155	154	153	156	155	156	153	157	160 (9)
RSP	10th highest 24-hour	86	85	84	82	81	82	82	82	82	82	81	81	82	82	82	100 (9)
	Annual	37	36	36	35	35	36	35	35	35	35	35	35	35	35	35	50
FSP	10th highest 24-hour [3]	65	64	63	62	61	62	62	62	62	62	61	61	62	62	62	75 (9)
	Annual [3]	26	25	25	25	25	25	25	25	25	25	25	25	25	25	25	35

Note:

Bolded value means exceedance of respective AQOs

[1] Values are given as highest 10-minute SO_2 concentrations.

[2] Values in ( ) indicate number of exceedance allowed under the AQO.

[3] FSP concentrations are estimated in accordance with EPD's "Guidelines on the Estimation of FSP for Air Quality Assessment in Hong Kong".

4.2.2.4 It can be seen from the above tables that, with the implementation of the emission reduction measures by both the Hong Kong and Guangdong Governments, future background air quality in Year 2020 would be improved from recent existing conditions. In particular, the annual background NO_2 concentration predicted in the vicinity of the Project site would be significantly improved to 28μg/m^3 (the highest concentration predicted) in Year 2020.

4.3 Air Sensitive Receivers

4.3.1.1 With reference to Clause 3.4.1.2 of the EIA Study Brief (i.e. ESB-246/2012), the study area for air quality impact assessment should be defined by a distance of 500m from the boundary of the Project site (including PDA and associated infrastructure). **Figure 4.2** and **Figure 4.3** illustrate the 500m study area from the boundary of the Project site during construction phase and operation phase respectively.

4.3.1.2 In accordance with Clause 2.1, Annex 12 of the EIAO-TM, Air Sensitive Receivers (ASRs) include domestic premises, hotel, hostel, hospital, clinic, nursery, temporary housing accommodation, school, educational institution, office, factory, shop, shopping centre, place of public worship, library, court of law, sports stadium or performing arts centre.

4.3.1.3 However, for other premises which are not stipulated above, including open space, farm land, and recreational uses (e.g. park, playground, basketball court, football field, etc.), reference shall be made to Clause 2.2, Annex 12 of the EIAO-TM, which stated that any other premises or place with which, in terms of duration or number of people affected, has a similar sensitivity to the air pollutants as the abovementioned premises and places are also considered as a sensitive receiver. Sensitivities of these places to air pollutant are not similar to those mentioned in Clause 2.1, Annex 12 of the EIAO-TM as it is reasonably considered that people will only stay at these places for short period of time and frequent occupancy by the same person is also unlikely. In consideration of short period of retention, these kinds of uses would therefore be sensitive to short-term air quality impact only (i.e. averaging time of 10-mintue, 1-hour, 8-hour and 24-hour, where applicable), and hence, only the short-term AQOs are applicable to these particular uses (i.e. open space, farm land, and recreational uses such as park, playground, basketball court, football field, etc.) in the assessment.

4.3.1.4 Representative ASRs within a distance of 500m from the boundary of the Project site (including the PDA and associated infrastructure) have been identified. These ASRs include both the existing and planned developments. Existing ASRs are identified by means of reviewing topographic maps, aerial photos, land status plans, supplemented by site inspections. They mainly include existing residential buildings with different storey height, educational institution etc.

4.3.1.5 Planned/committed ASRs have been identified by making reference to relevant Outline Zoning Plans (OZP), Outline Development Plans, Layout Plans and other published plans in the vicinity of the alignment, including:

- Yuen Long OZP (S/YL/23);
- Ping Shan OZP (S/YL-PS/17);
- Lam Tei & Yick Yuen OZP (S/TM-LTYY/9);
- Tong Yan San Tsuen OZP (S/YL-TYST/10); and
- Tai Tong OZP (S/YL-TT/16).

4.3.1.6 The locations of the representative ASRs for construction dust impact assessment and operational air quality impact assessment are illustrated in **Figure 4.2 and Figure 4.3** respectively, and are summarised in the tables below.

Table 4.4a Representative ASRs for construction dust impact assessment

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
Existing / Planned ASRs (outside PDA boundary)						
A2	House no. 40, Tai Tao Tsuen	R	3	30	1.5	-
A3	House no. 95, Tai Tao Tsuen	R	3	<10	1.5	-
A4	Sheltered Structure, Fui Sha Wai South Road	R	1	10	1.5	-
A6	House no. 176A, Fui Sha Wai	R	3	20	1.5	-
A7	Village House, Ping Tong Street South	R	2	10	1.5	-
A8	House no. 48, Tong Yan San Tsuen Road	R	1	10	1.5	-
A9	New Territories Assemblies of God Church	W	6	210	1.5	-
A10	Block 10, Jasper Court	R	4	10	1.5	-
A11	School	E	4	50	1.5	-
A12	Village House, Lam Yu Road	R	1	<10	1.5	-
A13	Elchk Lutheran Academy	E	8	80	1.5	-
A15	Village House, Lam Hei Road	R	1	50	1.5	-
A16	Village House, Lam Hei Road	R	2	70	1.5	-
A17	Sheltered Structure, Lam Hei Road	R	1	30	1.5	-
A18	Sheltered Structure no. 66 Kiu Hing Road	R	1	10	1.5	-
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	R	2	50	1.5	-
A20	House no. 89A, Lam Hau Tsuen	R	2	20	1.5	-
A21	House no. 324, Shan Ha Tsuen	R	3	10	1.5	-
A22	House no. 645, Shan Ha Tsuen	R	3	40	1.5	-
A23	House no. 193, Kung Um Road	R	3	20	1.5	-
A24	House no. 132, Sun Mei Garden	R	4	<10	1.5	-
A25	House no. 293, Kung Um Road	R	3	10	1.5	-
A26	Block 10, Chun Fai Garden	R	3	<10	1.5	-
A27	House no. 161, Kung Um Road	R	3	30	1.5	-
A28	Village House, Kung Um Road	R	3	20	1.5	-

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
A29	House no. 366A, Kung Um Road	R	3	10	1.5	-
A30	Sheltered Structure, Kung Um Road	R	3	<10	1.5	-
A31	House no. 31B, Kung Um Road	R	3	20	1.5	-
A32	House no. 241, Kiu Hing Road	R	3	10	1.5	-
A33	House no. 67A, Wong Nai Tun Tsuen	R	3	<10	1.5	-
A34	House no. 128, Kung Um Road	R	2	<10	1.5	-
A35	House no. 117, Kung Um Road	R	1	90	1.5	-
A36	Sheltered Structure no. 375, Tai To Tsuen	R	1	200	1.5	-
A37	Uptown Tower 1	R	19	40	1.5	-
A38	Chinese Mission Seminary	W	2	300	1.5	-
A39	Fui Sha Wai Playground	Rec	-	250	1.5	-
A40	Tower 8, Imperial Villas II	R	5	400	1.5	-
A41	Energy Industrial Centre Block B	I	3	240	1.5	-
A42	Block 1, Parkside Villa	R	11	20	1.5	-
A43	Block 2, Emerald Green	R	25	300	1.5	-
A44	House no. 49A, Lung Tin Tsuen	R	3	40	1.5	-
A45	House no. 139A	R	3	60	1.5	-
A46	House no. 101, Sham Chung Tsuen	R	3	<10	1.5	-
A47	Meadowlands Block 33	R	5	380	1.5	-
A48	Po Kok Branch School	E	3	350	1.5	-
A49	Treasure Court Block 7	R	6	370	1.5	-
A50	Beauty Court Block 2	R	11	430	1.5	-
A51	Park Nara Tower 1	R	12	260	1.5	-
A52	Hung Uk Garden	R	3	410	1.5	-
A53	Green Lodge House 16	R	3	190	1.5	-
A54	Ping Shan Garden Block 6	R	3	400	1.5	-
A55	Villa Sunshine Block 1	R	4	280	1.5	-
A56	Park Royale Block 10	R	11	330	1.5	-
A57	Park Royale Block 2	R	11	90	1.5	-
A58	Po Leung Kuk Law's Foundation School	E	2	90	1.5	-

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
A59	Yuen Long Public Secondary School	E	8	140	1.5	-
A60	Villa Art Deco Block 2	R	12	310	1.5	-
A61	Gertrude Simon Lutheran College	E	8	450	1.5	-
A62	Yuen Long Public Middle School Alumni Association Primary School	E	7	470	1.5	-
A63	Silver Field Garden Block 17	R	3	410	1.5	-
A64	La Grove Block 1	R	20	10	1.5	-
A65	Ma Tin Tsuen House 242	R	3	120	1.5	-
A66	Teng Lung Villa	R	3	200	1.5	-
A67	Yuen Long Baptist Church	W	2	440	1.5	-
A68	Fraser Village House 54	R	2	50	1.5	-
A69	The Brand	R	21	180	1.5	-
A70	Sereno Verde Block 2	R	14	110	1.5	-
A71	La Pradera Block 12	R	15	330	1.5	-
A72	The Reach Tower 2	R	25	180	1.5	-
A73	The Reach Tower 6	R	25	10	1.5	-
A74	Tai Kei Leng House 145	R	2	290	1.5	-
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	E	8	460	1.5	-
A76	Grand de Sol Block 15	R	12	340	1.5	-
A77	Grand de Sol Block 8	R	12	120	1.5	-
A78	Hoover Garden Block 4	R	3	40	1.5	-
A79	Ha Yau Tin Tsuen House 2	R	3	230	1.5	-
A80	Buddhist Wing Yan School	E	8	410	1.5	-
A81	Fortune Centre	R	22	400	1.5	-
A82	YOHO Town Block 6	R	34	30	1.5	-
A83	YOHO Town Block 9	R	35	130	1.5	-
A84 [7]	YOHO Midtown Block 5	R	33	50	1.5	-
A85	Ho Shun Yee Building Block 2	R	20	230	1.5	-
A86	Cheong Wai Building	R	19	390	1.5	-
A87	Kwong Ming Ying Loi School	E	7	20	1.5	-
A88	Sun Yuen Long Centre Block 5	R	28	200	1.5	-
A89	Tung Tau Tsuen House 2	R	3	430	1.5	-

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
A90	Tai Wai Tsuen House 30	R	4	280	1.5	-
A91	Small Traders New Village	R	2	420	1.5	-
A92	Pok Oi Hospital	H	12	290	1.5	-
A93	Yeung Uk Tsuen House 10	R	3	330	1.5	-
A94	Chuk San Tsuen House 17	R	3	90	1.5	-
A95	Greenfield Lodge Block 1	R	2	320	1.5	-
A96	Kong Tau San Tsuen House 5	R	3	80	1.5	-
A97	Tai Kei Leng House 414	R	1	40	1.5	-
A98	Shung Ching San Tseun House 49	R	3	30	1.5	-
A99	Silver Garden House 125B	R	3	240	1.5	-
A100	Tin Liu Tsuen House 32	R	3	10	1.5	-
A101	Pak Sha Tsuen	R	3	<10	1.5	-
A102	Wong Nai Tun Tsuen House 47C	R	3	<10	1.5	-
A103	Greenwood Gardens House 397	R	3	70	1.5	-
A104	Park Signature Block 1	R	23	50	1.5	-
A105 [6]	Market of Hung Fuk Estate	C	3	30	1.5	-
A106	Village House along Kung Um Road	R	3	<10	1.5	-
A107	Village House along Kung Um Road	R	3	<10	1.5	-
A108	Shan Ha Tsuen House No. 342	R	3	<10	1.5	-
A109	Shan Ha Tsuen House No. 343A	R	3	<10	1.5	-
A110	Shan Ha Tsuen House No. 345C	R	3	50	1.5	-
A111	Shan Ha Tsuen House No. 346	R	2	10	1.5	-
A112	Shan Ha Tsuen House No. 611	R	3	30	1.5	-
A113	Shan Ha Tsuen House No. 613F	R	3	<10	1.5	-
A114	Sheung Yau Tin Tsuen	R	1	20	1.5	-
A115	Le Regent	R	3	20	1.5	-
A116	Planned Long Bin Development	R	40	260	1.5	2024/25 and 2027/28
A117	Planned Long Bin Development	R	40	20	1.5	2024/25 and 2027/28
A118	Planned Long Bin Development	R	40	20	1.5	2024/25 and 2027/28
A119	Proposed Hung Shui Kiu Development	R	29	40	1.5	2030

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
A120	Yee Fung Garden Block A	R	34	440	1.5	-
A121	Ming Wan Court	R	11	520	1.5	-
A122	Yuen Long Villa House No. 252	R	3	290	1.5	-
A123	Sun Fai Court Block C	R	3	330	1.5	-
A124	Caritas Yuen Long Chan Chun Ha Secondary School	E	6	400	1.5	-
A125	Village House along Ma Tong Road	R	3	390	1.5	-
A126	Proposed Hung Shui Kiu Development	R	29	50	1.5	2030
A127	Uptown House 30	R	3	10	1.5	-
A128	The Woodside Tower 5	R	5	130	1.5	-
A129	Shung Tak Catholic English College	E	8	110	1.5	-
A130	Village House along Castle Peak Road	R	3	250	1.5	-
A131	Regent's Park	R	3	<10	1.5	-
A132	One Hyde Park House 7	R	3	<10	1.5	-
A133	Pak Sha Tsuen	R	3	10	1.5	-
A134	Shan Ha Tsuen House No. 330	R	3	90	1.5	-
A135	Lam Hau Tsuen House No. 110	R	3	110	1.5	-
A136	Evergreen Place Tower 5	R	4	310	1.5	-
A137	Proposed Hung Shui Kiu Development	R	29	230	1.5	2030
A138	Lok Kui Lau	R	5	510	1.5	-
A139	Ping Shan Home for The Aged	H	3	420	1.5	-
A140	Sheltered Structure along Tai Shu Ha Road East	R	2	<10	1.5	-
A141	Village House along Tai Shu Ha Road West	R	2	<10	1.5	-
A142	Sheltered Structure near Shap Pat Heung Interchange	R	2	40	1.5	-
A143	Kong Tau San Tsuen House No, 61	R	3	30	1.5	-
A144	Pok Oi Hos. Jockey Club Care and Attention Home	H	7	250	1.5	-
A145	Proposed Hung Shui Kiu Development	R	29	220	1.5	2030

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
A146	Hong Ping Villa Block 1	R	3	330	1.5	-
A147	Tai On Home for Aged	H	3	410	1.5	-
A148	Ming Sum Home for the Sen	H	3	370	1.5	-
A149	Hung Fuk Estate	R	24	90	1.5	-
A150 [7]	Grand Yoho Block 1	R	34	150	1.5	-
A151 [7]	Grand Yoho Block 5	R	40	170	1.5	-
A152 [7]	Grand Yoho Block 9	R	41	170	1.5	-
A412	San Sang Tsuen [2]	R	3	440	1.5	-
A413	Greenville Park [2]	R	3	100	1.5	-
A601	Tseung Kong Wai [2]	R	3	440	1.5	-
A602	Farm House [2]	C	1	130	1.5	-
A603	Farm House [2]	C	1	340	1.5	-
A701	Kau Lee Uk Tsuen [2]	R	3	50	1.5	-
A702	San Uk Tsuen [2]	R	3	20	1.5	-
A703	Sha Chau Lei Tsuen [2]	R	3	40	1.5	-
A704	Ha Tsuen Shi [2]	R	3	30	1.5	-
A705	Yan Wu Garden [2]	R	3	190	1.5	-
A706	Sik Kong Tsuen [2]	R	3	380	1.5	-
A707	Pui Shing Catholic Secondary School [2]	E	7	370	1.5	-
A708	Sik Kong Wai [2]	R	3	320	1.5	-
A808	Tang Siu Tong Secondary School [2]	E	7	200	1.5	-
A813	Block H, Tin Shing Court [2]	R	41	190	1.5	-
A1501	Ha Tsuen Weigh Station [2]	GIC	1	180	1.5	-
Existing ASRs (within PDA boundary) – PDA Area 1						
W-A1	House 33, Park Villa	R	3	Within Project Boundary	1.5	-
W-A2	House 11, Park Villa	R	3	Within Project Boundary	1.5	-
W-A3	Existing developments (under construction)	R	3	Within Project Boundary	1.5	-
W-A4			3	Within Project Boundary	1.5	-
W-A5	Block 1, Recours La Serre	R	5	Within Project Boundary	1.5	-
W-A6	Block 2, The Parkhill	R	4	Within Project Boundary	1.5	-

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
W-A7	Block 7, Greenville Residence	R	5	Within Project Boundary	1.5	-
W-A8	Block 6, Windsor Garden	R	5	Within Project Boundary	1.5	-
W-A9	Block 1, Marbella Garden	R	5	Within Project Boundary	1.5	-
W-A11	Kisland Villa Phase 2	R	5	Within Project Boundary	1.5	-
W-A13	Sha Tseng Tsuen	R	4	Within Project Boundary	1.5	-
W-A14			4	Within Project Boundary	1.5	-
Planned ASRs – PDA Area 2 and Area 3						
E-P1	Planned Residential	R	34	Within Project Boundary	1.5	End-2027 / 2028
E-P2	Planned Residential	R	32	Within Project Boundary	1.5	2033
E-P5	Planned Residential	R	34	Within Project Boundary	1.5	End-2027 / 2028
E-P6	Planned Residential	R	34	Within Project Boundary	1.5	2033
E-P7	Planned Residential	R	31	Within Project Boundary	1.5	2033
E-P8	Planned Residential	R	28	Within Project Boundary	1.5	2033
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	GIC	6	Within Project Boundary	1.5	2033
E-P15	Planned School	E	8	Within Project Boundary	1.5	2033
E-P17	Planned School	E	8	Within Project Boundary	1.5	2033
E-P19	Planned Residential	R	11	Within Project Boundary	1.5	2038 [3]
E-P20	Planned Residential	R	20	Within Project Boundary	1.5	2038 [3]
E-P21	Planned Residential	R	22	Within Project Boundary	1.5	2038 [3]
E-P23	Planned Residential	R	22	Within Project Boundary	1.5	2038 [3]
E-P25	Planned Fire Station & Ambulance Depot	GIC	9	Within Project Boundary	1.5	2038 [3]

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
E-P29	Planned Residential	R	11	Within Project Boundary	1.5	2038 [3]
E-P31	Planned Residential	R	22	Within Project Boundary	1.5	2038 [3]
E-P32	Planned Residential	R	20	Within Project Boundary	1.5	2038 [3]
E-P33	Planned Residential	R	17	Within Project Boundary	1.5	2038 [3]
E-P34	Planned Residential	R	20	Within Project Boundary	1.5	2038 [3]
E-P40	Planned Residential	R	18	Within Project Boundary	1.5	2038 [3]
E-P42	Planned Residential	R	11	Within Project Boundary	1.5	2038 [3]
E-P45	Planned School	E	8	Within Project Boundary	1.5	2038 [3]
E-P46	Planned Residential	R	8	Within Project Boundary	1.5	2038 [3]
E-P47	Planned Residential	R	9	Within Project Boundary	1.5	2038 [3]
Planned ASRs – PDA Area 1						
W-P2	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P4	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P6	Planned Residential	R	3	Within Project Boundary	1.5	2038 [3]
W-P7	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P9	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P10	Planned Residential	R	5	Within Project Boundary	1.5	2038 [3]
W-P11	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P12	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P14	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P16	Planned Residential	R	5	Within Project Boundary	1.5	2038 [3]

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)	Planned Population Intake Year [5]
W-P17	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P20	Planned Social Welfare Facility	GIC	4	Within Project Boundary	1.5	2038 [3]
W-P21	Planned Residential	R	12	Within Project Boundary	1.5	2038 [3]
W-P22	Planned Residential	R	12	Within Project Boundary	1.5	2038 [3]
W-P25	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P26	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P27	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
W-P29	Planned Residential	R	5	Within Project Boundary	1.5	2038 [3]
W-P31	Planned Residential	R	4	Within Project Boundary	1.5	2038 [3]
Planned ASRs (in the vicinity of the proposed utilities within Hung Shiu Kiu NDA) [4]						
P806	Planned Committed Commercial Area	C	12	150	1.5	N/A
P807	Planned Committed Commercial Area	C	12	140	1.5	N/A
P1036	Planned Port Back-up, Storage and Workshop	I	20	360	1.5	2026

Notes:

[1] R– Residential; E – Education; C – Commercial; I – Industrial; GIC – Government, Institution and Community; Rec – Recreational; W – Place of Public Worship; H – Hospital / Home for aged

[2] These ASRs are located within 500m assessment area from the alignment of proposed utilities connecting to the San Wai Sewage Treatment Plant. Except from the construction works associated with the utilities construction, which is tentatively scheduled from Year 2027 to 2029, there will be no other construction works under this Project located within 500m from these ASRs during the construction period.

[3] The planned ASRs will be in place in Year 2038. Since there will be superstructure works only in Year 2038, adverse construction dust impact are not anticipated.

[4] These planned ASRs under Hung Shui Kiu NDA are included in the construction dust impact assessment to determine the cumulative dust impact from the construction of proposed utilities connecting to the San Wai Sewage Treatment Plant during Year 2027 to 2029.

[5] "-" means existing development.

[6] The market at ground level is provided with central air-conditioning and the fresh air intake is located at 4mAG (i.e. no air sensitive use at 1.5mAG). However, for conservative construction dust impact assessment, assessment point is selected at 1.5mAG.

[7] The shopping mall at ground level is provided with central air-conditioning and the fresh air intake is located at least 6mAG (i.e. no air sensitive use at 1.5mAG). However, for conservative construction dust impact assessment, assessment point is selected at 1.5mAG.

[8] Dust sources are located at ground level and hence assessment height is assumed to be 1.5m for all ASRs for conservative assessment for construction stage.

Table 4.4b Representative ASRs for operational air quality impact assessment

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
Existing / Planned ASRs (outside PDA boundary)					
A2	House no. 40, Tai Tao Tsuen	R	3	30	1.5
A3	House no. 95, Tai Tao Tsuen	R	3	<10	1.5
A4	Sheltered Structure, Fui Sha Wai South Road	R	1	10	1.5
A6	House no. 176A, Fui Sha Wai	R	3	20	1.5
A7	Village House, Ping Tong Street South	R	2	10	1.5
A8	House no. 48, Tong Yan San Tsuen Road	R	1	10	1.5
A9	New Territories Assemblies of God Church	W	6	210	1.5
A10	Block 10, Jasper Court	R	4	10	1.5
A11	School	E	4	50	1.5
A12	Village House, Lam Yu Road	R	1	<10	1.5
A13	Elchk Lutheran Academy	E	8	80	1.5
A15	Village House, Lam Hei Road	R	1	50	1.5
A16	Village House, Lam Hei Road	R	2	70	1.5
A17	Sheltered Structure, Lam Hei Road	R	1	30	1.5
A18	Sheltered Structure no. 66 Kiu Hing Road	R	1	10	1.5
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	R	2	50	1.5
A20	House no. 89A, Lam Hau Tsuen	R	2	20	1.5
A21	House no. 324, Shan Ha Tsuen	R	3	10	1.5
A22	House no. 645, Shan Ha Tsuen	R	3	40	1.5
A23	House no. 193, Kung Um Road	R	3	20	1.5
A24	House no. 132, Sun Mei Garden	R	4	<10	1.5
A25	House no. 293, Kung Um Road	R	3	10	1.5
A26	Block 10, Chun Fai Garden	R	3	<10	1.5
A27	House no. 161, Kung Um Road	R	3	30	1.5
A28	Village House, Kung Um Road	R	3	20	1.5
A29	House no. 366A, Kung Um Road	R	3	10	1.5
A30	Sheltered Structure, Kung Um Road	R	3	<10	1.5
A31	House no. 31B, Kung Um Road	R	3	20	1.5
A32	House no. 241, Kiu Hing Road	R	3	10	1.5
A33	House no. 67A, Wong Nai Tun Tsuen	R	3	<10	1.5
A34	House no. 128, Kung Um Road	R	2	<10	1.5
A35	House no. 117, Kung Um Road	R	1	90	1.5
A36	Sheltered Structure no. 375, Tai To Tsuen	R	1	200	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
A37	Uptown Tower 1	R	19	40	1.5
A38	Chinese Mission Seminary	W	2	300	1.5
A39	Fui Sha Wai Playground	Rec	-	250	1.5
A40	Tower 8, Imperial Villas II	R	5	400	1.5
A41	Energy Industrial Centre Block B	I	3	240	1.5
A42	Block 1, Parkside Villa	R	11	20	5
A43	Block 2, Emerald Green	R	25	300	1.5
A44	House no. 49A, Lung Tin Tsuen	R	3	40	1.5
A45	House no. 139A	R	3	60	1.5
A46	House no. 101, Sham Chung Tsuen	R	3	<10	1.5
A47	Meadowlands Block 33	R	5	380	1.5
A48	Po Kok Branch School	E	3	350	1.5
A49	Treasure Court Block 7	R	6	370	1.5
A50	Beauty Court Block 2	R	11	430	1.5
A51	Park Nara Tower 1	R	12	260	1.5
A52	Hung Uk Garden	R	3	410	1.5
A53	Green Lodge House 16	R	3	190	1.5
A54	Ping Shan Garden Block 6	R	3	400	1.5
A55	Villa Sunshine Block 1	R	4	280	1.5
A56	Park Royale Block 10	R	11	330	5
A57	Park Royale Block 2	R	11	90	5
A58	Po Leung Kuk Law's Foundation School	E	2	90	1.5
A59	Yuen Long Public Secondary School	E	8	140	1.5
A60	Villa Art Deco Block 2	R	12	310	1.5
A61	Gertrude Simon Lutheran College	E	8	450	1.5
A62	Yuen Long Public Middle School Alumni Association Primary School	E	7	470	1.5
A63	Silver Field Garden Block 17	R	3	410	1.5
A64	La Grove Block 1	R	20	10	1.5
A65	Ma Tin Tsuen House 242	R	3	120	1.5
A66	Teng Lung Villa	R	3	200	1.5
A67	Yuen Long Baptist Church	W	2	440	1.5
A68	Fraser Village House 54	R	2	50	1.5
A69	The Brand	R	21	180	1.5
A70	Sereno Verde Block 2	R	14	110	1.5
A71	La Pradera Block 12	R	15	330	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
A72	The Reach Tower 2	R	25	180	1.5
A73	The Reach Tower 6	R	25	10	1.5
A74	Tai Kei Leng House 145	R	2	290	1.5
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	E	8	460	1.5
A76	Grand de Sol Block 15	R	12	340	1.5
A77	Grand de Sol Block 8	R	12	120	1.5
A78	Hoover Garden Block 4	R	3	40	1.5
A79	Ha Yau Tin Tsuen House 2	R	3	230	1.5
A80	Buddhist Wing Yan School	E	8	410	1.5
A81	Fortune Centre	R	22	400	1.5
A82	YOHO Town Block 6	R	34	30	1.5
A83	YOHO Town Block 9	R	35	130	1.5
A84	YOHO Midtown Block 5	R	33	50	15
A84a	Shopping Mall under YOHO Midtown (Fresh Air Intake) [2]	C	N/A	180	6
A84b	Shopping Mall under YOHO Midtown (Fresh Air Intake) [2]	C	N/A	80	15
A84c	Shopping Mall under YOHO Midtown (Fresh Air Intake) [2]	C	N/A	<10	20
A84d	Kindergarten under YOHO Midtown (Fresh Air Intake) [2]	E	N/A	<10	2
A85	Ho Shun Yee Building Block 2	R	20	230	1.5
A86	Cheong Wai Building	R	19	390	1.5
A87	Kwong Ming Ying Loi School	E	7	20	1.5
A88	Sun Yuen Long Centre Block 5	R	28	200	20
A89	Tung Tau Tsuen House 2	R	3	430	1.5
A90	Tai Wai Tsuen House 30	R	4	280	1.5
A91	Small Traders New Village	R	2	420	1.5
A92	Pok Oi Hospital	H	12	290	1.5
A93	Yeung Uk Tsuen House 10	R	3	330	1.5
A94	Chuk San Tsuen House 17	R	3	90	1.5
A95	Greenfield Lodge Block 1	R	2	320	1.5
A96	Kong Tau San Tsuen House 5	R	3	80	1.5
A97	Tai Kei Leng House 414	R	1	40	1.5
A98	Shung Ching San Tseun House 49	R	3	30	1.5
A99	Silver Garden House 125B	R	3	240	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
A100	Tin Liu Tsuen House 32	R	3	10	1.5
A101	Pak Sha Tsuen	R	3	<10	1.5
A102	Wong Nai Tun Tsuen House 47C	R	3	<10	1.5
A103	Greenwood Gardens House 397	R	3	70	1.5
A104	Park Signature Block 1	R	23	50	1.5
A105_FA	Market of Hung Fuk Estate (Fresh Air Intake) [3]	C	N/A	30	4
A105_D	Market of Hung Fuk Estate (Welfare Facilities) [3]	C	3	30	10
A106	Village House along Kung Um Road	R	3	<10	1.5
A107	Village House along Kung Um Road	R	3	<10	1.5
A108	Shan Ha Tsuen House No. 342	R	3	<10	1.5
A109	Shan Ha Tsuen House No. 343A	R	3	<10	1.5
A110	Shan Ha Tsuen House No. 345C	R	3	50	1.5
A111	Shan Ha Tsuen House No. 346	R	2	10	1.5
A112	Shan Ha Tsuen House No. 611	R	3	30	1.5
A113	Shan Ha Tsuen House No. 613F	R	3	<10	1.5
A114	Sheung Yau Tin Tsuen	R	1	20	1.5
A115	Le Regent	R	3	20	1.5
A116	Planned Long Bin Development	R	40	260	5
A117	Planned Long Bin Development	R	40	20	5
A118	Planned Long Bin Development	R	40	20	5
A119	Proposed Hung Shui Kiu Development	R	29	40	5
A120	Yee Fung Garden Block A	R	34	440	1.5
A121	Ming Wan Court	R	11	520	1.5
A122	Yuen Long Villa House No. 252	R	3	290	1.5
A123	Sun Fai Court Block C	R	3	330	1.5
A124	Caritas Yuen Long Chan Chun Ha Secondary School	E	6	400	1.5
A125	Village House along Ma Tong Road	R	3	390	1.5
A126	Proposed Hung Shui Kiu Development	R	29	50	5
A127	Uptown House 30	R	3	10	1.5
A128	The Woodside Tower 5	R	5	130	1.5
A129	Shung Tak Catholic English College	E	8	110	1.5
A130	Village House along Castle Peak Road	R	3	250	1.5
A131	Regent's Park	R	3	<10	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
A132	One Hyde Park House 7	R	3	<10	1.5
A133	Pak Sha Tsuen	R	3	10	1.5
A134	Shan Ha Tsuen House No. 330	R	3	90	1.5
A135	Lam Hau Tsuen House No. 110	R	3	110	1.5
A136	Evergreen Place Tower 5	R	4	310	1.5
A137	Proposed Hung Shui Kiu Development	R	29	230	5
A138	Lok Kui Lau	R	5	510	1.5
A139	Ping Shan Home for The Aged	H	3	420	1.5
A140	Sheltered Structure along Tai Shu Ha Road East	R	2	<10	1.5
A141	Village House along Tai Shu Ha Road West	R	2	<10	1.5
A142	Sheltered Structure near Shap Pat Heung Interchange	R	2	40	1.5
A143	Kong Tau San Tsuen House No, 61	R	3	30	1.5
A144	Pok Oi Hos. Jockey Club Care and Attention Home	H	7	250	1.5
A145	Proposed Hung Shui Kiu Development	R	29	220	5
A146	Hong Ping Villa Block 1	R	3	330	1.5
A147	Tai On Home for Aged	H	3	410	1.5
A148	Ming Sum Home for the Sen	H	3	370	1.5
A149	Hung Fuk Estate	R	24	90	1.5
A150	Grand Yoho Block 1	R	34	150	20
A150a	Shopping Mall under Grand Yoho (Fresh Air Intake) [2]	C	N/A	160	8
A151	Grand Yoho Block 5	R	40	170	20
A152	Grand Yoho Block 9	R	41	170	20
A152a	Shopping Mall under Grand Yoho (Fresh Air Intake) [2]	C	N/A	140	10
Existing ASRs (within PDA boundary) – PDA Area 1					
W-A1	House 33, Park Villa	R	3	Within Project Boundary	1.5
W-A2	House 11, Park Villa	R	3	Within Project Boundary	1.5
W-A3	Existing developments (under construction)	R	3	Within Project Boundary	1.5
W-A4			3	Within Project Boundary	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
W-A5	Block 1, Recours La Serre	R	5	Within Project Boundary	1.5
W-A6	Block 2, The Parkhill	R	4	Within Project Boundary	1.5
W-A7	Block 7, Greenville Residence	R	5	Within Project Boundary	1.5
W-A8	Block 6, Windsor Garden	R	5	Within Project Boundary	1.5
W-A9	Block 1, Marbella Garden	R	5	Within Project Boundary	1.5
W-A11	Kisland Villa Phase 2	R	5	Within Project Boundary	1.5
W-A13	Sha Tseng Tsuen	R	4	Within Project Boundary	1.5
W-A14			4	Within Project Boundary	1.5
Planned ASRs – PDA Area 2 and Area 3					
E-P1	Planned Residential	R	34	Within Project Boundary	1.5
E-P2	Planned Residential	R	32	Within Project Boundary	1.5
E-P5	Planned Residential	R	34	Within Project Boundary	1.5
E-P6	Planned Residential	R	34	Within Project Boundary	1.5
E-P7	Planned Residential	R	31	Within Project Boundary	1.5
E-P8	Planned Residential	R	28	Within Project Boundary	1.5
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	GIC	6	Within Project Boundary	1.5
E-P15	Planned School	E	8	Within Project Boundary	1.5
E-P17	Planned School	E	8	Within Project Boundary	1.5
E-P19	Planned Residential	R	11	Within Project Boundary	1.5
E-P20	Planned Residential	R	20	Within Project Boundary	1.5
E-P21	Planned Residential	R	22	Within Project Boundary	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
E-P23	Planned Residential	R	22	Within Project Boundary	1.5
E-P25	Planned Fire Station & Ambulance Depot	GIC	9	Within Project Boundary	1.5
E-P29	Planned Residential	R	11	Within Project Boundary	1.5
E-P31	Planned Residential	R	22	Within Project Boundary	1.5
E-P32	Planned Residential	R	20	Within Project Boundary	1.5
E-P33	Planned Residential	R	17	Within Project Boundary	1.5
E-P34	Planned Residential	R	20	Within Project Boundary	1.5
E-P40	Planned Residential	R	18	Within Project Boundary	1.5
E-P42	Planned Residential	R	11	Within Project Boundary	1.5
E-P45	Planned School	E	8	Within Project Boundary	1.5
E-P46	Planned Residential	R	8	Within Project Boundary	1.5
E-P47	Planned Residential	R	9	Within Project Boundary	1.5
Planned ASRs – PDA Area 1					
W-P2	Planned Residential	R	4	Within Project Boundary	1.5
W-P4	Planned Residential	R	4	Within Project Boundary	1.5
W-P6	Planned Residential	R	3	Within Project Boundary	1.5
W-P7	Planned Residential	R	4	Within Project Boundary	1.5
W-P9	Planned Residential	R	4	Within Project Boundary	1.5
W-P10	Planned Residential	R	5	Within Project Boundary	1.5
W-P11	Planned Residential	R	4	Within Project Boundary	1.5
W-P12	Planned Residential	R	4	Within Project Boundary	1.5

ASR ID	Location	Landuse [1]	No. of Storey	Approx. Separation Distance from Project Boundary (m)	Lowest Assessment Height (m)
W-P14	Planned Residential	R	4	Within Project Boundary	1.5
W-P16	Planned Residential	R	5	Within Project Boundary	1.5
W-P17	Planned Residential	R	4	Within Project Boundary	1.5
W-P20	Planned Social Welfare Facility	GIC	4	Within Project Boundary	1.5
W-P21	Planned Residential	R	12	Within Project Boundary	1.5
W-P22	Planned Residential	R	12	Within Project Boundary	1.5
W-P25	Planned Residential	R	4	Within Project Boundary	1.5
W-P26	Planned Residential	R	4	Within Project Boundary	1.5
W-P27	Planned Residential	R	4	Within Project Boundary	1.5
W-P29	Planned Residential	R	5	Within Project Boundary	1.5
W-P31	Planned Residential	R	4	Within Project Boundary	1.5

Notes:

[1] R– Residential; E – Education; C – Commercial; I – Industrial; GIC – Government, Institution and Community; Rec – Recreational; W – Place of Public Worship; H – Hospital / Home for aged

[2] The shopping malls and kindergarten under YOHO Midtown and shopping malls under Grand Yoho are all central air-conditioned. Locations of fresh air intake are identified for operational air quality impact assessment.

[3] According to the information provided by the Housing Authority, the market on G/F is central air-conditioned and its fresh air intake (i.e. A105_FA) is located at about 4mAG (i.e. no air sensitive use at 1.5m facing Hung Tin Road). There is a car park (i.e. not air sensitive use) on the 1/F (about 5mAG) and some social welfare facilities with openable windows (i.e. A105_D) on the 2/F (about 10mAG).

4.4 Construction Dust Impact Assessment

4.4.1 Assessment Area

4.4.1.1 With reference to Clause 3.4.1.2 of the EIA Study Brief, the study area for air quality impact assessment should be defined by a distance of 500m from the boundary of the Project site (including PDA and associated infrastructure). **Figure 4.2** illustrates the extent of the study area for construction dust impact assessment.

4.4.2 Identification of Pollution Sources and Emission Inventory

4.4.2.1 The key air pollution sources in the vicinity of the Project that may bear upon the air quality during construction phase include dust emission associated with the

construction activities due to the Project, particulate emission from the neighbouring roads such as Yuen Long Highway (YLH), various industries / Specified Processes (SPs) in the vicinity, and construction activities due to the concurrent projects (e.g. Hung Shui Kiu (HSK) New Development Area (NDA), etc.)

4.4.2.2 Other far-field emission sources further beyond which would also have certain influence on the background air quality level include territory wide vehicular emission, power plants, marine emission, as well as regional emission from PRD.

4.4.2.3 Specifically, the existing and potential near-field sources are described in the following sections:

Dust Emission associated with the Project

4.4.2.4 The tentative commencement year for the site formation works of this Project is 2022 with target full completion in Year 2038. Construction of the Project as well as occupancy will be taken place in 4 phases (i.e. Phase 1 – Year 2027, Phase 2 – Year 2033, and Phase 3 & 4 – Year 2038. Refer to **Figure 2.1** for the development phasing). A review on the construction methodology and tentative implementation programme has been conducted. In general, construction dust as the representative pollutants will be generated mainly from the at-grade construction works including the following activities:

- Demolition of existing buildings;
- Site clearance;
- Soil excavation;
- Site formation;
- Backfilling;
- Temporary storage, handling and transportation of material; and
- Wind erosion of open sites.

4.4.2.5 Since excavation and backfilling activities will involve large quantities of earthworks and material handling, it is anticipated that there may be dust impact as a result of these activities if mitigation measures are not implemented.

4.4.2.6 Fugitive dust impact assessments have been carried out based on conservative assumptions of general construction activities which include the following:

- Heavy construction activities including site clearance, ground excavation, site formation, construction of the associated facilities, etc.; and
- Wind erosion of all active open sites.

4.4.2.7 Construction programme as provided by the Project Engineer is adopted in order to assess the reasonable situation. The construction working periods of 30 days a month and 12 hours a day from 7:00am to 7:00pm.

Dust Emission associated with the Construction of Concurrent Projects

4.4.2.8 Concurrent projects in the vicinity of this Project, which may have cumulative environmental impacts, have been discussed in **Section 1.8**. Key concurrent projects of air quality concern during the construction phase of this Project have been identified and are summarised in the table below. The implementation programmes of these concurrent projects are provided by the respective project proponents.

Where information is not available, references have been made to the best available information such as EIA reports.

Table 4.5 Key concurrent projects for construction dust assessment

Key Concurrent Projects	Tentative Construction Commencement Year	Tentative Commissioning Year
Hung Shui Kiu New Development Area (HSK NDA) Planning and Engineering Study	2019	2024 (1st population intake) 2037 / 2038 (Full population intake)
Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station – Investigation, Design and Construction	Early 2018	Mid 2022
Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 1 Improvement Works – Design and Construction	2012	Late 2020
Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 2 Beautification Works – Design and Construction	2025	2027
Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long - Feasibility Study	No confirmed programme	2024/25 (1st population intake) 2027/28 (2nd population intake)
Site Formation and Infrastructural Works for the Development at Tan Kwai Tsuen, Yuen Long – Feasibility Study	No confirmed programme	2027/28 (2nd population intake)

4.4.2.9 It can be seen from the above table that the concurrent construction activities associated with the following concurrent projects are anticipated during the construction phase:

1) Hung Shui Kiu New Development Area Planning and Engineering Study;
2) Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station – Investigation, Design and Construction; and
3) Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 2 Beautification Works – Design and Construction.

Hung Shui Kiu New Development Area Planning and Engineering Study

4.4.2.10 According to the approved EIA study "*Hung Shui Kiu New Development Area*" (AEIAR-203/2016) (HSK NDA EIA), the major construction works are targeted to commence in Year 2019, while the first population intake is expected in Year 2024 and to be fully commissioned by Year 2037 / 2038. Hence, potential cumulative construction dust impact from its concurrent construction with the Project is anticipated.

Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station – Investigation, Design and Construction

4.4.2.11 According to the approved EIA study "*Elevated Pedestrian Corridor in Yuen Long Town Connecting with Long Ping Station*" (AEIAR-200/2016), its construction activities will only be confined within relatively small active area (about 100-150m^2) and scattered along the Yuen Long Town Nullah, and majority of the construction works will be conducted within the nullah which is about 4m to 5m lower than the ground level of the surrounding ASRs. No significant dust emission (due to small active area) as well as no dust dispersion from the work area to ASRs are anticipated. Hence, no adverse construction dust impact from this concurrent project is anticipated.

Improvement of Yuen Long Town Nullah (Town Centre Section) – Stage 2 Beautification Works – Design and Construction

4.4.2.12 As discussed in **Section 1.8**, the planned beautification works involve modification and reconstruction of the existing concrete nullah bed and wall. Similar to the construction of the Elevated Pedestrian Corridor in Yuen Long Town (refer to **Section 4.4.2.11**), the construction works will be within the nullah and lower than the ground level of surrounding ASRs, and hence dust dispersion from the work area to ASRs is considered unlikely. No adverse construction dust impact from this concurrent project is therefore anticipated.

Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long, Feasibility Study

4.4.2.13 The project area of the Long Bin development is about 10ha (100,000m^2) in size. There are no confirmed programme for the Long Bin development project. However, it is advised that there are two-phase developments. The tentative population intake year of the 1st phase is Year 2024/25 and the 2nd phase is Year 2027/28. The layout and phasing of the development is currently not available. With reference to the construction programme for the proposed public housing construction site under this Project, the site formation works and subsequent superstructure works and mechanical electrical plumbing (MEP) works would require about 24 months and 48 months respectively. Hence, it is anticipated that the superstructure works and MEP works for Long Bin development Phase 1 would be carried out from Year 2021/2022 to 2024/2025, and that for Phase 2 would be carried out from Year 2023/2024 to 2027/2028; while the site formation would be undertaken before 2021/2022 for Phase 1 and 2023/2024 for Phase 2. During the superstructure works and MEP works, the pre-cast fabrication work would unlikely generate significant dust emission and thus cumulative dust impacts are not anticipated. Besides, the Long Bin development site is currently a flat land and extensive site formation work is unlikely. However, some slope works are still expected and such works would still generate dust emission. The nearest ASRs within 500m of Long Bin development site include A8, A10, A11, A12, A42, A53 and A57.

4.4.2.14 The site formation work of Long Bin Development (i.e. likely before 2021/2022 for Phase 1 and 2023/2024 for Phase 2) would overlap with the site formation work of this Project from Year 2022 to 2024. During this overlapping period, construction works under this Project include Phase 1 and Phase 2 (see **Appendix 4.1** for location of workfronts and phasing), which are located at least 660m and 900m away from the Project boundary of Long Bin development and at least 500m from the concerned ASRs mentioned above. Given the large separation distance, cumulative dust impact

associated with the construction of Long Bin development is considered insignificant, and hence is not included in this quantitative construction dust impact assessment.

Site Formation and Infrastructural Works for the Development at Tan Kwai Tsuen

4.4.2.15 The project area of the Tan Kwai Tsuen (TKT) development is about 12.7ha (127,000m^2) in size. There are no confirmed programme for this potential concurrent project except the tentative population intake year in Year 2027/28. With reference to the construction programme for the proposed public housing construction site under this Project, the site formation works and subsequent superstructure works and mechanical electrical plumbing (MEP) works would require about 24 months and 48 months respectively. Hence, it is anticipated that the superstructure works and MEP works for TKT development would be carried out from Year 2023/2024 to 2027/2028; while the site formation would be undertaken before 2023/2024. During the superstructure works and MEP works, the pre-cast fabrication work would unlikely generate significant dust emission and thus cumulative dust impacts are not anticipated. Site formation and slope works are still expected and such works would still generate dust emission.

4.4.2.16 The site formation work of TKT development (i.e. likely before 2023/2024) would overlap with the site formation work of this Project from Year 2022 to 2024. As shown in **Figure 1.3**, its project boundary is located at about 500m away from this Project. The nearest ASR A36 (Sheltered Structure no. 375, Tai To Tsuen) and A47 (Meadowlands Block 33) are located about 330m and 370m away from the planned TKT development site, and are about 200m and 380m away from the YLS project site. However, given these ASRs are located at opposite downwind direction from the TKT project and YLS project, cumulative short-term dust impact from their possible concurrent construction is not anticipated. For cumulative annual dust impact, as presented in **Section 4.4.7**, the predicted annual RSP / FSP concentrations at these ASRs are far below the respective criteria. The contribution from YLS project to these ASRs is very small only (e.g. about 0.8µg/m^3 of annual RSP concentrations at A36). Given the TKT development is located farther away to these ASRs compared to YLS development, the contribution from TKT development is also expected to be small. Adverse cumulative annual dust impact from this concurrent project is therefore not anticipated.

Dust Emission associated with the Operation of Concurrent Sources

4.4.2.17 Apart from the concurrent projects as discussed above, particulate emissions, including TSP, RSP and/or FSP, would also be generated from nearby existing road networks, existing Specified Process (SP), and industrial chimneys. Details of these sources are described in **Section 4.5.1**. These sources have been considered in the cumulative construction dust impact assessment and are illustrated in **Figures 4.4** and **Figure 4.5**.

4.4.2.18 In particular, there are 2 concrete batching plants (i.e. SP) located within Project boundary, namely Golik Concrete and Redland Concrete, which will likely be operating until the proposed land resumption in July 2025. Dust emissions from these 2 SPs are included in the cumulative assessment for years on or before Year 2025. Locations of these 2 concrete batching plants are illustrated in **Figure 4.4**.

4.4.3 Key Representative Pollutants

4.4.3.1 According to Section 13.2.4.3 of USEPA AP-42, among all aerodynamic particle sizes (i.e. TSP), there are 47.3% of particles with an aerodynamic diameter of <10

μm (i.e. RSP). Hence, TSP and RSP are the most representative pollutants for construction phase assessment. However, upon the effect of the AQO from 1 January 2014, a new criteria pollutant, FSP, has been included in the AQO. As a conservative approach, FSP has also been assessed, notwithstanding that it only constitutes 7.2% of the total particles in fugitive dust. Hence, the 1-hour TSP, 24-hour RSP/ FSP, and annual RSP/ FSP concentrations at each identified ASR have been assessed and compared with the AQO or the requirements of EIAO-TM to determine their compliance.

4.4.3.2 Fuel combustion from the use of Powered Mechanical Equipment (PME) during construction works could be a source of NO_2, SO_2 and CO. To improve air quality and protect public health, EPD has introduced the Air Pollution Control (Non-road Mobile Machinery) (Emission) Regulation, which came in operation on 1 June 2015, to regulate emissions from machines and non-road vehicles. Starting from 1 December 2015, only approved or exempted non-road mobile machinery are allowed to be used in construction sites. Hence, with the effect of the Regulation, the emissions from PMEs are considered relatively small.

4.4.3.3 In addition, there is no source of Pb and O_3 emission during the construction phase. Hence, NO_2, SO_2, CO, Pb and O_3 are not considered as the key pollutants for quantitative dust assessment.

4.4.4 Assessment Methodology

Dust Emission associated with the Project

4.4.4.1 The prediction of dust emissions is based on typical values and emission factors from United States Environmental Protection Agency (USEPA) Compilation of Air Pollution Emission Factors (AP-42), 5th Edition. References of the dust emission factors for different dust generating activities are listed in **Table 4.6** below. Details are discussed in the following sections. **Appendix 4.1** presents the calculation of dust emission factors and locations of dust source.

Table 4.6 Key concurrent projects for construction dust assessment

Construction Site	Activities	Equations and Assumptions	Reference
All construction sites	Heavy construction activities including land clearance, ground excavation, cut and fill operations, construction of the facilities, haul road, etc.	$E_{(TSP)}$ = 1.2 tons/acre/month of activity or = 2.69 Mg/hectare/month of activity $E_{(RSP)} = E_{(TSP)} \times 0.473$ = 1.27 Mg/hectare/month of activity $E_{(FSP)} = E_{(TSP)} \times 0.072$ = 0.19 Mg/hectare/month of activity	USEPA AP42, S.13.2.3.3
	Wind erosion	$E_{(TSP)}$ = 0.85 Mg/hectare/year $E_{(RSP)} = E_{(TSP)} \times 0.473$ = 0.40 Mg/hectare/year $E_{(FSP)} = E_{(TSP)} \times 0.072$ = 0.06 Mg/hectare/year	USEPA AP42, S.11.9, Table 11.9.4

Note: RSP:TSP and FSP:TSP ratios are referenced from Section 13.2.4.3 of USEPA AP-42.

4.4.4.2 Dust emission from construction vehicle movement will generally be limited within the confined worksites and the emission factor given in AP-42 S.13.2.3.3 has taken

this factor into account. Watering facilities will be provided at every designated vehicular exit point. Since all vehicles will be washed at exit points and vehicle loaded with the dusty materials will be covered entirely by clean impervious sheeting before leaving the construction site, dust nuisance from construction vehicle movement outside the worksites is unlikely to be significant.

Determination of Worst Assessment Year

4.4.4.3 Quantities of dust generation during construction period associated with the Project for each year have been estimated to identify the worst assessment year and results are tabulated as in **Table 4.7** below. It can be seen that the highest dust generation for each phase would occur in Years 2024, 2028, 2034 and 2035 for Phases 1, 2, 3 and 4 respectively, thus they are selected as the worst assessment year. However, it should also be noted that, given the construction period will span more than 15 years, construction sites would be moving around, affecting different existing and planned ASRs at different times. Hence, Years 2023, 2025, 2026, 2027, 2029, 2033 and 2036 are also selected for construction dust impact assessment to evaluate the potential dust impact as there will be construction sites located close to existing and/or planned ASRs. For Years 2031, 2032, 2037 and 2038, since there will be superstructure works only, significant dust emission and adverse construction dust impact are not anticipated.

4.4.4.4 In addition, it should be noted site formation works in Phase 1 near the existing Tin Liu Tsuen during Year 2022 would also be carried out during Year 2023, while the site formation works in Phase 2 to be conducted near Tong Yan San Tsuen, Tai Tong and Wong Nai Tun Tsuen during Year 2030 would also be carried out during Year 2029. Hence, dust emission sources during Years 2022 and 2030 have been covered in Years 2023 and 2029 respectively and these years are therefore not selected for the construction dust impact assessment.

4.4.4.5 **Appendix 4.2** presents the tentative construction programme and calculations of the annual dust emission.

Table 4.7 Estimated dust emission from each phase during construction period

Year	Estimated Dust Emission (tonnes per year) Note [1]				Total Dust Emission (ton per year)
	Phase 1	Phase 2	Phase 3	Phase 4	
2022	55 #	-	-	-	55
2023 *	124	72	-	-	196
2024 *	**304**	197	-	-	500
2025 *	171	332	-	-	502
2026 *	16	266	-	-	282
2027 *	Note [2]	413	-	-	413
2028 *	Note [2]	**632**	-	-	**632**
2029 *	Note [2]	448	-	-	448
2030	-	71 #	-	-	71
2031	-	Note [2]	-	-	Note [2]
2032	-	Note [2]	-	-	Note [2]
2033 *	-	Note [2]	99	-	99
2034 *	-	-	**277**	107	383
2035 *	-	-	231	**195**	426
2036 *	-	-	Note [2]	164	164

Year	Estimated Dust Emission (tonnes per year) Note [1]				Total Dust Emission (ton per year)
	Phase 1	Phase 2	Phase 3	Phase 4	
2037	-	-	Note [2]	Note [2]	Note [2]
2038	-	-	Note [2]	Note [2]	Note [2]

Notes:

Values presented are rounded to the nearest integer.

Values in bold are the maximum for each phase within the construction period.

* Selected years for the construction dust impact assessment

Emission sources in Year 2022 are covered in Years 2023, while those in Year 2030 are covered in Year 2029

[1] Dust emission generated from heavy construction works and wind erosion.

[2] According to the current construction programme, major heavy construction works, such as excavation, site formation, backfilling, etc. will be completed. There will be superstructures works, road paving works or landscaping works only in these years. Significant dust emission is not anticipated.

Dust Dispersion Modelling Approach

4.4.4.6 Dust impact assessment is undertaken using the EPD approved AMS/EPA Regulatory Model (AERMOD). It is a well-known model designed for computing air dispersion. Modelling parameters including dust emission factors, particles size distributions, surface roughness, etc. are referred to EPD's "*Guidelines on Choice of Models and Model Parameters*" and USEPA AP-42. The density of dust is assumed to be 2.5g/cm^3, with reference to the "*Coal Mining Emission Factor Development and Modelling Study*" (USEPA-AP42). Construction activities include heavy construction activities (including site clearance, soil excavation, etc.) and wind erosion of all active open sites. Particle size distribution is estimated based on S13.2.4.3 of USEPA AP-42. **Table 4.8** presents the particle size distribution of TSP, RSP and FSP adopted in the assessment.

Table 4.8 Particle size distribution assumed in AERMOD

Particle Size (µm)	Average Particle Size (µm)	Particle Size Distribution		
		TSP	RSP	FSP
0 – 2.5	1.25	7%	15%	100%
2.5 – 5	3.75	20%	42%	-
5 – 10	7.5	20%	43%	-
10 – 15	12.5	18%	-	-
15 – 30	22.5	35%	-	-
	Total	**100%**	**100%**	**100%**

4.4.4.7 As the site formation works of this Project will be from Year 2022 to Year 2038, hourly air quality data from PATH-2016 model for Year 2020 is used as the background concentrations for conservative assessment. In addition, hourly meteorological data (including wind direction, wind speed, temperature and mixing height) for Year 2010 extracted from the PATH-2016 model are used. Mixing heights from the PATH-2016 which are lower than the minimum mixing height recorded by the Hong Kong Observatory (HKO) in Year 2010 (i.e. 121m) are capped at 121m. For the treatment of calm hours, the wind speeds are capped at 1m/s for those from PATH-2016 below 1m/s.

4.4.4.8 During daytime working hours (7am to 7pm), it is assumed that dust emissions would be generated from all dust generating activities. During night-time non-working hours (7pm to 7am of the next day), dust emission source would include wind erosion only as construction activities during these hours are ceased.

4.4.4.9 Fugitive dust impacts are modelled for ASR heights at 1.5m, 5m, 10m, 15m, 20m, 50m, 80m and 120m above ground. Since all the dust generating sources associated with the Project are located on ground level, these assessment levels would therefore cover the worst-case scenario. Both the unmitigated and mitigated scenarios are presented. A 50x50m grid is used to generate the pollution contours at the worst hit level for the worst year in order to investigate the pollutant dispersion.

4.4.4.10 A summary of AERMOD modelling parameters that have been adopted in the construction dust assessment are given in **Table 4.9** below:

Table 4.9 Modelling parameters adopted in AERMOD

Parameters	Input
Background Concentration	Hourly RSP concentrations from PATH-2016 (Year 2020)
Meteorological Data	2010 hourly meteorological data adopted in PATH-2016
Anemometer Height	9m (According to EPD's Guidelines on Choice of Models and Model Parameters)
Albedo	0.14 (Within 10km x 10km region from the Project site, landuses comprise of 45%, 48%, 5% and 2% of Urban, Forest, Agricultural land and Water areas respectively. Refer to **Appendix 4.3.**)
Bowen ratio	0.9 (Within 10km x 10km region from the Project site, landuses comprise of 45%, 48%, 5% and 2% of Urban, Forest, Agricultural land and Water areas respectively. Refer to **Appendix 4.3.**)
Landuse and Surface Roughness	Refer to **Appendix 4.3** for surface characteristic within 1km for each PATH grid
Emission Period	General construction activities during daytime working hours (7 am to 7 pm) Wind erosion during night-time (7pm to 7am of the next day)
Assessment Heights	1.5m, 5m, 10m, 15m, 20m, 50m, 80m and 120m

4.4.4.11 It is understood that construction activities will not be taken place on the entire work sites at the same time, but to be undertaken at moving multiple work fronts spread across the work sites. Notwithstanding this, a "Two Tiers" assessment approach has been adopted. An initial screening test, namely, "Tier 1 Screening Test", which assumes that all the worksites would be active, has been undertaken. The Tier 1 screening test is conservative and has represented the worst case situation.

4.4.4.12 The purpose of the Tier 1 screening test is to identify the potentially affected areas where construction dust may accumulate. The hot spot areas identified in the Tier 1 assessment have been subsequently assessed by a more focused Tier 2 test. Under the Tier 2 test, an active construction site with a size of 50m x 50m (or 2,500m^2) is

assumed to be positioned closest to the potentially affected ASR to reflect the realistic worst case scenarios, while all other construction sites located relatively further away from the ASRs remain unchanged as per Tier 1. The active construction site of 2,500m^2 is assumed based on the advice from Project Engineer that any active construction at any one time in each phase would not be more than 50m x 50m (or 2,500m^2) in reality. It is considered as a reasonable size for the operation of the necessary Powered Mechanical Equipment (PME) for the construction activities. In addition, in order to achieve the current construction programme, site formation works would be completed at a maximum rate of about 9ha (i.e. 90,000m^2) by 3 months, which is about 1,200m^2 per day (see **Appendix 4.2**). The assumed active works area of 2,500m^2 would therefore be conservative.

Dust Emission associated with the Construction of Concurrent Projects

4.4.4.13 As discussed in **Section 4.4.2.10**, potential cumulative construction dust impact from the concurrent construction of planned HSK NDA is anticipated. Dust emission sources within 500m assessment area from the Project boundary and their dust emissions as presented in the approved HSK EIA (AEIAR-203/2016) are therefore included in the construction dust impact assessment. Details are given in **Appendix 4.1**.

Dust Emission associated with the Operation of Concurrent Sources

4.4.4.14 Approach for prediction of particulate emissions from open road is the same as operational phase assessment as described in **Section 4.5.3**, in which the dispersion model, CALINE4 is used to assess the vehicular emission impact. Particulate emissions from open road are determined using the latest EMFAC-HK based on the projected traffic data for Years 2038 coupled with emission factors for Year 2022. Traffic data in Year 2038, which is the final year of construction period, is considered to be conservative as it would be the highest within the construction phase, while the RSP / FSP emission factors for Year 2022 would also be the highest. **Appendix 4.4** presents the hourly RSP and FSP emission factors for each road link. It should be noted that using traffic data in Year 2038 and emission factors in Year 2022 is a very conservative approach and would not occur in reality.

4.4.4.15 For chimney emissions from various industries and SPs, the prediction approach is the same as the operational air quality assessment as described in **Section 4.5.3**, in which the dispersion model, AERMOD, is used to assess the chimney emission impact.

Far-field Source Contribution (i.e. Future Background Air Quality)

4.4.4.16 Details are given in **Section 4.5.3**. Hourly pollutant concentration data predicted by PATH-2016 for Year 2020 provided by EPD are directly adopted as the background concentration.

4.4.4.17 FSP concentrations are not available from PATH model. According to EPD's "*Guidelines on the Estimation of PM2.5 for Air Quality Assessment in Hong Kong*", the conservative correction as shown in **Table 4.10** are adopted to determine the background FSP concentrations. For hourly background TSP concentrations, it is considered reasonable to assume the hourly RSP concentrations from PATH as the ambient TSP background concentrations, since the particulates of sizes larger than 10μm generated from far-field dust sources would have been largely settled before reaching the ASRs, and hence most of the particulates contributed from far-field sources affecting the ASRs will likely be of less than or equal to 10μm in size (i.e. RSP).

Table 4.10 Conversion factors for RSP / FSP

Daily Concentration (µg/m3)	Annual Concentration (µg/m3)
FSP = 0.75 x RSP	FSP = 0.71 x RSP

Prediction of the Cumulative Construction Dust Impact

4.4.4.18 The cumulative construction dust impact is a combination of the emission impacts contributed from the near field and far field sources (i.e. at local scale and background air quality impact from other concurrent and regional sources) on an hourly basis.

4.4.4.19 In consideration of the number of exceedance allowance of the daily AQOs (refer to **Table 4.1**), the pollutant concentrations beyond the AQO's allowance limits (i.e. the 10th highest 24-hour RSP/ FSP concentrations) are presented. The predicted annual RSP/ FSP concentrations are also assessed and all predicted levels are then compared with the AQOs. Besides, the 1-hour TSP concentration as stipulated under Annex 4 of EIAO-TM is also determined at each ASR.

4.4.5 Assessment Results (Unmitigated)

4.4.5.1 The predicted maximum unmitigated 1-hour TSP concentrations, 10th highest 24-hour and annual RSP / FSP concentrations for Years 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2033, 2034, 2035 and 2036 are presented in the **Table 4.11** below and detailed in **Appendix 4.5**. Exceedances of the TSP, RSP and/or FSP criteria are predicted at both existing and planned ASRs. Mitigation measures are therefore required to reduce the potential air quality impact during construction phase. It should be noted that the predicted concentrations are based on very conservative assumptions, such as assuming 100% active area, adopting vehicular emission in Year 2022 but coupling with maximum traffic data in Year 2038.

4.4.5.2 As mentioned in **Table 4.7**, highest dust emission from the construction of the Project is expected in Year 2028. Comparing to the detailed results presented in **Appendix 4.5**, it is also indicated that the highest predicted pollutant concentrations would mostly occur in Year 2028, and the worst affected height is identified at 1.5m above ground. To this end, contours for the cumulative unmitigated 1-hour TSP concentrations, and 10th highest 24-hour and annual RSP / FSP concentrations at 1.5m above ground for Year 2028 are plotted in **Figure 4.6a** to **Figure 4.6e**.

Table 4.11 Unmitigated cumulative TSP, RSP and FSP concentrations

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
Existing / Planned ASRs (outside PDA boundary)								
A2	House no. 40, Tai Tao Tsuen	2024 / 2025	1.5 / 10	**2287**	**159**	**71**	72	32

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A3	House no. 95, Tai Tao Tsuen	2025 / 2036	1.5 / 10	**1481**	**137**	**63**	70	31
A4	Sheltered Structure, Fui Sha Wai South Road	2027 / 2028	1.5 / 5	**1590**	**130**	**51**	67	29
A6	House no. 176A, Fui Sha Wai	2027 / 2028	1.5 / 5	**2156**	**166**	**76**	**76**	33
A7	Village House, Ping Tong Street South	2027 / 2028	1.5 / 10	**1871**	**163**	**76**	73	33
A8	House no. 48, Tong Yan San Tsuen Road	2027 / 2028	1.5	**2079**	**137**	**69**	67	32
A9	New Territories Assemblies of God Church	2027 / 2028 / 2034	1.5 / 10 / 15	**863**	90	43	64	27
A10	Block 10, Jasper Court	2028 / 2034	1.5 / 10	**1617**	**121**	**59**	65	30
A11	School	2027 / 2028	1.5 / 10	**1314**	**110**	50	64	29
A12	Village House, Lam Yu Road	2028	1.5	**2641**	**202**	**93**	71	**36**
A13	Elchk Lutheran Academy	2024	1.5 / 10 / 15	**1051**	**109**	45	66	27
A15	Village House, Lam Hei Road	2024	1.5 / 5	**1213**	**117**	50	67	29
A16	Village House, Lam Hei Road	2024 / 2027	1.5 / 10	**1701**	**123**	50	68	29
A17	Sheltered Structure, Lam Hei Road	2024 / 2027	1.5 / 5	**1326**	**163**	**79**	74	34
A18	Sheltered Structure no. 66 Kiu Hing Road	2024 / 2025	1.5	**2882**	**201**	**84**	**79**	35
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	2028	1.5 / 10	**964**	**103**	43	64	27
A20	House no. 89A, Lam Hau Tsuen	2027 / 2034	1.5	**2003**	**113**	45	67	27
A21	House no. 324, Shan Ha Tsuen	2025 / 2027 / 2033	1.5 / 10	**875**	97	45	65	27

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A22	House no. 645, Shan Ha Tsuen	2025 / 2028	1.5 / 5	**1645**	**123**	**58**	69	30
A23	House no. 193, Kung Um Road	2024 / 2025 / 2027	1.5 / 10	**2139**	**156**	**54**	**76**	29
A24	House no. 132, Sun Mei Garden	2026	1.5	**3141**	**226**	**91**	71	35
A25	House no. 293, Kung Um Road	2028 / 2029	1.5	**4165**	**242**	**61**	**81**	30
A26	Block 10, Chun Fai Garden	2026 / 2028 / 2029	1.5 / 5	**2618**	**133**	47	68	27
A27	House no. 161, Kung Um Road	2028 / 2029	1.5 / 5	**2452**	**122**	48	68	28
A28	Village House, Kung Um Road	2026 / 2029	1.5 / 5	**1645**	**127**	42	70	26
A29	House no. 366A, Kung Um Road	2028 / 2033	1.5 / 5	**2121**	**123**	44	67	27
A30	Sheltered Structure, Kung Um Road	2034	1.5	**5248**	**279**	**94**	**78**	**36**
A31	House no. 31B, Kung Um Road	2026 / 2027 / 2034	1.5 / 5	**1925**	**136**	47	70	28
A32	House no. 241, Kiu Hing Road	2026 / 2027 / 2034	1.5 / 5	**2828**	**144**	48	74	28
A33	House no. 67A, Wong Nai Tun Tsuen	2034	1.5	**2972**	**145**	48	68	28
A34	House no. 128, Kung Um Road	2035	1.5	**3375**	**186**	48	**79**	27
A35	House no. 117, Kung Um Road	2035	1.5 / 5	**1417**	**120**	40	69	27
A36	Sheltered Structure no. 375, Tai To Tsuen	2024 / 2025	1.5 / 5	**992**	100	42	67	27
A37	Uptown Tower 1	2025 / 2028	1.5 / 10 / 15	**898**	98	43	65	27

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A38	Chinese Mission Seminary	2024 / 2025 / 2028	1.5 / 10	**787**	98	41	64	26
A39	Fui Sha Wai Playground	2027 / 2028	1.5 / 5	**964**	**109**	48	66	28
A40	Tower 8, Imperial Villas II	2028 / 2029	1.5 / 10 / 15	**835**	**101**	43	65	27
A41	Energy Industrial Centre Block B	2027 / 2028 / 2029	1.5	**1092**	**108**	**51**	68	29
A42	Block 1, Parkside Villa	2024 / 2028	1.5	**1736**	**119**	**51**	64	29
A43	Block 2, Emerald Green	2024	1.5 / 10 / 15	**795**	95	41	64	27
A44	House no. 49A, Lung Tin Tsuen	2024 / 2025	1.5 / 5 / 10	**1278**	**113**	50	66	29
A45	House no. 139A	2024	1.5 / 10	**1039**	99	46	65	28
A46	House no. 101, Sham Chung Tsuen	2024 / 2025	1.5 / 10	**1114**	**126**	**53**	70	29
A47	Meadowlands Block 33	2024 / 2025	1.5 / 10 / 15	**779**	92	40	63	27
A48	Po Kok Branch School	2024 / 2025	1.5 / 10 / 15	**872**	97	40	63	27
A49	Treasure Court Block 7	2024 / 2025	1.5 / 15	**761**	93	40	62	26
A50	Beauty Court Block 2	2025 / 2028	1.5 / 10 / 20	**711**	91	39	62	26
A51	Park Nara Tower 1	2025 / 2028	1.5 / 20	**763**	99	42	63	27
A52	Hung Uk Garden	2027 / 2028	1.5 / 10	**628**	98	41	64	26
A53	Green Lodge House 16	2027 / 2028	1.5 / 10	**769**	91	43	63	27
A54	Ping Shan Garden Block 6	2027 / 2028	1.5 / 10	**556**	89	39	63	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A55	Villa Sunshine Block 1	2024 / 2028	1.5 / 10 / 15	**539**	88	39	63	26
A56	Park Royale Block 10	2024 / 2028 / 2029	1.5 / 20	446	85	37	62	26
A57	Park Royale Block 2	2024 / 2028	1.5 / 15	**648**	89	39	63	26
A58	Po Leung Kuk Law's Foundation School	2028	1.5 / 10	**898**	91	40	63	26
A59	Yuen Long Public Secondary School	2024 / 2028	1.5 / 10 / 15	**818**	93	40	63	26
A60	Villa Art Deco Block 2	2024 / 2028	1.5 / 10 / 15	**670**	92	39	64	26
A61	Gertrude Simon Lutheran College	2024	1.5 / 10 / 15	**642**	91	39	63	26
A62	Yuen Long Public Middle School Alumni Association Primary School	2024 / 2028	1.5 / 10 / 20	**585**	90	40	63	26
A63	Silver Field Garden Block 17	2024 / 2025 / 2027	1.5 / 10	**646**	90	40	63	26
A64	La Grove Block 1	2024	1.5	**2044**	**200**	**98**	**78**	**37**
A65	Ma Tin Tsuen House 242	2024	1.5 / 10	**915**	88	40	63	26
A66	Teng Lung Villa	2024 / 2028	1.5 / 10	**977**	87	38	62	26
A67	Yuen Long Baptist Church	2025 / 2028 / 2029	1.5 / 10	**601**	88	37	63	26
A68	Fraser Village House 54	2024 / 2025 / 2028	1.5 / 10	**806**	92	39	63	26
A69	The Brand	2028 / 2029	1.5 / 10 / 20	**726**	92	39	64	26
A70	Sereno Verde Block 2	2028 / 2029	1.5 / 10	**698**	**103**	43	65	27

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A71	La Pradera Block 12	2028 / 2029	1.5 / 10 / 20	**605**	90	38	62	26
A72	The Reach Tower 2	2028 / 2029	1.5 / 15	436	90	38	62	26
A73	The Reach Tower 6	2028 / 2029	1.5	**907**	**106**	45	66	27
A74	Tai Kei Leng House 145	2028 / 2029	1.5 / 10	457	89	38	63	26
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	2028 / 2029	1.5 / 20	**513**	86	37	62	26
A76	Grand de Sol Block 15	2028 / 2029	1.5 / 10 / 20	**505**	87	37	62	26
A77	Grand de Sol Block 8	2028 / 2029	1.5 / 15	410	91	38	62	26
A78	Hoover Garden Block 4	2028 / 2029	1.5	**743**	**117**	47	68	28
A79	Ha Yau Tin Tsuen House 2	2028 / 2029	1.5 / 10	334	91	38	63	26
A80	Buddhist Wing Yan School	2028 / 2029	1.5 / 10 / 20	470	88	37	62	26
A81	Fortune Centre	2025 / 2028 / 2029	1.5 / 20	452	85	37	62	26
A82	YOHO Town Block 6	2028 / 2029	1.5	**860**	**119**	49	68	28
A83	YOHO Town Block 9	2028 / 2029	1.5 / 5	**505**	92	39	63	26
A84	YOHO Midtown Block 5	2028 / 2029	1.5 / 5	**933**	**103**	44	65	28
A85	Ho Shun Yee Building Block 2	2024 / 2028 / 2029	1.5 / 20	415	87	38	63	26
A86	Cheong Wai Building	2025 / 2028 / 2029	1.5 / 10 / 20	402	86	37	63	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A87	Kwong Ming Ying Loi School	2028 / 2029	1.5 / 10	**966**	95	39	65	26
A88	Sun Yuen Long Centre Block 5	2028 / 2029	1.5 / 10	432	88	38	63	27
A89	Tung Tau Tsuen House 2	2028 / 2029	1.5 / 10	276	85	36	62	26
A90	Tai Wai Tsuen House 30	2028 / 2029	1.5 / 10 / 15	**515**	87	37	62	26
A91	Small Traders New Village	2028 / 2029	1.5 / 10	383	83	36	62	26
A92	Pok Oi Hospital	2028 / 2029	1.5 / 15	374	83	36	62	26
A93	Yeung Uk Tsuen House 10	2028 / 2029 / 2035	1.5 / 10	352	83	35	61	25
A94	Chuk San Tsuen House 17	2028 / 2029	1.5 / 10	**914**	87	36	62	25
A95	Greenfield Lodge Block 1	2028 / 2029	1.5 / 10	433	86	36	62	25
A96	Kong Tau San Tsuen House 5	2028 / 2029	1.5 / 10	**841**	92	38	65	26
A97	Tai Kei Leng House 414	2028 / 2029	1.5 / 5	**960**	**113**	42	70	27
A98	Shung Ching San Tseun House 49	2028 / 2029	1.5 / 10	**1196**	**120**	44	69	27
A99	Silver Garden House 125B	2024 / 2028 / 2029	1.5 / 10	**909**	91	38	63	26
A100	Tin Liu Tsuen House 32	2024 / 2026 / 2028	1.5 / 10	**1591**	**108**	47	67	28
A101	Pak Sha Tsuen	2026	1.5	**1755**	**163**	48	**77**	28
A102	Wong Nai Tun Tsuen House 47C	2026 / 2027	1.5	**1679**	**158**	47	75	27

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A103	Greenwood Gardens House 397	2026 / 2028 / 2034	1.5 / 10	**1010**	90	37	62	25
A104	Park Signature Block 1	2024	1.5 / 15	**1142**	**129**	**60**	69	31
A105	Market of Hung Fuk Estate	2025 / 2028 / 2029	1.5 / 10	**619**	91	42	63	27
A106	Village House along Kung Um Road	2026 / 2028 / 2029	1.5	**5000**	**198**	**58**	**78**	29
A107	Village House along Kung Um Road	2026 / 2027	1.5	**2990**	**231**	**63**	**85**	30
A108	Shan Ha Tsuen House No. 342	2025 / 2033	1.5 / 10	**1373**	**113**	**54**	66	29
A109	Shan Ha Tsuen House No. 343A	2025	1.5	**2138**	**146**	**65**	69	31
A110	Shan Ha Tsuen House No. 345C	2025 / 2028 / 2035	1.5 / 5 / 10	**1607**	**101**	43	64	27
A111	Shan Ha Tsuen House No. 346	2025 / 2035	1.5	**2130**	**115**	44	64	27
A112	Shan Ha Tsuen House No. 611	2027 / 2028	1.5	**3237**	**213**	**63**	**82**	31
A113	Shan Ha Tsuen House No. 613F	2025 / 2027	1.5	**3762**	**237**	**102**	**86**	**38**
A114	Sheung Yau Tin Tsuen	2028 / 2029	1.5 / 5	**550**	93	40	63	26
A115	Le Regent	2024 / 2025 / 2028	1.5 / 10	**1060**	**112**	49	66	28
A116	Planned Long Bin Development	2024 / 2028	1.5 / 15	481	87	38	63	26
A117	Planned Long Bin Development	2028	1.5 / 10	**906**	93	43	64	27
A118	Planned Long Bin Development	2028	1.5	**2424**	**198**	**91**	72	**37**

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A119	Proposed Hung Shui Kiu Development	2035 / 2036	1.5 / 10 / 15	**660**	92	39	63	26
A120	Yee Fung Garden Block A	2024 / 2028	1.5 / 20	**593**	85	38	62	26
A121	Ming Wan Court	2024 / 2028	1.5 / 10 / 20	**521**	85	37	62	26
A122	Yuen Long Villa House No. 252	2024 / 2028	1.5 / 10	**588**	85	38	62	26
A123	Sun Fai Court Block C	2024	1.5 / 10	**568**	85	38	62	26
A124	Caritas Yuen Long Chan Chun Ha Secondary School	2024 / 2028	1.5 / 10 / 20	**579**	85	37	62	26
A125	Village House along Ma Tong Road	2024 / 2028	1.5 / 10	**542**	86	37	62	26
A126	Proposed Hung Shui Kiu Development	2035 / 2036	1.5 / 10	489	90	38	62	26
A127	Uptown House 30	2024 / 2025	1.5 / 10	**1151**	**111**	48	65	28
A128	The Woodside Tower 5	2024 / 2025	1.5 / 15	**898**	98	42	63	26
A129	Shung Tak Catholic English College	2024 / 2025	1.5 / 15	**887**	98	41	63	26
A130	Village House along Castle Peak Road	2024 / 2025 / 2028	1.5 / 10	**821**	**101**	42	64	27
A131	Regent's Park	2026	1.5	**1995**	**195**	**58**	**79**	29
A132	One Hyde Park House 7	2026	1.5	**2341**	**199**	**59**	**81**	30
A133	Pak Sha Tsuen	2026 / 2027	1.5 / 10	**1540**	**137**	45	73	27
A134	Shan Ha Tsuen House No. 330	2025 / 2028 / 2033 / 2034	1.5 / 10	**1036**	**102**	43	65	27
A135	Lam Hau Tsuen House No. 110	2027 / 2028	1.5 / 10	**1674**	**134**	48	70	28

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A136	Evergreen Place Tower 5	2027 / 2028 / 2034	1.5 / 10 / 15	**736**	88	41	63	27
A137	Proposed Hung Shui Kiu Development	2035 / 2036	1.5 / 10 / 20	473	86	37	62	26
A138	Lok Kui Lau	2028 / 2034	1.5 / 15	**560**	89	39	63	26
A139	Ping Shan Home for The Aged	2027 / 2028	1.5 / 10	**505**	89	39	63	26
A140	Sheltered Structure along Tai Shu Ha Road East	2028 / 2029	1.5	**2341**	**206**	**82**	**81**	34
A141	Village House along Tai Shu Ha Road West	2028 / 2029	1.5	**3005**	**214**	**79**	**81**	34
A142	Sheltered Structure near Shap Pat Heung Interchange	2028 / 2029	1.5	**1039**	**139**	**55**	71	29
A143	Kong Tau San Tsuen House No, 61	2028 / 2029	1.5 / 5	**941**	100	39	65	26
A144	Pok Oi Hos. Jockey Club Care and Attention Home	2028 / 2029	1.5 / 10	397	84	36	62	26
A145	Proposed Hung Shui Kiu Development	2035 / 2036	1.5 / 15	**509**	87	38	62	26
A146	Hong Ping Villa Block 1	2024 / 2028	1.5 / 10	458	87	38	63	26
A147	Tai On Home for Aged	2027 / 2028 / 2029	1.5 / 15	**867**	95	43	65	27
A148	Ming Sum Home for the Sen	2028 / 2029	1.5 / 15	381	84	37	63	26
A149	Hung Fuk Estate	2025 / 2028 / 2035	1.5 / 20	**664**	90	41	62	26
A150	Grand Yoho Block 1	2028 / 2029	1.5 / 10	**603**	91	39	63	27
A151	Grand Yoho Block 5	2028 / 2029	1.5 / 10 / 15	**638**	90	38	63	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A152	Grand Yoho Block 9	2028 / 2029	1.5 / 10	**583**	85	38	62	26
A412	San Sang Tsuen	2028	1.5 / 10	313	83	36	61	25
A413	Greenville Park	2027 / 2028 / 2029	1.5 / 10	478	87	37	62	26
A601	Tseung Kong Wai	2027 / 2028	1.5 / 10	288	86	37	62	25
A602	Farm House	2027 / 2028	1.5	395	**101**	44	66	27
A603	Farm House	2027 / 2028	1.5	261	88	38	64	26
A701	Kau Lee Uk Tsuen	2027 / 2029	1.5 / 10	318	84	37	62	25
A702	San Uk Tsuen	2029	1.5 / 10	342	87	39	62	26
A703	Sha Chau Lei Tsuen	2028 / 2029	1.5 / 5 / 10	308	85	37	62	25
A704	Ha Tsuen Shi	2027 / 2028 / 2029	1.5 / 10	290	87	39	61	26
A705	Yan Wu Garden	2028	1.5 / 10	283	83	36	61	25
A706	Sik Kong Tsuen	2027 / 2028	1.5 / 10	286	82	36	61	25
A707	Pui Shing Catholic Secondary School	2028 / 2029	1.5 / 20	326	83	36	61	25
A708	Sik Kong Wai	2027 / 2028	1.5 / 10	267	83	36	61	25
A808	Tang Siu Tong Secondary School	2027 / 2028 / 2029	1.5 / 15 / 20	316	86	37	62	26
A813	Block H, Tin Shing Court	2028 / 2029	1.5 / 15 / 20	313	85	37	61	25
A1501	Ha Tsuen Weigh Station	2027 / 2028	1.5	298	91	40	65	27
Existing ASRs (within PDA boundary)								
W-A1	House 33, Park Villa	2036	1.5	**3171**	**280**	**70**	**92**	32

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
W-A2	House 11, Park Villa	2024 / 2025	1.5	**2617**	**173**	**53**	**76**	28
W-A3	Existing developments (under construction)	2036	1.5	**2430**	**282**	**68**	**90**	31
W-A4		2028 / 2029	1.5	**3936**	**326**	**117**	**93**	**40**
W-A5	Block 1, Recours La Serre	2028 / 2029	1.5 / 10	**2092**	**136**	**52**	70	28
W-A6	Block 2, The Parkhill	2035	1.5	**2501**	**133**	**53**	68	29
W-A7	Block 7, Greenville Residence	2035	1.5 / 10	**1510**	**109**	48	65	28
W-A8	Block 6, Windsor Garden	2035	1.5	**2466**	**132**	**60**	67	30
W-A9	Block 1, Marbella Garden	2028 / 2035	1.5 / 10	**1615**	**124**	50	65	28
W-A11	Kisland Villa Phase 2	2035	1.5	**2999**	**184**	**57**	69	29
W-A13	Sha Tseng Tsuen	2027 / 2028 / 2034	1.5 / 5 / 10	**1695**	**111**	46	66	28
W-A14		2034	1.5	**4871**	**325**	**75**	**86**	32
Planned ASRs (PDA Area 2 and Area 3)								
E-P1	Planned Residential	2028	1.5 / 10	**1428**	**122**	46	66	27
E-P2	Planned Residential	2033 / 2034	1.5 / 10	**539**	84	37	62	26
E-P5	Planned Residential	2028 / 2029	1.5 / 5	**3125**	**201**	**54**	**79**	29
E-P6	Planned Residential	2033	1.5 / 10	**1298**	91	39	65	26
E-P7	Planned Residential	2033	1.5 / 5	**2281**	**124**	46	70	28
E-P8	Planned Residential	2033 / 2034	1.5 / 10	**1206**	86	39	62	26
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	2033 / 2034	1.5 / 5	**2136**	**116**	47	63	28
E-P15	Planned School	2033 / 2034	1.5 / 10	**1632**	86	37	62	26
E-P17	Planned School	2033	1.5 / 10	**1847**	91	37	62	26
Planned ASRs (within Hung Shui Kiu NDA)								

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
P806	Planned Committed Commercial Area	2027 / 2028	1.5	418	93	40	63	26
P807	Planned Committed Commercial Area	2027 / 2028 / 2029	1.5 / 20	399	92	40	62	26
P1036	Planned Port Back-up, Storage and Workshop	2027 / 2028	1.5 / 5	413	94	43	64	27

Note:
[1] Value in bold means exceedance of its respective criteria.

4.4.6 Recommended Mitigation Measures

4.4.6.1 In order to reduce the dust emission from the Project and achieve compliances of relevant criteria at ASRs, regular watering under a good site practice should be adopted. In accordance with the "*Control of Open Fugitive Dust Sources*" (USEPA AP-42) as given in **Appendix 4.6**, watering once per hour on exposed worksites and haul road is proposed to achieve dust removal efficiency of 91.7%. These dust suppression efficiencies are derived based on the average haul road traffic of 40 per hour, average evaporation, etc. (see **Appendix 4.6**). Any potential dust impact and watering mitigation would be subject to the actual site conditions. For example, for a construction activity that produces inherently wet conditions or in cases under rainy weather, the above water application intensity may not be unreservedly applied. While the above watering frequencies are to be followed, the extent of watering may vary depending on actual site conditions. The dust levels would be monitored and managed under an Environmental Monitoring and Audit (EM&A) programme as specified in the EM&A Manual.

4.4.6.2 In addition, the Contractor is also obliged to follow the procedures and requirements given in the Air Pollution Control (Construction Dust) Regulation. It stipulates the construction dust control requirements for both Notifiable (e.g. site formation) and Regulatory (e.g. road opening) Works to be carried out by the Contractor. The following dust suppression measures should be incorporated by the Contractor to control the dust nuisance throughout the construction phase:

- Any excavated or stockpile of dusty material should be covered entirely by impervious sheeting or sprayed with water to maintain the entire surface wet and then removed or backfilled or reinstated where practicable within 24 hours of the excavation or unloading;
- Any dusty materials remaining after a stockpile is removed should be wetted with water and cleared from the surface of roads;
- A stockpile of dusty material should not be extended beyond the pedestrian barriers, fencing or traffic cones;

- The load of dusty materials on a vehicle leaving a construction site should be covered entirely by impervious sheeting to ensure that the dusty materials do not leak from the vehicle;
- Where practicable, vehicle washing facilities with high pressure water jet should be provided at every discernible or designated vehicle exit point. The area where vehicle washing takes place and the road section between the washing facilities and the exit point should be paved with concrete, bituminous materials or hardcores;
- When there are open excavation and reinstatement works, hoarding of not less than 2.4m high should be provided as far as practicable along the site boundary with provision for public crossing. Good site practice shall also be adopted by the Contractor to ensure the conditions of the hoardings are properly maintained throughout the construction period;
- The portion of any road leading only to construction site that is within 30m of a vehicle entrance or exit should be kept clear of dusty materials;
- Surfaces where any pneumatic or power-driven drilling, cutting, polishing or other mechanical breaking operation takes place should be sprayed with water or a dust suppression chemical continuously;
- Any area that involves demolition activities should be sprayed with water or a dust suppression chemical immediately prior to, during and immediately after the activities so as to maintain the entire surface wet;
- Where a scaffolding is erected around the perimeter of a building under construction, effective dust screens, sheeting or netting should be provided to enclose the scaffolding from the ground floor level of the building, or a canopy should be provided from the first floor level up to the highest level of the scaffolding;
- Any skip hoist for material transport should be totally enclosed by impervious sheeting;
- Every stock of more than 20 bags of cement or dry pulverised fuel ash (PFA) should be covered entirely by impervious sheeting or placed in an area sheltered on the top and the 3 sides;
- Cement or dry PFA delivered in bulk should be stored in a closed silo fitted with an audible high level alarm which is interlocked with the material filling line and no overfilling is allowed;
- Loading, unloading, transfer, handling or storage of bulk cement or dry PFA should be carried out in a totally enclosed system or facility, and any vent or exhaust should be fitted with an effective fabric filter or equivalent air pollution control system; and
- Exposed earth should be properly treated by compaction, turfing, hydroseeding, vegetation planting or sealing with latex, vinyl, bitumen, shortcrete or other suitable surface stabiliser within six months after the last construction activity on the construction site or part of the construction site where the exposed earth lies.

4.4.7 Assessment Results (Mitigated)

Tier 1 Screening Test

4.4.7.1 With implementation of the abovementioned mitigation measures, the maximum mitigated 1-hour TSP concentrations, and 10th highest 24-hour and annual RSP / FSP concentrations among Years 2023, 2024, 2025, 2026, 2027, 2028, 2029, 2033, 2034, 2035 and 2036 are calculated and presented in **Table 4.12** below and detailed in **Appendix 4.7**. Results indicate that there are no exceedances of respective criteria predicted at all ASRs, except the following:

- Exceedance of 1-hour TSP criterion at :
  - A106 (Village House along Kung Um Road) at 1.5mAG in Year 2028
- Exceedance of 10th highest 24-hour RSP criterion at :
  - A30 (Village House, Kung Um Road) at 1.5mAG in Year 2034;
  - A34 (House no. 128, Kung Um Road) at 1.5mAG in Year 2035; and
  - W-A3 (Existing developments (under construction)) at 1.5mAG in Year 2036.

4.4.7.2 According to the assessment results presented in **Appendix 4.7** and **Table 4.12**, the worst construction dust impact on identified ASRs generally occur at ground level (i.e. 1.5m above ground) due to the at-grade construction site. For example, at ASR A106 (Village House along Kung Um Road), the predicted mitigated 1-hour TSP concentrations at 1.5m above ground in Year 2028 is 509μg/m^3, and would significantly drop to 461μg/m^3 and 343μg/m^3 at 5m and 10m above ground respectively. ASRs located at higher level would generally be less affected as particulates would likely settle before reaching higher level.

4.4.7.3 Contours of mitigated 1-hour TSP concentrations, and 10th highest and annual RSP / FSP concentrations at 1.5m above ground for Year 2028 are illustrated in **Figures 4.7a-e**. In addition, as mentioned in **Section 4.4.7.1**, exceedances at ASRs A30, A34, and W-A3 are also predicted in different years. Marginal concentrations of the cumulative 10th highest 24-hour RSP are also predicted ASR W-A1 and W-A4. To this end, the following contours are also plotted to illustrate the pollutant concentrations:

- **Figure 4.7f** : 10th highest 24-hour RSP concentrations at 1.5mAG in Year 2034 (near ASR A30)
- **Figure 4.7g** : 10th highest 24- hour RSP concentrations at 1.5mAG in Year 2035 (near ASR A34)
- **Figure 4.7h** : 10th highest 24- hour RSP concentrations at 1.5mAG in Year 2036 (near ASR W-A1 and W-A3)
- **Figure 4.7i** : 10th highest 24- hour RSP concentrations at 1.5mAG in Year 2029 (near ASR W-A4)

4.4.7.4 In summary, contours indicate that there are no exceedances on air sensitive uses except the abovementioned ASRs and pollutant parameters.

Table 4.12 Mitigated cumulative TSP, RSP and FSP concentrations (Tier 1)

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
Existing / Planned ASRs (outside PDA boundary)								
A2	House no. 40, Tai Tao Tsuen	2024 / 2025	1.5 / 10	260	88	41	63	27
A3	House no. 95, Tai Tao Tsuen	2034 / 2035	1.5	245	89	41	63	27
A4	Sheltered Structure, Fui Sha Wai South Road	2025 / 2028 / 2036	1.5 / 5	214	86	39	62	26
A6	House no. 176A, Fui Sha Wai	2027 / 2028	1.5 / 5	297	92	42	63	27
A7	Village House, Ping Tong Street South	2028	1.5 / 5	279	91	42	64	27
A8	House no. 48, Tong Yan San Tsuen Road	2027 / 2028	1.5 / 5	231	88	42	64	27
A9	New Territories Assemblies of God Church	2028 / 2034 / 2035	1.5 / 10	215	85	38	63	26
A10	Block 10, Jasper Court	2028 / 2035	1.5 / 5	220	88	40	64	27
A11	School	2028 / 2035	1.5 / 5	217	86	38	63	26
A12	Village House, Lam Yu Road	2028	1.5	279	89	44	64	27
A13	Elchk Lutheran Academy	2024 / 2027 / 2028	1.5	213	86	37	63	26
A15	Village House, Lam Hei Road	2024	1.5	219	86	38	63	26
A16	Village House, Lam Hei Road	2024 / 2027	1.5 / 10	227	86	38	63	26
A17	Sheltered Structure, Lam Hei Road	2024	1.5	236	91	42	63	27
A18	Sheltered Structure no. 66 Kiu Hing Road	2024	1.5	322	96	42	64	27

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	2028 / 2034 / 2035	1.5 / 5 / 10	213	85	37	63	26
A20	House no. 89A, Lam Hau Tsuen	2034 / 2035	1.5	220	88	40	64	26
A21	House no. 324, Shan Ha Tsuen	2025 / 2027	1.5 / 5	223	87	38	64	26
A22	House no. 645, Shan Ha Tsuen	2024 / 2028 / 2029	1.5 / 10	226	89	40	64	26
A23	House no. 193, Kung Um Road	2024 / 2027 / 2028	1.5 / 5	252	90	38	64	26
A24	House no. 132, Sun Mei Garden	2026 / 2028	1.5	296	85	43	62	27
A25	House no. 293, Kung Um Road	2028 / 2029	1.5	435	99	43	64	27
A26	Block 10, Chun Fai Garden	2028 / 2029	1.5 / 5	299	91	38	62	26
A27	House no. 161, Kung Um Road	2028 / 2029	1.5 / 5	300	90	38	62	26
A28	Village House, Kung Um Road	2029	1.5 / 5	227	88	37	62	25
A29	House no. 366A, Kung Um Road	2034	1.5 / 5	232	88	39	63	26
A30	Sheltered Structure, Kung Um Road	2034	1.5	465	**103**	48	66	28
A31	House no. 31B, Kung Um Road	2026 / 2034	1.5 / 10	230	88	38	63	26
A32	House no. 241, Kiu Hing Road	2026 / 2034	1.5 / 5	294	89	37	63	26
A33	House no. 67A, Wong Nai Tun Tsuen	2034	1.5	298	90	38	63	26
A34	House no. 128, Kung Um Road	2035	1.5	392	**103**	38	65	26
A35	House no. 117, Kung Um Road	2026 / 2035	1.5	228	93	38	65	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A36	Sheltered Structure no. 375, Tai To Tsuen	2024 / 2025 / 2034	1.5 / 5	227	86	38	63	26
A37	Uptown Tower 1	2025	1.5 / 10	291	87	38	63	26
A38	Chinese Mission Seminary	2025 / 2035	1.5 / 10	211	85	37	62	26
A39	Fui Sha Wai Playground	2025 / 2028 / 2029	1.5 / 5	210	85	38	62	26
A40	Tower 8, Imperial Villas II	2025 / 2028 / 2029 / 2035	1.5 / 10	227	85	37	62	26
A41	Energy Industrial Centre Block B	2025 / 2028	1.5	461	93	42	64	28
A42	Block 1, Parkside Villa	2028 / 2035	1.5	218	86	39	63	26
A43	Block 2, Emerald Green	2024 / 2028 / 2035	1.5	219	83	37	62	26
A44	House no. 49A, Lung Tin Tsuen	2024 / 2028	1.5	219	85	38	62	26
A45	House no. 139A	2024 / 2028	1.5	219	84	38	62	26
A46	House no. 101, Sham Chung Tsuen	2024	1.5 / 5	221	87	38	63	26
A47	Meadowlands Block 33	2024 / 2025	1.5 / 15	218	84	37	62	26
A48	Po Kok Branch School	2025	1.5 / 10	213	84	37	62	26
A49	Treasure Court Block 7	2025	1.5	201	84	37	62	26
A50	Beauty Court Block 2	2025	1.5 / 5	204	84	37	62	25
A51	Park Nara Tower 1	2025	1.5 / 5	264	89	39	63	26
A52	Hung Uk Garden	2025 / 2035	1.5	208	84	37	62	25
A53	Green Lodge House 16	2028	1.5	213	85	37	63	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A54	Ping Shan Garden Block 6	2023 / 2028 / 2034	1.5	208	83	36	62	26
A55	Villa Sunshine Block 1	2028 / 2035	1.5 / 15	212	83	36	62	26
A56	Park Royale Block 10	2023 / 2028 / 2029	1.5	208	82	36	62	25
A57	Park Royale Block 2	2028 / 2029 / 2035	1.5 / 15	211	83	36	61	26
A58	Po Leung Kuk Law's Foundation School	2023 / 2028	1.5	212	84	37	63	26
A59	Yuen Long Public Secondary School	2023 / 2028	1.5	212	84	37	63	26
A60	Villa Art Deco Block 2	2023 / 2024 / 2028	1.5	213	84	37	63	26
A61	Gertrude Simon Lutheran College	2023 / 2024 / 2028	1.5	219	83	37	62	26
A62	Yuen Long Public Middle School Alumni Association Primary School	2023 / 2024 / 2028	1.5	219	83	37	62	26
A63	Silver Field Garden Block 17	2024 / 2028	1.5	219	83	37	62	26
A64	La Grove Block 1	2024	1.5	274	93	44	64	27
A65	Ma Tin Tsuen House 242	2024 / 2028	1.5	219	84	37	62	26
A66	Teng Lung Villa	2024 / 2028	1.5 / 10	219	83	36	62	26
A67	Yuen Long Baptist Church	2028 / 2029	1.5	219	83	36	62	26
A68	Fraser Village House 54	2028 / 2029 / 2034	1.5	219	84	37	62	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A69	The Brand	2028 / 2029	1.5	219	83	36	62	26
A70	Sereno Verde Block 2	2028 / 2029	1.5	231	84	37	62	26
A71	La Pradera Block 12	2028 / 2029	1.5	231	83	36	62	25
A72	The Reach Tower 2	2028 / 2029	1.5	231	83	36	62	25
A73	The Reach Tower 6	2028 / 2029	1.5 / 5	231	85	37	62	26
A74	Tai Kei Leng House 145	2028 / 2029	1.5	231	83	36	62	26
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	2028 / 2029	1.5	231	82	36	62	25
A76	Grand de Sol Block 15	2028 / 2029	1.5	231	83	36	62	25
A77	Grand de Sol Block 8	2028 / 2029	1.5 / 15	231	83	36	62	25
A78	Hoover Garden Block 4	2028 / 2029	1.5 / 5	232	85	37	62	26
A79	Ha Yau Tin Tsuen House 2	2028 / 2029	1.5 / 10	231	83	36	62	25
A80	Buddhist Wing Yan School	2028 / 2029	1.5	231	83	36	62	25
A81	Fortune Centre	2023 / 2028 / 2029	1.5	223	83	36	62	25
A82	YOHO Town Block 6	2028 / 2029	1.5 / 5	232	86	37	62	26
A83	YOHO Town Block 9	2028 / 2029	1.5	223	84	36	62	25
A84	YOHO Midtown Block 5	2023 / 2028 / 2029	1.5	223	85	37	64	26
A85	Ho Shun Yee Building Block 2	2028 / 2029	1.5	223	83	36	62	25

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A86	Cheong Wai Building	2023 / 2028 / 2029	1.5	223	83	36	63	26
A87	Kwong Ming Ying Loi School	2023 / 2028 / 2029	1.5	223	84	37	63	26
A88	Sun Yuen Long Centre Block 5	2023 / 2028 / 2029	1.5	223	83	37	62	26
A89	Tung Tau Tsuen House 2	2023 / 2027 / 2029 / 2033	1.5 / 10	222	82	36	62	25
A90	Tai Wai Tsuen House 30	2023 / 2028 / 2029 / 2033	1.5 / 15	222	82	36	62	25
A91	Small Traders New Village	2023 / 2028 / 2029	1.5	232	82	36	62	25
A92	Pok Oi Hospital	2023 / 2028 / 2029	1.5	232	82	36	62	26
A93	Yeung Uk Tsuen House 10	2023 / 2029 / 2035	1.5	232	82	35	61	25
A94	Chuk San Tsuen House 17	2023 / 2028 / 2029	1.5 / 10	238	83	36	62	25
A95	Greenfield Lodge Block 1	2023 / 2028 / 2029	1.5	238	82	35	62	25
A96	Kong Tau San Tsuen House 5	2028 / 2029	1.5 / 5	231	83	36	62	25
A97	Tai Kei Leng House 414	2028 / 2029	1.5 / 5	233	86	36	63	26
A98	Shung Ching San Tseun House 49	2028 / 2029	1.5 / 5	244	87	37	63	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m³) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A99	Silver Garden House 125B	2028 / 2029	1.5 / 10	228	84	36	62	25
A100	Tin Liu Tsuen House 32	2024 / 2026 / 2028	1.5 / 5	220	85	37	63	26
A101	Pak Sha Tsuen	2026	1.5	252	91	37	63	26
A102	Wong Nai Tun Tsuen House 47C	2026	1.5	244	90	37	63	25
A103	Greenwood Gardens House 397	2028 / 2034 / 2035	1.5	228	83	36	62	25
A104	Park Signature Block 1	2024	1.5 / 5	220	87	39	63	26
A105	Market of Hung Fuk Estate	2025 / 2033 / 2035	1.5 / 10	306	87	39	62	26
A106	Village House along Kung Um Road	2028 / 2029	1.5	**509**	96	42	63	26
A107	Village House along Kung Um Road	2026 / 2027	1.5	307	97	39	64	26
A108	Shan Ha Tsuen House No. 342	2025 / 2027	1.5 / 10	224	88	39	64	26
A109	Shan Ha Tsuen House No. 343A	2025 / 2034	1.5 / 10	227	90	41	65	27
A110	Shan Ha Tsuen House No. 345C	2028 / 2035	1.5 / 5	230	85	37	63	26
A111	Shan Ha Tsuen House No. 346	2028 / 2035	1.5	262	87	38	63	26
A112	Shan Ha Tsuen House No. 611	2027 / 2028	1.5	343	94	41	65	27
A113	Shan Ha Tsuen House No. 613F	2025 / 2027	1.5	376	97	44	66	27
A114	Sheung Yau Tin Tsuen	2028 / 2029	1.5 / 5	231	84	37	62	26
A115	Le Regent	2034 / 2035	1.5	212	88	40	63	26
A116	Planned Long Bin Development	2028 / 2035	1.5 / 15	212	83	36	62	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A117	Planned Long Bin Development	2028 / 2035	1.5 / 15	215	86	38	63	26
A118	Planned Long Bin Development	2028	1.5	265	90	45	64	28
A119	Proposed Hung Shui Kiu Development	2034 / 2035	1.5 / 10	205	84	37	62	26
A120	Yee Fung Garden Block A	2028 / 2029	1.5	219	83	36	62	26
A121	Ming Wan Court	2023 / 2028 / 2029	1.5	215	83	36	62	26
A122	Yuen Long Villa House No. 252	2024 / 2028	1.5	219	83	36	62	26
A123	Sun Fai Court Block C	2024 / 2028	1.5	219	83	36	62	26
A124	Caritas Yuen Long Chan Chun Ha Secondary School	2028 / 2029	1.5	219	83	36	62	26
A125	Village House along Ma Tong Road	2028 / 2029	1.5	219	83	36	62	26
A126	Proposed Hung Shui Kiu Development	2033 / 2035	1.5 / 10	244	84	37	62	26
A127	Uptown House 30	2035 / 2036	1.5	225	89	39	63	26
A128	The Woodside Tower 5	2025	1.5 / 5	253	86	38	62	26
A129	Shung Tak Catholic English College	2025	1.5 / 5	231	85	37	62	26
A130	Village House along Castle Peak Road	2025 / 2035	1.5 / 10	209	85	37	62	26
A131	Regent's Park	2026	1.5	272	94	38	64	26
A132	One Hyde Park House 7	2026	1.5	270	94	38	63	26
A133	Pak Sha Tsuen	2026 / 2034	1.5 / 5	229	88	37	63	26
A134	Shan Ha Tsuen House No. 330	2025 / 2027 / 2028 / 2034	1.5 / 10	224	87	38	64	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A135	Lam Hau Tsuen House No. 110	2027 / 2028	1.5 / 10	245	88	38	63	26
A136	Evergreen Place Tower 5	2023 / 2028 / 2035	1.5	213	85	37	63	26
A137	Proposed Hung Shui Kiu Development	2033 / 2035	1.5 / 15	211	83	36	61	25
A138	Lok Kui Lau	2028 / 2034 / 2035	1.5 / 10 / 15	211	83	37	62	26
A139	Ping Shan Home for The Aged	2023 / 2028 / 2034	1.5 / 10	208	83	36	62	26
A140	Sheltered Structure along Tai Shu Ha Road East	2028 / 2029	1.5	285	95	42	64	27
A141	Village House along Tai Shu Ha Road West	2028 / 2029	1.5	334	93	41	64	27
A142	Sheltered Structure near Shap Pat Heung Interchange	2028 / 2029	1.5 / 5	232	89	38	63	26
A143	Kong Tau San Tsuen House No, 61	2028 / 2029	1.5 / 5	231	85	36	62	25
A144	Pok Oi Hos. Jockey Club Care and Attention Home	2023 / 2028 / 2029	1.5	232	83	36	62	26
A145	Proposed Hung Shui Kiu Development	2035	1.5	200	83	36	61	25
A146	Hong Ping Villa Block 1	2028 / 2035	1.5 / 10	212	83	36	62	26
A147	Tai On Home for Aged	2028 / 2035	1.5 / 10	236	85	38	62	26
A148	Ming Sum Home for the Sen	2023 / 2028 / 2029	1.5	232	83	36	62	26
A149	Hung Fuk Estate	2025 / 2033 / 2035	1.5 / 10	244	86	39	62	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	500	100	50	75	35
A150	Grand Yoho Block 1	2023 / 2028 / 2029	1.5	222	83	37	63	26
A151	Grand Yoho Block 5	2023 / 2028 / 2029 / 2035	1.5	222	82	36	62	26
A152	Grand Yoho Block 9	2023 / 2028 / 2029	1.5	223	83	37	62	26
A412	San Sang Tsuen	2027 / 2028	1.5	200	82	36	61	25
A413	Greenville Park	2027 / 2028	1.5	200	83	36	61	25
A601	Tseung Kong Wai	2027 / 2028	1.5 / 10	288	86	37	62	25
A602	Farm House	2027 / 2028	1.5	395	97	44	66	27
A603	Farm House	2027 / 2028	1.5	234	87	37	64	26
A701	Kau Lee Uk Tsuen	2027 / 2028	1.5	199	83	36	61	25
A702	San Uk Tsuen	2027 / 2028	1.5 / 5	199	83	36	61	25
A703	Sha Chau Lei Tsuen	2027 / 2028	1.5	199	83	36	61	25
A704	Ha Tsuen Shi	2027 / 2028	1.5	199	83	36	61	25
A705	Yan Wu Garden	2027 / 2028	1.5	199	82	35	61	25
A706	Sik Kong Tsuen	2027 / 2028	1.5	199	81	35	61	25
A707	Pui Shing Catholic Secondary School	2027 / 2028	1.5	199	81	35	61	25
A708	Sik Kong Wai	2027 / 2028	1.5 / 10	215	82	35	61	25

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
A808	Tang Siu Tong Secondary School	2027 / 2028	1.5	203	82	36	61	25
A813	Block H, Tin Shing Court	2027 / 2028	1.5 / 10	203	81	36	61	25
A1501	Ha Tsuen Weigh Station	2028	1.5	252	91	39	65	27
Existing ASRs (within PDA boundary)								
W-A1	House 33, Park Villa	2025 / 2036	1.5	362	100	39	65	26
W-A2	House 11, Park Villa	2024 / 2025 / 2036	1.5	305	90	39	63	26
W-A3	Existing developments (under construction)	2025 / 2029 / 2036	1.5	315	**101**	39	64	26
W-A4		2028 / 2029 / 2036	1.5	410	100	46	65	27
W-A5	Block 1, Recours La Serre	2028 / 2029 / 2035	1.5 / 10	245	87	38	63	26
W-A6	Block 2, The Parkhill	2035	1.5	253	94	41	64	26
W-A7	Block 7, Greenville Residence	2035	1.5 / 5	223	90	40	64	26
W-A8	Block 6, Windsor Garden	2035	1.5	263	96	42	64	26
W-A9	Block 1, Marbella Garden	2028 / 2035	1.5 / 10	240	91	41	64	26
W-A11	Kisland Villa Phase 2	2035	1.5	298	93	44	64	27
W-A13	Sha Tseng Tsuen	2027 / 2035	1.5 / 10	242	89	40	64	26
W-A14		2034 / 2035	1.5	456	99	48	65	28
Planned ASRs (PDA Area 2 and Area 3)								
E-P1	Planned Residential	2028 / 2029	1.5 / 5	221	86	38	63	26
E-P2	Planned Residential	2034	1.5	220	84	37	62	26

ASR ID	Location	Worst affected Year	Worst affected Height above Ground	Pollutant Concentration (μg/m^3) Note [1]				
				TSP	RSP		FSP	
				Max. 1-hour	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
			Criteria	**500**	**100**	**50**	**75**	**35**
E-P5	Planned Residential	2028 / 2029	1.5 / 5	341	95	41	64	26
E-P6	Planned Residential	2034	1.5 / 5	224	88	38	64	26
E-P7	Planned Residential	2033 / 2034	1.5 / 5	282	91	41	65	26
E-P8	Planned Residential	2034	1.5	227	85	37	62	26
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	2034	1.5	227	89	39	63	26
E-P15	Planned School	2033 / 2034	1.5	227	85	36	62	25
E-P17	Planned School	2033 / 2034	1.5	227	84	36	62	25
Planned ASRs (within Hung Shiu Kiu NDA)								
P806	Planned Committed Commercial Area	2027 / 2028	1.5	418	89	39	62	26
P807	Planned Committed Commercial Area	2027 / 2028	1.5	348	87	39	61	26
P1036	Planned Port Back-up, Storage and Workshop	2028	1.5 / 5	356	94	43	64	27

Note:
[1] Value in bold means exceedance of its respective criteria.

4.4.7.5 Since the Tier 1 screening test has adopted very conservative assumptions that all the worksites would be active, which would unlikely occur in reality, a more focused Tier 2 assessment is therefore conducted for these ASRs in order to predict the construction dust impact under more practicable assumptions.

Tier 2 Assessment

4.4.7.6 As discussed in previous section, exceedance of 1-hour TSP and 10th highest 24-hour RSP concentrations are predicted at some ASRs under Tier 1 screening test. A more focused Tier 2 assessment has been conducted for these ASRs. Tier 2 assessment adopts a more practicable assumption that any active construction site at any one time in each phase would be 50m x 50m (or 2,500m^2) in size. This assumption is provided and confirmed by the Project Engineer.

4.4.7.7 The maximum Tier 2 1-hour TSP concentrations and 10th highest 24-hour RSP concentrations are presented in the following table and details are presented in **Appendix 4.7**. Results show that the predicted 1-hour TSP concentrations and 10th highest 24-hour RSP concentrations would comply with the criteria of 500μg/m^3 and

100μg/m^3 respectively. Hence, adverse short-term construction dust impact is not anticipated.

Table 4.13 Mitigated cumulative TSP and RSP concentrations (Tier 2)

ASR ID	Location	Worst affected Year	Worst affected Height	TSP Conc. (μg/m^3)	RSP Conc. (μg/m^3)
				Max. 1-hour	24-hour (10th highest)
			Criteria	500	100
Existing ASRs (outside PDA boundary)					
A30	Village House, Kung Um Road	2034	1.5	Note [1]	91s
A34	House no. 128, Kung Um Road	2035	1.5	Note [1]	93
A106	Village House along Kung Um Road	2028	1.5	366	Note [1]
Existing ASRs (within PDA boundary)					
W-A1	House 33, Park Villa	2036	1.5	Note [1]	94
W-A3	Existing Development (under Construction)	2036	1.5	Note [1]	91
W-A4	Existing Development (under Construction)	2028	1.5	Note [1]	90

Note:
[1] No exceedance is predicted under Tier 1 screening test and hence the Tier 2 assessment is not conducted for this parameter.

4.4.7.8 The worst hit level under Tier 2 assessment at all concerned ASRs is 1.5m above ground. Contours of Tier 2 1-hour TSP concentrations at 1.5m above ground are therefore plotted and presented in **Figures 4.7a. 4.7b, 4.7f, 4.7g, 4.7h,** and **4.7i**. Contour plots indicate that there are no air sensitive uses located within the area of exceedance, and hence adverse dust impact is not anticipated.

4.4.8 Residual Impacts

4.4.8.1 With the implementation of the mitigation measures as stipulated in the Air Pollution Control (Construction Dust) Regulation, dust control measures, including watering once per hour on exposed worksites and haul road, and good site practices, the predicted 1-hour TSP, 24-hour and annual RSP / FSP concentrations on all sensitive uses in the vicinity of the construction sites would comply with the respective criteria. Hence, no adverse residual air quality impact during construction phase is anticipated.

4.5 Operational Air Quality Impact Assessment

4.5.1 Identification of Pollution Sources

4.5.1.1 The key existing air pollution sources within the assessment area that may bear upon the air quality during operational phase include the vehicular emission from project roads and induced traffic, neighbouring roads such as Yuen Long Highway, and industrial emissions.

4.5.1.2 Other far-field emission sources outside the assessment area which would have certain influence on the background air quality level include territory wide vehicular

emission, power plants, marine emission, as well as regional emission from Pearl River Delta (PRD).

4.5.1.3 Other than the existing air pollution sources, it is anticipated that the future developments in the vicinity of the Project would induce additional traffic and hence incur additional emission burden which may also cause potential air quality impacts on the ASRs.

4.5.1.4 Specifically, the existing and potential near-field sources are described in the following sections below:

Vehicular Emission from Open Road

4.5.1.5 Major air pollution source in the vicinity of the Project during operational phase would be tailpipe emission generated from traffic along open road. Vehicular emissions from the existing road networks, and those arising from the concurrent projects due to planned road network or induced traffic, including the HSK NDA, planned housing sites in the vicinity of the Project, TMWB etc. that would have cumulative air quality impact on nearby ASRs have also been addressed. **Figure 4.9a** and **Figure 4.9b** illustrate the road networks within 500m assessment area which are considered as near-field sources in the operational air quality assessment.

Vehicular Emission from Portal of the Proposed Underpass and Proposed Full Enclosure

4.5.1.6 Other than vehicular emission along open road, those emitted from the portals of the proposed underpass near TYST Interchange and proposed full enclosure along Long Ha Road as noise mitigation measures under this Project would also cause cumulative air quality impact. **Figure 4.10** illustrates the locations of the proposed underpass and proposed full enclosures to be considered in the near-field model.

Industrial Emission

4.5.1.7 Chimney surveys (conducted in December 2012, January 2013, March 2013, October 2014, September 2015 and December 2016) and desktop study have been conducted to identify existing and planned, if any, chimneys within 500m study area. The chimney information, including fuel consumption rate, stack height, gas exhaust velocity, exhaust temperature and the internal diameter of the stack etc. have been collected from the respective operators where available. References have also been made to specified process (SP) licences.

4.5.1.8 As described in **Section 3**, there are no planned industrial uses under the current RODP, and only storage area and workshop uses are proposed adjacent to YLH near TYST. Hence, planned industrial emission from the Project is therefore not anticipated. **Table 4.14** lists the identified chimneys. **Figure 4.11** illustrates the locations of these chimneys.

Table 4.14 Identified chimneys within the 500m from the PDA boundary and associated road infrastructure

Source ID	Description
PC_CH01 – PC_CH10	Pun Chun Sauce & Preserved Fruit Factory Ltd.
HSCM_CH12	Hang Sun Chemical Mfg. Ltd. (HSCM)
WKDR_CH13	Wing Kai Destruction & Recycle Co.
HHOF_CH14	Hop Hing Oil Factory Ltd.

Source ID	Description
HHCF_CH16 – HFCF_CH27	Hang Heung Cake Factory [1]
HKCC_EP1A – HKCC_EP9	Specified Process – Hong Kong Concrete Company Ltd
RED_EP1 – RED_EP14	Specified Process – Redland Concrete Company Ltd [2]
GOL_EP1 – GOL_EP11	Specified Process – Golik Concrete Company Ltd [2]
POH_EP1 – POH_EP5	Pok Oi Hospital

Note:
[1] According to the information provided by the operator, only electricity is used to power all boiler within the factory. Hence, no industrial emission from these chimney is anticipated.
[2] These SPs will only operate until the proposed land resumption in July 2025.

4.5.2 Determination of Representative Air Pollutants

4.5.2.1 As discussed in **Section 4.1**, the APCO (Cap 311) and its subsidiary regulations define statutory AQOs for seven common air pollutants including NO_2, SO_2, RSP, FSP, CO, O_3 and lead. According to Appendix B, Clause 5 (ii) of the EIA Study Brief, the key / representative air pollutant parameters for the project shall be identified, including the types of pollutants and the averaging time concentration.

4.5.2.2 The air quality pollutant source during the operational phase of the project would be the emission from the vehicles travelling on the new and existing roads. The tailpipe emission would comprise a number of pollutants, including Nitrogen Oxides (NO_X), Respirable Suspended Particulates (RSP), Fine Suspended Particulates (FSP), Sulphur Dioxides (SO_2), Toxic Air Pollutants (TAP), Lead (Pb) etc. Determination of representative air pollutants for this Project is discussed in the following:

i) Nitrogen Dioxide (NO_2)

4.5.2.3 Nitrogen oxides (NO_x) is known to be one of the pollutants emitted by vehicles. Together with VOC and in the presence of O_3 under sunlight, NO_x would be transformed to NO_2. According to the "*2015 Hong Kong Emission Inventory Report*" published by EPD which is the latest available information at the time of preparing this report, the dominant source of NO_x generated in HK is the navigation which constitutes about 37% of the total in 2015. Road transport is the third largest NO_x emission group, accounting for about 18% of the total (see table below).

Table 4.15 The emission percentage and the amount of NOx in Hong Kong (2015)

Pollutant Source Categories	NO_x Emission %[1]	NO_x Emission (tons) [1]
Public Electricity Generation	28%	26,090
Road Transport	18%	16,200
Navigation	37%	33,900
Civil Aviation	5%	5,000
Other Combustion	11%	10,450
Non-combustion	N/A	N/A
Biomass Burning	<1%	60
Total	**100%**	**91,700**

Note:

[1] Figures extracted from 2015 Hong Kong Emission Inventory Report

4.5.2.4 Upon population intake of the proposed development, there would be an increase in the traffic flow and hence the NOx emission and subsequently the NO_2 concentrations near to the roadside. Hence, NO_2 is one of the key representative pollutants for the operational air quality assessment of the Project.

4.5.2.5 The 1-hour and annual average NO_2 concentrations at each identified ASR are assessed and compared with the AQO to determine their compliance.

ii) Respirable Suspended Particulates (RSP) and Fine Suspended Particulates (FSP)

4.5.2.6 RSP refers to suspended particulates with a nominal aerodynamic diameter of 10μm or less. According to the latest statistics of "*2015 Hong Kong Emission Inventory Report*", road transport only accounted for 9% and 10% of the total RSP / FSP emissions while navigation accounted for 34% and 39% respectively.

Table 4.16 The emission percentage and the amount of RSP in Hong Kong (2015)

Pollutant Source Categories	RSP		FSP	
	Emission (%)	Emission (tons)	Emission (%)	Emission (tons)
Public Electricity Generation	11%	580	7%	290
Road Transport	9%	490	10%	450
Navigation	34%	1,860	39%	1,690
Civil Aviation	1%	50	1%	50
Other Combustion	15%	800	17%	740
Non-combustion	17%	910	11%	470
Biomass Burning	14%	740	14%	600
Total	100%	**5,430**	100%	**4,300**

Note:

[1] Figures extracted from 2015 Hong Kong Emission Inventory Report

4.5.2.7 Upon population intake of the proposed development, there would be an increase in the traffic flow and hence the RSP concentrations near to the roadside. Hence, RSP is also one of the key representative pollutants for the operational air quality assessment of the Project.

4.5.2.8 FSP refers to suspended particulates with a nominal aerodynamic diameter of 2.5μm or less. Similar to RSP, FSP is sourced from fuel combustion, road vehicles, etc., and is also considered as one of the key representative pollutants for the operational air quality assessment of the Project.

4.5.2.9 The 24-hour and annual average RSP/ FSP concentrations at each identified ASR are assessed and compared with the AQOs to determine their compliance.

iii) Sulphur Dioxide (SO_2)

4.5.2.10 According to the latest statistics of "*2015 Hong Kong Emission Inventory Report*", the dominant source of SO_2 in Hong Kong is from navigation, which constitutes 59% of the total emissions. Road transport and other combustion only contribute to less than 1% of the total SO_2 emissions respectively. The introduction of ultra-low sulphur diesel for vehicle fleet and implementation of the Air Pollution Control (Fuel

Restriction) Regulations have reduced the SO_2 emission from road transport and fuel combustion in Hong Kong.

Table 4.17 The emission percentage and the amount of SO_2 in Hong Kong (2015)

Pollutant Source Categories	SO_2 Emission %[1]	SO_2 Emission (tons) [1]
Public Electricity Generation	37%	7,280
Road Transport	<1%	40
Navigation	59%	11,460
Civil Aviation	3%	510
Other Combustion	1%	240
Non-combustion	N/A	N/A
Biomass Burning	<1%	10
Total	100%	**19,540**

Note:

[1] Figures extracted from 2015 Hong Kong Emission Inventory Report

4.5.2.11 As discussed in **Section 4.2**, the latest 5-year average of 4th highest 24-hour SO_2 concentration in Yuen Long is only 20% of the AQOs, and future predicted 24-hour SO_2 background concentrations in the study area would be less than 25% of the AQOs in Year 2020. This clearly indicates that the AQOs for SO_2 could be well achieved with great margin in the study area. Given that road transport and other fuel combustion only contribute a very small amount of SO_2 and there is still a large margin to the AQOs compared to the other pollutants such as RSP and NO_2, it is appropriate to consider that SO_2 is not the key pollutant for quantitative assessment for the operational phase of the Project.

iv) Ozone (O_3)

4.5.2.12 Unlike other pollutants such as NOx, O_3 is not a primary pollutant emitted from man-made sources but is formed by a set of complex chain reactions between various chemical species, including NOx and VOC, in the presence of sunlight. According to Sun et al. (2010) the rate of formation of O_3, also known as Ozone Production Efficiency, depends not only on NOx and VOC levels, but atmospheric oxidation, temperature, radiation, and other meteorological factors in the atmosphere of different regions. With reference to EPD's "*Air Quality in Hong Kong 2015*", the formation of O_3 generally takes several hours to proceed and therefore O_3 recorded locally could be attributed to emissions generated from places afar.

4.5.2.13 According to EPD's "*A Study to Review Hong Kong's Air Quality Objectives*" (EPD, 2009), due to the abundance of its precursors (VOC and NOx) from a great variety of sources such as motor vehicles, industries, power plants and consumer products, etc., ozone can be widely formed in the region and can be transported over long distance. The general rising trend of ozone levels in Hong Kong over the past years reflects an aggravation in the photochemical smog problem on a regional scale. All these indicate that local traffic emission is not a dominant controlling factor in O_3 formation.

4.5.2.14 In addition, the EPD's "*Air Quality in Hong Kong 2015*" report stated that NOx emissions from motor vehicles and chimneys have the potential to react with and remove O_3 in the air, and regions with heavy traffic normally have lower ozone levels than areas with light traffic. It is therefore possible that the Project may contribute to

a decrease in O_3 in the immediate area along main roads. O_3 is therefore not considered as a key parameter in this assessment.

v) Lead (Pb)

4.5.2.15 As leaded petrol has been banned in Hong Kong since in 1999, it is no longer considered as a primary source in Hong Kong. According to the "*Air Quality in Hong Kong 2015*" by EPD, the measured annual averaged lead level ranged from 22ng/m^3 (at Central/Western, Kwun Tong, Shum Shui Po, Kwai Chung and Mong Kok) to 29ng/m^3 (Yuen Long), which were well below the annual AQO of 500ng/m^3. Therefore, lead is not considered as a key / representative air pollutant for the operational air quality assessment.

vi) Carbon Monoxide (CO)

4.5.2.16 Carbon Monoxide (CO) is one of the primary pollutants emitted by road transport. According to the latest statistics of "*2015 Hong Kong Emission Inventory Report*" CO emissions from road transport contributed about 51% of total CO emission in 2015 (see the table below).

Table 4.18 The emission percentage and the amount of CO in Hong Kong (2015)

Pollutant Source Categories	CO Emission % [1]	CO Emission (tonnes)[1]
Public Electricity Generation	6%	3,580
Road Transport	51%	29,700
Navigation	23%	13,280
Civil Aviation	7%	3,950
Other Combustion	10%	5,920
Non-combustion	N/A	N/A
Biomass Burning	3%	1,720
Total	**100%**	**58,150**

Note:

[1] Figures extracted from 2015 Hong Kong Emission Inventory Report

4.5.2.17 Despite road transportation being the dominant source of CO emission, however, the air quality impact from CO is relatively minor considering its monitoring stations data records. In Year 2016, the highest 1-hour concentration of 3,130µg/m^3 and the highest 8-hour concentration of 2,339µg/m^3 were recorded at Causeway Bay roadside station and Tsuen Wan general station respectively, both of which were well below the AQO standard of 30,000 (1-hour) and 10,000 (8-hour), respectively. It is therefore not a key parameter for assessment.

vii) Toxic Air Pollutants (TAPs)

4.5.2.18 There are six kinds of Toxic Air Pollutants (TAPs) routinely monitored in HK, including polychlorinated biphenyls (PCBs), dioxins, polycyclic aromatic hydrocarbons (PAHs), volatile organic compounds (VOCs), carbonyls, and toxic elemental species.

4.5.2.1 Dioxins, carbonyls, PCBs and most toxic elemental species are not considered primary sources of vehicular emissions (http://www.epd.gov.hk/epd/english/environmentinhk/air/studyrpts/assessment_of_tap_measurements.html and http://www.eea.europa.eu/publications/EMEPCORINAIR5/Sources_of_PCB_emissions.pdf/view), and hence, these three TAPs (i.e. Dioxins, carbonyls and PCBs) are

not considered as key / representative air pollutants for the operational air quality assessment.

4.5.2.2 Vehicular emissions may be a source of diesel particulate matters, PAHs and VOCs. Elemental carbon, which constitutes a large portion of diesel particulate matters mass, is commonly used as a surrogate for diesel particulate matter. According to the data from EPD, the elemental carbon showed a significant decrease in concentration in Mong Kok by 47.5% from 2001 to 2009, and Tsuen Wan by 51.3% from 1999 to 2009. This is because the implementation of EURO III vehicle emission standard to goods vehicle and bus in 2001 and EURO IV standard to all types of vehicle in 2006-2007 (http://www.epd.gov.hk/epd/english/environmentinhk/air/data/emission_inve.html). Hence, diesel particulate matters are not considered as a key air pollutant for the operational air quality assessment.

4.5.2.3 Currently, no ambient air quality standards have been set for PAHs. However, with reference to US and European Community air quality guidelines, the European commission has a very stringent guideline concentration for PAHs. According to the "*Air Quality in Hong Kong 2015*" published by EPD, the concentration of PAHs level (Benzo[a]pyrene, BaP) in Hong Kong were 0.12ng/m^3 and 0.06ng/m^3 monitored at the Tsuen Wan and Central/Western stations respectively in 2015 which was still much lower than the guidelines of European Communities of 1ng/m^3.

Table 4.19 Comparison of TAPs concentration in Hong Kong (2015) and the EU Air Quality Standards

Air Pollutants	Guidelines / Standards (ng/m^3)	Annual Avg Conc at Tsuen Wan station (ng/m^3)	Annual Avg Conc at Central/Western station (ng/m^3)	Compliance
	EU			EU
PAHs (BaP)	1 (Annual Average)[1]	0.12 [2]	0.06 [2]	Well Achieved

Note:

[1] Referenced from http://ec.europa.eu/environment/air/quality/standards.htm

[2] Referenced from http://www.aqhi.gov.hk/api_history/english/report/files/AQR2015e_final.pdf

4.5.2.4 There are different standards for different VOC compounds. According to the "*Air Quality in Hong Kong 2015*" published by EPD, benzene, 1-3 butadiene, formaldehyde and perchloroethylene are the VOCs that may have more health concern, and the USEPA also identified benzene and 1-3 butadiene are carcinogenic.

Table 4.20 Comparison of VOCs concentration in Hong Kong (2015) and the EU Air Quality Standards

TAP	Guidelines / Standards (µg/m^3)	Annual Avg Conc at Tsuen Wan station (µg/m^3)	Annual Avg Conc at Central/Western station (µg/m^3)	Compliance
Benzene	5 (Annual Average) [1]	2.21	1.11	Well Achieved
1-3 butadiene	2.25 (Running Annual Average) [1]	0.12	0.06	Well Achieved

TAP	Guidelines / Standards (µg/m^3)	Annual Avg Conc at Tsuen Wan station (µg/m^3)	Annual Avg Conc at Central/Western station (µg/m^3)	Compliance
Formaldehyde	9 (Annual Average) [3]	3.73	- [2]	Well Achieved
Perchloroethylene	40 (Annual Average) [4]	0.46	0.64	Well Achieved

Note:

[1] Referenced from the UK National Air Quality Strategy (NAQS) (http://www.medway.gov.uk/environmentandplanning/environmentalhealth/airquality/airqualityfordevelopers.aspx)

[2] The measurement of formaldehyde at Central/Western Station was affected by influence from construction works at Sai Ying Pun MTR Station. Hence, the measurement result is not reported in 2015.

[3] Referenced from the Office of Environmental Health Hazard Assessment (OEHHA) Toxicity Criteria Database, California, USA (http://www.oehha.ca.gov/air/Allrels.html)

[4] Referenced from the Integrated Risk Information System (IRIS), USEPA (http://cfpub.epa.gov/ncea/iris2/chemicalLanding.cfm?substance_nmbr=106)

4.5.2.5 As shown in the above table, the measured VOCs concentration in Hong Kong urban area is far below the UK and US standards. Also, according to "*2015 Hong Kong Air Pollutants Emission Inventory*", the VOCs level has dropped by approximately 65% in 2015 since 1997 due to the EPD progressive improvement of EURO standard vehicles over the past two decades. In addition, vehicular emission is also not the primary source of VOCs, accounting for about 18% of the total in Hong Kong. Besides, according to another study - "*Seasonal and diurnal variations of volatile organic compounds (VOCs) in the atmosphere of Hong Kong*", benzene, and 1-3 butadiene only contributed about 6-13% of overall vehicular emission VOCs. In other words, only 1.1-2.3% of the overall VOC emissions in Hong Kong are benzene and 1-3 butadiene contributed by vehicular emission.

4.5.2.6 The historical monitoring data showed that the concentrations of PAHs and VOCs were only in small amount. It is also reasonably believed that the emission of PAHs and VOCs should be significantly decreased after the implementation of EURO V standard vehicles in 2013 and the phasing out of the pre-EURO IV diesel commercial vehicles. The TAPs is also not specified under the current AQOs. Based on above reasons, TAPs is not considered as a key air pollutant for the operational air quality assessment.

viii) Conclusion

4.5.2.7 As discussed in the above sections, only NO_2, RSP and FSP are considered the key air quality pollutant for this project and the concentrations of the other pollutants are very low and hence are not considered as the key pollutants for the purposes of this operational air quality assessment.

4.5.3 Assessment Methodology

General

4.5.3.1 The area for air quality impact assessment should be defined by a distance of 500m from the boundary of the Project site (including PDA and associated infrastructure).

4.5.3.2 The assessment has evaluated the impacts arising from three classes of emission sources depending on their distance from the project site, including:

(1) Project induced contribution;

(2) Pollutant-emitting activities in the immediate neighbourhood; and

(3) Other contributions from pollution not accounted for by (1) and (2).

4.5.3.3 All sources within 500m assessment area (i.e. (1) and (2)) are considered as near-field source impacts and are predicted using local-scale models. These sources include vehicular emission from existing road network and proposed roads within the Project site, as well as industrial emission from identified chimneys.

4.5.3.4 Other far-field pollution source impacts (3) which are beyond 500m from the Project (i.e. background concentration), are predicted using regional scale model – Pollutant in the Atmosphere and the Transport over Hong kong, PATH. In PATH model, all major emission sources including public electricity generation, road transport, navigation, civil aviation, industries, other fuel combustion and non-combustion sources covering both HKSAR and Pearl River Delta Economic Zone (PRDEZ) are considered.

4.5.3.5 The cumulative operational air quality impact is then a combination of the contributions from the near-field and far-field sources.

Determination of Worst Case Scenario for Operational Air Quality Assessment

4.5.3.6 According to Appendix B, Clause 5 (iv) of the EIA Study Brief for the Project, the air pollution impacts of future road traffic shall be calculated based on the highest emission strength from road within the next 15 years upon commissioning of the proposed development. The selected assessment year should represent the highest emission scenario, given the combination of emission factors and traffic flow for the selected year.

4.5.3.7 Vehicular tailpipe emissions from open roads are calculated based on the EMFAC-HK v3.3 model available at the time of preparing this report. Based on the current tentative implementation programme, population intake will be in phases in Years 2027, 2033 and 2038. Therefore, EMFAC-HK model runs have been carried out for Year 2027 (first population intake i.e. commissioning), 2029 (interim year), 2033 (second population intake), 2038 (full population intake), and 2042 (15 years after commissioning) to determine the highest emission scenario and the worst assessment year. The traffic forecast data provided by the Project Traffic Engineer, which has been endorsed by the Transport Department, are given in **Appendix 4.8**. The methodology, key model assumptions and results (including emission factors) are presented in **Appendix 4.9**.

4.5.3.8 The total vehicular emissions within the assessment areas predicted by EMFAC-HK based on the projected traffic flows for all years are summarized in the table below. Results indicate that the highest NOx, RSP and FSP emission scenario occurs in Year 2027 and hence is the worst assessment year for operational air quality assessment.

4.5.3.9 However, it should also be noted that the planned YLS development would be in place by phases with some Project roads to be commissioned in different years (from Year 2033 to 2038). Operational air quality assessment has also been conducted for the highest emission scenario with all project roads in operation and within 15 year upon commissioning of the proposed development. Based on the results, the total

emissions in Year 2042 are found to be higher than that in Year 2033 and Year 2038 (**Table 4.21a**).

4.5.3.10 It should be noted that although most of the Project roads will be in place during population intake of Phase 2 in Year 2033, Phases 3 and 4 are still not occupied and hence induced traffic on the Project roads are much lower than in Year 2038 and 2042. The emission of Project roads only is also the highest in Year 2042 compared to Year 2033 and Year 2038. The emission for the most critical pollutant NOx is 34 tonnes per year in Year 2042, 22 tonnes per year in Year 2033 and is 33 tonnes per year in Year 2038.

4.5.3.11 Besides, when comparing the emission for the individual DP roads, Year 2042 again is the highest. The emission for the most critical pollutant NOx is shown in **Table 4.21b** below.

Table 4.21a Summary of total pollutant emissions (all roads within 500m from the PDA boundary and associated road infrastructure)

Year	Vehicular Emission Burden (tonnes per year)		
	NOx	RSP	FSP
2027	**171**	**8.3**	**7.6**
2029	150	7.1	6.5
2033	110	4.5	4.2
2038	117	4.5	4.1
2042	**123**	**4.8**	**4.4**

Table 4.21b Comparison of NOx emissions for DP roads

Year	Vehicular Emission Burden (tonnes per year)			
	Road D1	Road D2	Tong Yan San Tsuen Interchange	Tin Shui Wai Interchange
2033	1.1	0.9	1.7	1.0
2038	4.5	0.9	2.0	1.6
2042	**4.7**	**1.0**	**2.1**	**1.7**

4.5.3.12 Based on the results above, two scenarios have been conducted for the operational air quality impact assessment (i.e. Scenario 2027 and Scenario 2042). Year 2027 represents the highest emission scenario within 15 years upon commissioning of the proposed development and Year 2042 represents the highest emission scenario with all project roads in operation and within 15 years upon commissioning of the proposed development.

Vehicular Emission from Open Road

4.5.3.13 The EMFAC-HK calculates the hourly vehicular emission (in tonne) for each road category. The hourly emission rates for each vehicle class (in gram per mile per vehicle) are obtained by dividing the hourly emissions calculated in the EMFAC-HK by the Vehicle Miles Travelled (VMT) for the respective hour. The calculation of the NOx, RSP and FSP emission factors for different road groups are given in **Appendix 4.9**. The composite vehicle emission factors for each road link in Year 2027 (based on 2027 traffic forecast and 2027 emission factors) and Year 2042 (based on 2042 traffic forecast and 2042 emission factors) are given in **Appendix 4.10**.

4.5.3.14 The USEPA approved near field air dispersion model, CALINE4 developed by the California Department of Transport is used to assess vehicular emissions impact from all existing and planned open road network.

4.5.3.15 Grid-specific composite real meteorological data extracted from EPD's PATH-2016 model is adopted in CALINE4 model, including relevant temperature, wind speed, direction and mixing height. The stability classes are estimated from PCRAMMET model. The mixing height is capped to 121m as per the real meteorological data. For the treatment of calm hours, the wind speeds are capped at 1m/s for those from PATH-2016 are lower than 1m/s.

4.5.3.16 The surface roughness height is closely related to the land use characteristics, and the surface roughness is estimated as 10% of the average height of physical structures within 1km study area. A surface roughness of 50cm, 100cm or 370cm are assumed to represent the rural nature, low-rise developments, and mid-/high-rise developments in Yuen Long and the proposed development respectively. The wind standard deviation is estimated in accordance with the "*Guideline on Air Quality Models (Revised), 1984 (USEPA), Table 9-2*", as summarized in the table below.

Table 4.22 Summary of Wind Standard Deviation

Stability Class	Wind Standard Deviation (roughness = 50cm)	Wind Standard Deviation (roughness = 100cm)	Wind Standard Deviation (roughness = 370cm)
A	28.6	32.9	42.7
B	28.6	32.9	42.7
C	22.3	25.6	33.2
D	15.9	18.3	23.7
E	9.5	11.0	14.2
F	4.8	5.6	7.2

4.5.3.17 Since the vehicular emission sources are at local ground level, the impact at ASRs are assessed at 1.5m, 5m, 10m, 15m and 20m above ground. However, assessment heights of 50m, 80m and 120m above ground are also included to confirm the compliance of AQOs at higher levels.

4.5.3.18 Owing to the limitation of CALINE4 model, road elevation is limited to 10 metres above ground (mAG) which may underestimate the pollutant concentrations at ASRs located 10mAG or above. Since the road heights within 500m from the PDA boundary and associated road infrastructure range from 0mAG to about 16mAG, roads are therefore grouped into 3 separate models respectively, namely the "At-grade" model, "Bridge-1" model and "Bridge-2" model. Under the "At-grade" model (i.e. road elevation lower than 10mAG), both actual heights of ASR and road are adopted. For the "Bridge-1" model (i.e. road elevation and ASR with actual height higher than 10mAG), a correction has been made by adjusting both heights of road and ASR so that height variation of roads higher than 10mAG can be reflected in the dispersion model. For ASR with actual height with actual height equal or lower than 10mAG, road with height above 10mAG are capped at 10mAG and no adjustment is made to ASR heights in the "Bridge-2" model for a reasonably conservative assessment. Total contributions from all roads are the summation of those from "At-grade" and "Bridge-1" / "Bridge-2" models. **Table 4.23** below illustrates the assessment heights adopted in "At-grade", "Bridge-1" and "Bridge-2" models.

Table 4.23 Assessment height adopted in "At-grade", "Bridge-1" and "Bridge-2" models

Actual ASR Height above Ground (m)	Adjusted ASR Height in "At-grade" model (m)	Adjusted ASR Height in "Bridge-1" model [1] (m)	Adjusted ASR Height in "Bridge-2" model [2] (m)
1.5	1.5	-	1.5
5	5	-	5
10	10	-	10
15	15	5	-
20	20	10	-
50	50	40	-
80	80	70	-
120	120	110	-

Note:

[1] For road and ASR with actual height above 10mAG, both heights of road and ASR are reduced by 10m.

[2] Road with height above 10mAG are capped at 10mAG in this CALINE4 model.

4.5.3.19 In addition, for barriers along existing roads or proposed noise barriers (see **Figure 5.7**) as a noise mitigation measures, the line source is modelled at the tip of the barrier and the mixing width is limited to the actual uncovered road width in order to address the associated secondary environmental impact. There are some noise barriers located more than 5m away from the road kerb and some even with more than 10-20m from the road kerb. Besides, the ASRs are located at about 7m to 100m behind these barriers. It is anticipated that these noise barriers would not significantly affect the dispersion of pollutant to the nearby ASRs. A sensitivity test has been carried out on the representative ASRs which are located close to these noise barriers. Results indicate that without the barrier effect (assuming at grade line source), the predicted concentrations from the concerned road section at ground level (i.e. 1.5mAG) are higher than that of with the barrier effect (by modelling the source at the tip of barrier) and are considered conservative. Taking into account the effect of the proposed noise barrier, the predicted concentrations from the concerned road section at ground level (i.e. 1.5mAG) are still higher than that predicted at 5mAG and 10mAG, and the difference between with and without the proposed noise barrier effect is only less than $0.1\mu g/m^3$ for annual NO_2, and about $1\mu g/m^3$ for 1-hour NO_2. Hence, for the proposed noise barrier located more than 5m away from the road kerb, the corresponding roads are therefore modelled as "at-grade" as a more conservative approach. For few scattered short noise barriers (with modelled road length shorter than the modelled road width of source) and discontinuous noise barriers (where the opening with modelled road length shorter than the modelled road width of source), due to the limitation of CALINE4 model, they are modelled as a continuous road link assuming without short barrier and as continuous barriers respectively.

4.5.3.20 According to EPD's "*Guidelines on Choice of Models and Model Parameters*", the individual initial tailpipe NO_2/NO_x ratios of each EMFAC-HK vehicle type have been adopted to calculate the initial NO_2 and residual NO_x tailpipe emission rates. Ozone Limiting Method (OLM) is adopted for conversion of residual NO_x to NO_2, using the predicted O_3 levels from PATH-2016 model.

Vehicular Emission from Portal of the Proposed Underpass and Proposed Full Enclosure

4.5.3.21 The USEPA approved model, AERMOD, is adopted to model the vehicular emission from portals of the proposed underpass and full enclosures. Portal emissions are

modelled in accordance with the Permanent International Association of Road Congress Report (PIARC, 1991), where it is assumed that the pollutant will be ejected from the portal as a portal jet such that 2/3 of the total emission will be dispersed within first 50m from the portal, and 1/3 of the total emission within the second 50m. To take into account the horizontal jet effect, portal emission is modelled as "Volume" source. Detailed calculations of portal emission is given in **Appendix 4.11**.

4.5.3.22 Similarly, the individual initial tailpipe NO_2/NOx ratios of each EMFAC-HK vehicle type have been adopted to calculate the initial NO_2 and residual NOx tailpipe emission rates. OLM is adopted for conversion of residual NOx to NO_2, using the predicted O_3 levels from PATH-2016 model.

Industrial Emission

4.5.3.23 All chimneys being in-use within 500m from the PDA boundary and associated road infrastructure as identified in **Section 4.5.1.7** are included in the near-field dispersion model. Latest information on the fuel consumption rate, stack height, diameter, exit velocity and temperature was provided by the chimney operator. The NOx, RSP and FSP emissions (in g/s) are calculated based on the respective emission factors in accordance with the USEPA AP-42 or those stipulated in the SP Licence. In addition, for the concrete batching plant, in accordance with "*A Guidance Note on the Technical Management and Monitoring Requirements for Specified Process – Cement Works (Concrete Batching Plant)*"(BPM 3/2 (16), emission of particulate matter from fixed emission point of bag filters shall meet the concentration limit of 10mg/m^3 (design standard) by 1 January 2018 for all plants. Hence, particulate matter emissions from the concrete batching plant are estimated with reference to the BPM 3/2 (16). **Appendix 4.12** presents the detailed calculations of industrial emission.

4.5.3.24 Potential air quality impact associated with the industrial emissions is assessed by the EPD approved dispersion model, AERMOD. Chimneys are modelled as "Point" source, while haul road and loading / unloading area are modelled as "Area" sources in the model. OLM is adopted for conversion of NOx to NO_2, using the predicted O_3 and NO_2 levels from PATH-2016 model. The in-stack NO_2:NOx ratio for the industrial chimneys is assumed to be 10% in accordance with EPD's "*Guidelines on Choice of Models and Model Parameters*".

Far-field Source Contribution (i.e. Future Background Air Quality)

4.5.3.25 PATH (Pollutants in the Atmosphere and their Transport over Hongkong) is a regional air quality model developed by EPD to simulate air quality over Hong Kong against the Pearl River Delta (PRD) as background. It simulates wind field, pollutant emissions, transportation and chemical transformation and outputs pollutant concentrations over Hong Kong and the PRD region at a fine grid size of 1km.

4.5.3.26 PATH-2016 model is used to quantify the future background air quality. Far-field emission sources (i.e. all those outside 500m assessment area) including roads, marine, airports, power plants and industries within the Pearl River Delta Economic Zone and Hong Kong were considered in the PATH-2016 model. Details of the PATH-2016 model and related emission inventory can be found in EPD's website.

4.5.3.27 The Project site is located inland and road emission is anticipated as the major source to the background concentrations. Considering that the emission control technology will be progressively improving in future years, use of the territory wide emission

inventory for Year 2020 for assessing the future background concentrations for Years 2027 and 2042 is considered to be very conservative.

4.5.3.28 On the other hand, since the vehicular emission at local-scale (i.e. the road network within the 500m study area as illustrated in **Figures 4.9a-b**) is modelled by the near-field dispersion model CALINE4, the respective emission has been removed from the concerned grids to avoid overestimation.

4.5.3.29 PATH model has been re-run based on the above consideration, and the hourly concentrations of NO_2, RSP and O_3 predicted by this updated PATH-2016 model have been adopted in the calculation of cumulative air quality impact for Years 2027 and 2042 in this assessment.

4.5.3.30 It is understood that FSP concentrations are not available from PATH model. According to EPD's "*Guidelines on the Estimation of FSP for Air Quality Assessment in Hong Kong*", the conservative corrections as shown in the following table are adopted to determine the background FSP concentrations.

Table 4.24 Conversion factors for RSP/FSP

Annual (µg/m^3)	Daily (µg/m^3)
FSP = 0.71 x RSP	FSP = 0.75 x RSP

Prediction of the Cumulative Operational Air Quality Impact

4.5.3.31 The cumulative operational air quality is a combination of the emission impacts contributed from the near-field and far-field sources (i.e. at local scale and background air quality impact from other concurrent and regional sources) on an hourly basis.

4.5.3.32 OLM is used for conversion of NOx to NO_2 based on the O_3 level from PATH-2016 directly. As a conservative approach, the OLM is applied separately to the following groups of emission sources:

- Group A – All open roads and portal emission from the proposed full enclosure / underpass; and
- Group B – All industrial sources

4.5.3.33 In consideration of the number of exceedance allowance of the hourly and daily AQOs (refer to **Table 4.1**), the pollutant concentrations after the AQOs' allowance limits (i.e. the 19th highest 1-hour NO_2 concentrations and 10th highest 24-hour RSP/ FSP concentrations) are determined. The annual predicted concentrations are also assessed and all predicted levels are then compared with the AQOs.

4.5.3.34 Moreover, a 50x50m grid is used to generate pollution contours in order to investigate the pollutant dispersion.

Air Quality Implication due to the Recommended Direct Technical Noise Remedies

4.5.3.35 As mentioned in **Sections 4.5.3.19 and 4.5.3.21**, proposed noise mitigation measures (i.e. direct technical noise remedies, see **Figure 5.7**), including noise barriers, full enclosure and semi-enclosures, have been taken into account in the cumulative operational air quality impact assessment for Year 2042. According to Appendix B, Clause 5(v) of the EIA Study Brief, the air quality implication due to the recommended direct technical noise remedies shall be assessed. To this end, an

additional scenario without the proposed noise mitigation measures has been conducted for Year 2042.

4.5.4 Prediction and Evaluation of Impacts

Assessment Results in Year 2027

4.5.4.1 The 19th highest 1-hour and annual NO_2 concentrations, and 10th highest 24-hour and annual RSP/ FSP concentrations predicted in Year 2027 are presented in **Table 4.25**. Detailed results are presented in **Appendix 4.13**. It should be noted that this operational air quality impact assessment has adopted the territory wide emission inventory for Year 2020 for assessing the future background concentrations for Year 2027, which is more conservative.

Table 4.25 Cumulative NO_2, RSP and FSP concentrations (Year 2027)

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
Existing / Planned ASRs (outside PDA Boundary)								
A2	House no. 40, Tai Tao Tsuen	1.5 / 5	116	33.0	82	35.4	62	25.3
A3	House no. 95, Tai Tao Tsuen	1.5	120	34.8	82	35.5	61	25.4
A4	Sheltered Structure, Fui Sha Wai South Road	1.5 / 5	119	32.1	81	35.4	61	25.3
A6	House no. 176A, Fui Sha Wai	5 / 10	120	30.5	82	35.4	61	25.2
A7	Village House, Ping Tong Street South	1.5 / 10	121	31.7	82	35.6	61	25.3
A8	House no. 48, Tong Yan San Tsuen Road	1.5 / 5	122	31.5	83	35.6	62	25.4
A9	New Territories Assemblies of God Church	1.5	109	28.1	83	35.4	62	25.2
A10	Block 10, Jasper Court	1.5 / 5	122	31.9	83	35.6	63	25.4
A11	School	1.5	119	29.7	83	35.4	62	25.3
A12	Village House, Lam Yu Road	5	123	31.1	83	35.5	62	25.3
A13	Elchk Lutheran Academy	1.5	112	28.4	83	35.4	62	25.2
A15	Village House, Lam Hei Road	1.5	121	32.9	82	35.6	61	25.4
A16	Village House, Lam Hei Road	1.5	121	33.1	82	35.6	61	25.4
A17	Sheltered Structure, Lam Hei Road	1.5	122	35.7	82	35.7	61	25.5
A18	Sheltered Structure no. 66 Kiu Hing Road	1.5	124	34.3	82	35.6	62	25.4
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	1.5	106	25.0	82	35.2	62	25.0
A20	House no. 89A, Lam Hau Tsuen	1.5 / 10	118	28.8	83	35.4	62	25.2
A21	House no. 324, Shan Ha Tsuen	1.5	96	20.9	84	35.7	63	25.4
A22	House no. 645, Shan Ha Tsuen	1.5	95	21.0	84	35.7	63	25.4

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A23	House no. 193, Kung Um Road	1.5	118	29.3	82	35.3	62	25.2
A24	House no. 132, Sun Mei Garden	1.5	122	33.5	82	35.8	62	25.6
A25	House no. 293, Kung Um Road	1.5	111	22.7	82	35.3	61	25.1
A26	Block 10, Chun Fai Garden	1.5	108	22.7	82	35.3	61	25.1
A27	House no. 161, Kung Um Road	1.5	108	21.7	82	35.3	61	25.1
A28	Village House, Kung Um Road	1.5	108	20.6	82	35.2	61	25.0
A29	House no. 366A, Kung Um Road	1.5	109	21.1	82	35.2	61	25.0
A30	Sheltered Structure, Kung Um Road	1.5	90	19.4	85	35.7	64	25.4
A31	House no. 31B, Kung Um Road	1.5	106	21.3	82	35.5	61	25.2
A32	House no. 241, Kiu Hing Road	1.5	105	20.5	82	35.4	61	25.2
A33	House no. 67A, Wong Nai Tun Tsuen	1.5	102	19.3	82	35.4	61	25.1
A34	House no. 128, Kung Um Road	1.5	90	19.0	85	35.7	64	25.4
A35	House no. 117, Kung Um Road	1.5	95	19.5	86	36.8	64	26.2
A36	Sheltered Structure no. 375, Tai To Tsuen	1.5	107	32.8	83	36.3	63	25.9
A37	Uptown Tower 1	1.5	119	34.5	82	35.5	62	25.3
A38	Chinese Mission Seminary	1.5	112	28.0	81	35.1	61	25.0
A39	Fui Sha Wai Playground	1.5	113	28.7	81	35.2	61	25.1
A40	Tower 8, Imperial Villas II	1.5	109	28.1	82	35.4	61	25.1
A41	Energy Industrial Centre Block B	1.5	108	28.5	90	40.1	64	26.7
A42	Block 1, Parkside Villa	5	123	31.1	83	35.5	62	25.3
A43	Block 2, Emerald Green	1.5	119	29.6	82	35.3	61	25.2
A44	House no. 49A, Lung Tin Tsuen	1.5	122	33.4	82	35.5	61	25.3
A45	House no. 139A	1.5 / 5	123	34.0	82	35.6	61	25.4
A46	House no. 101, Sham Chung Tsuen	1.5	123	31.8	82	35.5	62	25.3
A47	Meadowlands Block 33	1.5	103	29.1	82	36.0	62	25.6
A48	Po Kok Branch School	1.5	108	29.8	82	36.1	62	25.6
A49	Treasure Court Block 7	1.5	111	29.3	81	35.0	61	24.9
A50	Beauty Court Block 2	1.5	115	29.5	81	35.0	61	25.0
A51	Park Nara Tower 1	1.5	116	30.7	81	35.1	61	25.0
A52	Hung Uk Garden	1.5	105	25.4	81	35.0	61	24.9
A53	Green Lodge House 16	1.5	108	27.9	83	35.3	62	25.2
A54	Ping Shan Garden Block 6	1.5	117	31.9	82	35.2	61	25.1

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (μg/m³)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A55	Villa Sunshine Block 1	1.5	118	33.8	82	35.3	62	25.2
A56	Park Royale Block 10	5	120	30.7	82	35.1	61	25.0
A57	Park Royale Block 2	5	120	32.3	82	35.2	61	25.1
A58	Po Leung Kuk Law's Foundation School	1.5	117	28.0	83	35.3	62	25.1
A59	Yuen Long Public Secondary School	1.5	115	27.6	83	35.3	62	25.1
A60	Villa Art Deco Block 2	1.5	113	28.7	83	35.3	62	25.2
A61	Gertrude Simon Lutheran College	1.5	124	32.8	82	35.5	61	25.3
A62	Yuen Long Public Middle School Alumni Association Primary School	1.5	127	34.9	82	35.5	61	25.3
A63	Silver Field Garden Block 17	1.5	122	31.0	82	35.4	61	25.2
A64	La Grove Block 1	1.5	122	35.9	82	35.6	62	25.4
A65	Ma Tin Tsuen House 242	1.5	120	30.6	82	35.4	61	25.2
A66	Teng Lung Villa	1.5	119	30.0	82	35.4	61	25.2
A67	Yuen Long Baptist Church	1.5	120	34.1	82	35.5	62	25.3
A68	Fraser Village House 54	1.5	122	35.8	82	35.6	62	25.4
A69	The Brand	1.5	119	31.0	82	35.4	61	25.2
A70	Sereno Verde Block 2	1.5	114	25.9	81	35.2	61	25.1
A71	La Pradera Block 12	1.5	111	23.8	81	35.0	61	25.0
A72	The Reach Tower 2	1.5	112	24.8	81	35.1	61	25.0
A73	The Reach Tower 6	1.5	115	29.3	81	35.3	61	25.2
A74	Tai Kei Leng House 145	1.5	114	28.3	82	35.2	61	25.1
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	1.5	116	27.8	81	35.2	61	25.1
A76	Grand de Sol Block 15	1.5	113	25.0	81	35.1	61	25.0
A77	Grand de Sol Block 8	1.5	114	25.1	81	35.1	61	25.0
A78	Hoover Garden Block 4	1.5	117	28.6	82	35.3	61	25.2
A79	Ha Yau Tin Tsuen House 2	1.5	113	24.1	81	35.1	61	25.0
A80	Buddhist Wing Yan School	1.5	113	24.9	81	35.1	61	25.0
A81	Fortune Centre	1.5	113	27.8	82	35.2	62	25.1
A82	YOHO Town Block 6	1.5	112	25.8	81	35.2	61	25.1
A83	YOHO Town Block 9	1.5	112	25.7	82	35.1	62	25.0
A84	YOHO Midtown Block 5	15	117	27.2	82	35.2	62	25.1

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (μg/m³)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A84a	Shopping Mall under YOHO Midtown (Fresh Air Intake)	6	129	34.3	82	35.5	62	25.4
A84b	Shopping Mall under YOHO Midtown (Fresh Air Intake)	15	119	27.0	82	35.2	62	25.1
A84c	Shopping Mall under YOHO Midtown (Fresh Air Intake)	20	114	25.6	82	35.1	62	25.0
A84d	Kindergarten under YOHO Midtown (Fresh Air Intake)	2	122	33.3	83	35.5	62	25.4
A85	Ho Shun Yee Building Block 2	1.5	112	27.8	82	35.2	62	25.1
A86	Cheong Wai Building	1.5	125	33.9	82	35.5	62	25.4
A87	Kwong Ming Ying Loi School	1.5	118	33.2	82	35.5	62	25.4
A88	Sun Yuen Long Centre Block 5	20	118	26.1	82	35.1	62	25.0
A89	Tung Tau Tsuen House 2	1.5	119	26.2	82	35.2	62	25.0
A90	Tai Wai Tsuen House 30	1.5	122	26.9	82	35.2	62	25.1
A91	Small Traders New Village	1.5	133	28.3	82	35.2	61	25.2
A92	Pok Oi Hospital	1.5	133	31.4	82	35.4	61	25.3
A93	Yeung Uk Tsuen House 10	1.5	102	19.5	82	34.8	61	24.7
A94	Chuk San Tsuen House 17	1.5	111	19.7	82	35.2	62	25.1
A95	Greenfield Lodge Block 1	1.5	96	17.5	82	35.1	61	25.0
A96	Kong Tau San Tsuen House 5	1.5	112	21.5	81	34.9	61	24.9
A97	Tai Kei Leng House 414	1.5	115	24.7	82	35.1	62	25.0
A98	Shung Ching San Tseun House 49	1.5	120	30.8	82	35.4	62	25.3
A99	Silver Garden House 125B	1.5	112	23.7	82	35.4	61	25.2
A100	Tin Liu Tsuen House 32	1.5	119	28.7	82	35.3	62	25.1
A101	Pak Sha Tsuen	1.5	105	21.0	82	35.5	62	25.2
A102	Wong Nai Tun Tsuen House 47C	1.5	103	19.4	82	35.4	61	25.1
A103	Greenwood Gardens House 397	1.5	102	19.4	82	35.4	61	25.1
A104	Park Signature Block 1	1.5	120	32.6	82	35.5	61	25.3
A105_FA	Market of Hung Fuk Estate (Fresh Air Intake)	4	129	37.2	82	35.4	61	25.3
A105_D	Market of Hung Fuk Estate (Welfare Facilities)	10	122	32.7	81	35.2	61	25.1
A106	Village House along Kung Um Road	1.5	116	25.4	82	35.4	62	25.2
A107	Village House along Kung Um Road	1.5	114	24.8	82	35.4	61	25.2

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A108	Shan Ha Tsuen House No. 342	1.5	95	20.7	84	35.7	63	25.4
A109	Shan Ha Tsuen House No. 343A	1.5	95	20.6	84	35.7	63	25.4
A110	Shan Ha Tsuen House No. 345C	1.5	101	23.4	82	35.1	62	24.9
A111	Shan Ha Tsuen House No. 346	1.5	102	23.6	82	35.1	62	25.0
A112	Shan Ha Tsuen House No. 611	1.5	96	21.6	84	35.7	63	25.4
A113	Shan Ha Tsuen House No. 613F	1.5	96	21.5	84	35.7	63	25.4
A114	Sheung Yau Tin Tsuen	1.5	119	30.7	82	35.4	61	25.3
A115	Le Regent	1.5	121	34.1	82	35.5	61	25.3
A116	Planned Long Bin Development	5	123	36.1	82	35.4	62	25.3
A117	Planned Long Bin Development	5	120	35.9	83	35.7	62	25.5
A118	Planned Long Bin Development	5	121	36.0	83	35.8	63	25.6
A119	Proposed Hung Shui Kiu Development	5	128	36.3	82	35.4	61	25.3
A120	Yee Fung Garden Block A	1.5	121	31.2	82	35.4	61	25.2
A121	Ming Wan Court	1.5	123	31.8	82	35.6	62	25.3
A122	Yuen Long Villa House No. 252	1.5	118	29.5	82	35.3	61	25.2
A123	Sun Fai Court Block C	1.5	120	30.7	82	35.4	61	25.2
A124	Caritas Yuen Long Chan Chun Ha Secondary School	1.5	122	32.8	82	35.5	61	25.3
A125	Village House along Ma Tong Road	1.5	120	30.2	82	35.4	61	25.2
A126	Proposed Hung Shui Kiu Development	5	124	34.4	82	35.3	61	25.2
A127	Uptown House 30	1.5	123	34.6	82	35.5	62	25.3
A128	The Woodside Tower 5	1.5	114	31.6	81	35.2	61	25.1
A129	Shung Tak Catholic English College	1.5	112	29.4	81	35.1	61	25.0
A130	Village House along Castle Peak Road	1.5	116	29.9	81	35.2	61	25.1
A131	Regent's Park	1.5	114	24.2	82	35.4	61	25.2
A132	One Hyde Park House 7	1.5	108	23.0	82	35.6	62	25.3
A133	Pak Sha Tsuen	1.5	104	20.5	82	35.4	61	25.2
A134	Shan Ha Tsuen House No. 330	1.5	96	20.9	84	35.7	63	25.4
A135	Lam Hau Tsuen House No. 110	1.5	106	24.8	82	35.2	62	25.0
A136	Evergreen Place Tower 5	1.5	107	27.3	83	35.3	62	25.1
A137	Proposed Hung Shui Kiu Development	5	110	28.0	81	35.0	61	24.9
A138	Lok Kui Lau	1.5	127	34.7	82	35.4	61	25.2

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (μg/m³)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
A139	Ping Shan Home for The Aged	1.5	125	33.2	82	35.3	61	25.2
A140	Sheltered Structure along Tai Shu Ha Road East	1.5	122	32.5	81	35.5	61	25.4
A141	Village House along Tai Shu Ha Road West	1.5	125	37.4	82	35.7	62	25.5
A142	Sheltered Structure near Shap Pat Heung Interchange	1.5	119	31.2	82	35.4	61	25.3
A143	Kong Tau San Tsuen House No, 61	1.5	113	22.1	81	35.0	61	24.9
A144	Pok Oi Hos. Jockey Club Care and Attention Home	1.5	135	31.3	82	35.4	62	25.3
A145	Proposed Hung Shui Kiu Development	5	116	30.2	81	35.1	61	25.0
A146	Hong Ping Villa Block 1	1.5	130	36.0	82	35.4	61	25.3
A147	Tai On Home for Aged	1.5	112	29.6	82	35.7	61	25.3
A148	Ming Sum Home for the Sen	1.5	128	32.3	82	35.5	61	25.4
A149	Hung Fuk Estate	1.5	117	31.4	81	35.1	61	25.1
A150	Grand Yoho Block 1	20	118	25.9	82	35.1	62	25.0
A150a	Shopping Mall under Grand Yoho (Fresh Air Intake)	8	126	33.8	82	35.5	62	25.4
A151	Grand Yoho Block 5	20	119	26.0	82	35.1	62	25.0
A152	Grand Yoho Block 9	20 / 80	126	29.1	82	35.3	62	25.2
A152a	Shopping Mall under Grand Yoho (Fresh Air Intake)	10	132	38.0	83	35.8	62	25.6
Existing ASRs (within PDA Boundary)								
W-A1	House 33, Park Villa	1.5	110	24.2	81	34.9	61	24.9
W-A2	House 11, Park Villa	1.5	99	22.5	81	34.8	61	24.8
W-A3	Existing developments (under construction)	1.5	105	23.7	81	34.9	61	24.8
W-A4		1.5	101	23.3	81	34.9	61	24.8
W-A5	Block 1, Recours La Serre	1.5 / 10	100	22.6	81	34.8	61	24.8
W-A6	Block 2, The Parkhill	1.5	100	23.3	82	35.1	62	24.9
W-A7	Block 7, Greenville Residence	1.5	101	23.5	82	35.1	62	24.9
W-A8	Block 6, Windsor Garden	1.5	100	23.1	82	35.1	62	24.9
W-A9	Block 1, Marbella Garden	1.5	101	23.8	82	35.1	62	25.0
W-A11	Kisland Villa Phase 2	1.5	102	24.4	82	35.2	62	25.0
W-A13	Sha Tseng Tsuen	1.5	108	26.5	82	35.3	62	25.1

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3)					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
W-A14		1.5 / 5	118	28.8	83	35.4	62	25.2
Planned ASRs (PDA Area 2 and Area 3) [1]								
E-P1	Planned Residential	1.5	115	27.7	82	35.3	61	25.1
E-P5	Planned Residential	1.5	107	21.8	82	35.3	61	25.1

Note:

[1] Planned ASRs to be occupied in Year 2027 only include E-P1 and E-P5. Assessment results for other planned ASRs to be occupied in Year 2033 or Year 2038 are presented in Table 4.26.

4.5.4.2 It can be seen from the table and **Appendix 4.13** that the predicted cumulative concentrations of 19th 1-hour and annual NO_2, 10th highest 24-hour and annual RSP/ FSP are within the respective criteria at all height levels.

4.5.4.3 Higher pollutant concentrations (e.g. annual NO_2) are generally predicted at existing / planned ASRs located adjacent to the major roads (including YLH, Long Tin Road, Hung Tin Road, Castle Peak Road, etc.). In particular, the predicted cumulative annual NO_2 concentrations at the following ASRs are greater than 36µg/m^3 (i.e. >90% of the AQOs):

- A105_FA (Fresh Air Intake at Market of Hung Fuk Estate) at 4mAG;
- A116 (Planned Long Bin Development) at 5mAG;
- A118 (Planned Long Bin Development) at 5mAG;
- A119 (Proposed Hung Shui Kiu Development) at 5mAG;
- A141 (Village House along Tai Shu Ha Road West) at 1.5mAG; and
- A152a (Fresh Air Intake at Shopping Mall under Grand Yoho) at 10mAG.

4.5.4.4 According to the assessment results for Year 2027, the worst hit level is generally found at 1.5mAG, contours of 19th highest 1-hour and annual NO_2 concentrations, 10th highest 24-hour and annual RSP/FSP concentrations at 1.5mAG are therefore plotted in **Figure 4.12a - Figure 4.12f**. However, for some ASRs near Hung Fuk Estate, Long Bin and Pok Oi Interchange, worst affected level is observed at higher level. To this end, the following contours of cumulative annual NO_2 concentration near these areas are plotted. As discussed in **Section 4.5.4.2**, considering that the predicted concentrations of other parameters are well within their respective criteria, contours of other pollutant concentrations are only presented at 1.5mAG.

- 5mAG for Grid 22_46 (**Figure 4.12g**);
- 5mAG for Grids 24_46 and 24_47 (**Figure 4.12h**); and
- 6mAG and 10mAG for Grid 26_47 (**Figures 4.12i-j**).

4.5.4.5 According to the contours plots in **Figure 4.12a - Figure 4.12j**, it is concluded that there is no exceedance at all air sensitive uses.

Assessment Results in Year 2042

4.5.4.6 The 19th highest 1-hour and annual NO_2 concentrations, and 10th highest 24-hour and annual RSP/ FSP concentrations predicted in Year 2042 are presented in **Table 4.26**. Detailed results are presented in **Appendix 4.13**. It can be seen from the table that all the predicted NO_2/ RSP/ FSP concentrations are within the respective criteria. It should also be noted that this operational air quality impact assessment has adopted the territory wide emission inventory for Year 2020 for assessing the future background concentrations for Year 2042, which is more conservative.

Table 4.26 Cumulative NO_2, RSP and FSP concentrations (Year 2042)

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
Existing / Planned ASRs (outside PDA boundary)								
A2	House no. 40, Tai Tao Tsuen	1.5	103	26.9	81	35.0	61	24.9
A3	House no. 95, Tai Tao Tsuen	1.5	108	28.4	81	35.1	61	25.0
A4	Sheltered Structure, Fui Sha Wai South Road	1.5 / 5	109	27.8	81	35.1	61	25.0
A6	House no. 176A, Fui Sha Wai	1.5 / 5	110	27.8	81	35.1	61	25.0
A7	Village House, Ping Tong Street South	1.5 / 5 / 10	113	28.3	82	35.3	61	25.0
A8	House no. 48, Tong Yan San Tsuen Road	1.5	120	33.0	83	35.5	62	25.3
A9	New Territories Assemblies of God Church	1.5	105	26.7	83	35.3	62	25.1
A10	Block 10, Jasper Court	1.5	112	29.0	83	35.4	62	25.2
A11	School	1.5	109	26.7	82	35.2	62	25.1
A12	Village House, Lam Yu Road	5	115	27.5	83	35.3	62	25.1
A13	Elchk Lutheran Academy	1.5	107	26.2	82	35.2	62	25.0
A15	Village House, Lam Hei Road	1.5	120	30.3	82	35.4	61	25.2
A16	Village House, Lam Hei Road	1.5	119	30.1	82	35.3	61	25.2
A17	Sheltered Structure, Lam Hei Road	1.5	120	32.2	82	35.4	61	25.2
A18	Sheltered Structure no. 66 Kiu Hing Road	1.5	121	30.9	82	35.4	61	25.2
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	1.5	102	24.6	82	35.1	62	25.0
A20	House no. 89A, Lam Hau Tsuen	1.5	114	28.1	83	35.3	62	25.1
A21	House no. 324, Shan Ha Tsuen	1.5	99	22.3	84	35.7	63	25.4
A22	House no. 645, Shan Ha Tsuen	1.5	96	22.0	84	35.7	63	25.4
A23	House no. 193, Kung Um Road	1.5	115	28.1	82	35.2	61	25.1

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A24	House no. 132, Sun Mei Garden	1.5	114	26.4	82	35.4	61	25.2
A25	House no. 293, Kung Um Road	1.5	106	22.4	82	35.3	61	25.1
A26	Block 10, Chun Fai Garden	1.5	107	22.2	82	35.3	61	25.1
A27	House no. 161, Kung Um Road	1.5	107	21.5	82	35.2	61	25.0
A28	Village House, Kung Um Road	1.5	107	20.4	81	35.2	61	25.0
A29	House no. 366A, Kung Um Road	1.5	105	21.1	82	35.2	61	25.0
A30	Sheltered Structure, Kung Um Road	1.5	96	20.5	85	35.8	64	25.4
A31	House no. 31B, Kung Um Road	1.5	105	21.1	82	35.4	61	25.2
A32	House no. 241, Kiu Hing Road	1.5	103	21.0	82	35.4	61	25.2
A33	House no. 67A, Wong Nai Tun Tsuen	1.5	101	19.6	82	35.4	61	25.1
A34	House no. 128, Kung Um Road	1.5	92	19.1	85	35.7	64	25.4
A35	House no. 117, Kung Um Road	1.5	95	19.4	86	36.8	64	26.1
A36	Sheltered Structure no. 375, Tai To Tsuen	1.5	96	25.8	83	35.9	62	25.5
A37	Uptown Tower 1	1.5	106	28.0	81	35.0	61	24.9
A38	Chinese Mission Seminary	1.5	100	24.7	81	34.9	61	24.8
A39	Fui Sha Wai Playground	1.5	104	25.7	81	35.0	61	24.9
A40	Tower 8, Imperial Villas II	1.5	100	25.4	82	35.2	61	24.9
A41	Energy Industrial Centre Block B	1.5	101	26.2	90	39.9	64	26.5
A42	Block 1, Parkside Villa	5	113	27.3	82	35.2	62	25.1
A43	Block 2, Emerald Green	1.5	116	27.8	82	35.2	61	25.0
A44	House no. 49A, Lung Tin Tsuen	1.5	120	30.7	82	35.3	61	25.1
A45	House no. 139A	1.5 / 5	118	30.3	82	35.4	61	25.2
A46	House no. 101, Sham Chung Tsuen	1.5	118	29.8	82	35.3	62	25.1
A47	Meadowlands Block 33	1.5	103	27.9	82	35.9	61	25.5
A48	Po Kok Branch School	1.5	107	28.5	82	35.9	61	25.5
A49	Treasure Court Block 7	1.5	107	27.7	81	34.8	61	24.8
A50	Beauty Court Block 2	1.5	106	27.1	81	34.8	61	24.8
A51	Park Nara Tower 1	1.5	108	27.5	81	34.9	61	24.8
A52	Hung Uk Garden	1.5	97	23.4	81	34.9	61	24.8
A53	Green Lodge House 16	1.5	105	25.7	82	35.2	62	25.0
A54	Ping Shan Garden Block 6	1.5	111	28.5	82	35.0	61	24.9

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A55	Villa Sunshine Block 1	1.5	112	29.2	82	35.0	61	24.9
A56	Park Royale Block 10	5	113	27.6	81	34.9	61	24.8
A57	Park Royale Block 2	5	114	28.8	81	35.0	61	24.9
A58	Po Leung Kuk Law's Foundation School	1.5	108	25.6	82	35.1	62	25.0
A59	Yuen Long Public Secondary School	1.5	107	25.5	82	35.1	62	25.0
A60	Villa Art Deco Block 2	1.5	105	26.4	82	35.2	62	25.0
A61	Gertrude Simon Lutheran College	1.5	120	30.9	82	35.3	61	25.1
A62	Yuen Long Public Middle School Alumni Association Primary School	1.5	124	32.7	82	35.4	61	25.2
A63	Silver Field Garden Block 17	1.5	120	29.2	82	35.3	61	25.1
A64	La Grove Block 1	1.5	119	32.2	82	35.4	61	25.2
A65	Ma Tin Tsuen House 242	1.5	118	29.1	82	35.3	61	25.1
A66	Teng Lung Villa	1.5	117	28.1	82	35.2	61	25.1
A67	Yuen Long Baptist Church	1.5	115	30.2	82	35.3	61	25.1
A68	Fraser Village House 54	1.5	119	32.2	82	35.4	61	25.2
A69	The Brand	1.5	117	28.4	82	35.2	61	25.1
A70	Sereno Verde Block 2	1.5	111	22.6	81	34.9	61	24.9
A71	La Pradera Block 12	1.5	108	21.5	81	34.9	61	24.8
A72	The Reach Tower 2	1.5	107	22.1	81	34.9	61	24.8
A73	The Reach Tower 6	1.5	113	24.8	81	35.0	61	24.9
A74	Tai Kei Leng House 145	1.5	111	25.2	81	35.0	61	24.9
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	1.5	111	24.8	81	35.0	61	24.9
A76	Grand de Sol Block 15	1.5	109	22.2	81	34.9	61	24.8
A77	Grand de Sol Block 8	1.5	109	21.8	81	34.9	61	24.8
A78	Hoover Garden Block 4	1.5	110	23.0	81	35.0	61	24.9
A79	Ha Yau Tin Tsuen House 2	1.5	106	21.0	81	34.9	61	24.8
A80	Buddhist Wing Yan School	1.5	110	22.1	81	34.9	61	24.8
A81	Fortune Centre	1.5	109	24.7	82	35.0	61	24.9
A82	YOHO Town Block 6	1.5	104	21.7	81	34.9	61	24.8
A83	YOHO Town Block 9	1.5	106	22.2	82	34.9	61	24.8
A84	YOHO Midtown Block 5	15	107	23.2	82	35.0	61	24.9

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A84a	Shopping Mall under YOHO Midtown (Fresh Air Intake)	6	114	28.0	82	35.1	62	25.0
A84b	Shopping Mall under YOHO Midtown (Fresh Air Intake)	15	107	23.1	82	35.0	61	24.9
A84c	Shopping Mall under YOHO Midtown (Fresh Air Intake)	20	105	22.1	82	34.9	61	24.8
A84d	Kindergarten under YOHO Midtown (Fresh Air Intake)	2	111	26.6	82	35.1	62	25.0
A85	Ho Shun Yee Building Block 2	1.5	109	24.0	82	35.0	61	24.9
A86	Cheong Wai Building	1.5	112	27.4	82	35.1	62	25.0
A87	Kwong Ming Ying Loi School	1.5	107	25.7	82	35.1	62	25.0
A88	Sun Yuen Long Centre Block 5	20	110	22.9	82	34.9	61	24.9
A89	Tung Tau Tsuen House 2	1.5	117	23.5	82	34.9	61	24.9
A90	Tai Wai Tsuen House 30	1.5	115	23.0	82	35.0	61	24.9
A91	Small Traders New Village	1.5	117	22.7	82	34.9	61	24.8
A92	Pok Oi Hospital	1.5	118	26.5	82	35.0	61	24.9
A93	Yeung Uk Tsuen House 10	1.5	98	18.0	82	34.7	61	24.6
A94	Chuk San Tsuen House 17	1.5	100	18.2	82	35.1	61	25.0
A95	Greenfield Lodge Block 1	1.5	92	16.9	82	35.1	61	24.9
A96	Kong Tau San Tsuen House 5	1.5	106	19.6	81	34.8	61	24.8
A97	Tai Kei Leng House 414	1.5	106	21.6	81	34.9	61	24.8
A98	Shung Ching San Tseun House 49	1.5	118	28.8	82	35.3	62	25.1
A99	Silver Garden House 125B	1.5	107	22.5	82	35.3	61	25.1
A100	Tin Liu Tsuen House 32	1.5	115	27.6	82	35.2	61	25.0
A101	Pak Sha Tsuen	1.5	102	21.1	82	35.4	61	25.2
A102	Wong Nai Tun Tsuen House 47C	1.5	103	19.7	82	35.4	61	25.1
A103	Greenwood Gardens House 397	1.5	101	19.6	82	35.4	61	25.1
A104	Park Signature Block 1	1.5	118	29.8	82	35.3	61	25.1
A105_FA	Market of Hung Fuk Estate (Fresh Air Intake)	4	110	29.4	81	34.9	61	24.9
A105_D	Market of Hung Fuk Estate (Welfare Facilities)	10	109	27.1	81	34.9	61	24.8
A106	Village House along Kung Um Road	1.5	109	23.5	82	35.3	61	25.1

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
A107	Village House along Kung Um Road	1.5	112	23.2	82	35.3	61	25.1
A108	Shan Ha Tsuen House No. 342	1.5	101	23.1	84	35.7	63	25.4
A109	Shan Ha Tsuen House No. 343A	1.5	104	23.9	84	35.8	63	25.4
A110	Shan Ha Tsuen House No. 345C	1.5	102	24.0	82	35.1	62	24.9
A111	Shan Ha Tsuen House No. 346	1.5	102	24.4	82	35.1	62	24.9
A112	Shan Ha Tsuen House No. 611	1.5	97	22.8	84	35.7	63	25.4
A113	Shan Ha Tsuen House No. 613F	1.5	97	23.2	84	35.8	63	25.4
A114	Sheung Yau Tin Tsuen	1.5	113	24.0	81	35.0	61	25.0
A115	Le Regent	1.5	109	27.6	81	35.0	61	24.9
A116	Planned Long Bin Development	5	115	30.1	82	35.0	61	24.9
A117	Planned Long Bin Development	5	112	28.9	83	35.3	62	25.1
A118	Planned Long Bin Development	5	114	29.7	83	35.3	62	25.2
A119	Proposed Hung Shui Kiu Development	5	110	29.3	81	34.9	61	24.9
A120	Yee Fung Garden Block A	1.5	118	29.8	82	35.3	61	25.1
A121	Ming Wan Court	1.5	121	30.2	82	35.5	62	25.2
A122	Yuen Long Villa House No. 252	1.5	117	28.2	82	35.2	61	25.0
A123	Sun Fai Court Block C	1.5	115	29.0	82	35.2	61	25.1
A124	Caritas Yuen Long Chan Chun Ha Secondary School	1.5	120	30.4	82	35.3	61	25.1
A125	Village House along Ma Tong Road	1.5	118	28.3	82	35.2	61	25.0
A126	Proposed Hung Shui Kiu Development	5	106	28.3	81	34.9	61	24.8
A127	Uptown House 30	1.5	110	27.8	81	35.0	61	24.9
A128	The Woodside Tower 5	1.5	107	27.0	81	34.9	61	24.8
A129	Shung Tak Catholic English College	1.5	107	26.3	81	34.8	61	24.8
A130	Village House along Castle Peak Road	1.5	104	25.6	81	34.9	61	24.8
A131	Regent's Park	1.5	111	22.7	82	35.3	61	25.1
A132	One Hyde Park House 7	1.5	107	22.6	82	35.5	62	25.2
A133	Pak Sha Tsuen	1.5	102	20.7	82	35.4	61	25.2
A134	Shan Ha Tsuen House No. 330	1.5	97	21.9	84	35.7	63	25.4
A135	Lam Hau Tsuen House No. 110	1.5	101	24.6	82	35.1	62	25.0
A136	Evergreen Place Tower 5	1.5	103	25.5	82	35.2	62	25.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
A137	Proposed Hung Shui Kiu Development	5	103	25.5	81	34.8	61	24.7
A138	Lok Kui Lau	1.5	118	30.4	82	35.1	61	24.9
A139	Ping Shan Home for The Aged	1.5	117	29.2	81	35.0	61	24.9
A140	Sheltered Structure along Tai Shu Ha Road East	1.5	118	28.7	81	35.2	61	25.1
A141	Village House along Tai Shu Ha Road West	1.5	117	28.7	81	35.2	61	25.1
A142	Sheltered Structure near Shap Pat Heung Interchange	1.5	114	26.1	81	35.1	61	25.0
A143	Kong Tau San Tsuen House No, 61	1.5	106	19.8	81	34.8	61	24.8
A144	Pok Oi Hos. Jockey Club Care and Attention Home	1.5	122	26.3	82	35.0	61	24.9
A145	Proposed Hung Shui Kiu Development	5	109	26.8	81	34.8	61	24.8
A146	Hong Ping Villa Block 1	1.5	119	30.5	82	35.0	61	24.9
A147	Tai On Home for Aged	1.5	105	26.7	82	35.5	61	25.1
A148	Ming Sum Home for the Sen	1.5	109	24.2	82	35.0	61	24.9
A149	Hung Fuk Estate	1.5	103	26.5	81	34.8	61	24.8
A150	Grand Yoho Block 1	20	111	22.6	82	34.9	61	24.8
A150a	Shopping Mall under Grand Yoho (Fresh Air Intake)	8	121	28.4	82	35.1	62	25.0
A151	Grand Yoho Block 5	20	111	22.4	82	34.9	61	24.8
A152	Grand Yoho Block 9	20 / 80	125	23.9	82	35.0	61	24.9
A152a	Shopping Mall under Grand Yoho (Fresh Air Intake)	10	121	29.0	82	35.2	62	25.1
Existing ASRs (within PDA boundary) – PDA Area 1								
W-A1	House 33, Park Villa	1.5	105	25.0	81	34.9	61	24.8
W-A2	House 11, Park Villa	1.5	98	22.7	81	34.8	61	24.7
W-A3	Existing developments (under construction)	1.5	102	24.4	81	34.9	61	24.8
W-A4		1.5	102	26.5	81	34.9	61	24.9
W-A5	Block 1, Recours La Serre	1.5	101	23.9	81	34.8	61	24.8
W-A6	Block 2, The Parkhill	1.5	100	24.9	82	35.1	62	24.9
W-A7	Block 7, Greenville Residence	1.5	101	25.2	82	35.1	62	25.0
W-A8	Block 6, Windsor Garden	1.5	101	24.8	82	35.1	62	24.9

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**
W-A9	Block 1, Marbella Garden	1.5	101	24.7	82	35.1	62	25.0
W-A11	Kisland Villa Phase 2	1.5	105	25.7	82	35.1	62	25.0
W-A13	Sha Tseng Tsuen	1.5	107	26.9	82	35.2	62	25.0
W-A14		1.5	109	27.7	83	35.3	62	25.1
Planned ASRs – PDA Area 2 and Area 3								
E-P1	Planned Residential	1.5	114	27.5	82	35.2	61	25.0
E-P2	Planned Residential	1.5	116	28.8	82	35.2	61	25.1
E-P5	Planned Residential	1.5	108	23.4	82	35.3	61	25.1
E-P6	Planned Residential	1.5	100	25.0	84	35.8	63	25.5
E-P7	Planned Residential	1.5	98	23.7	84	35.8	63	25.4
E-P8	Planned Residential	1.5	107	22.7	82	35.3	61	25.1
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	1.5	109	23.0	82	35.3	61	25.1
E-P15	Planned School	1.5	107	21.5	82	35.2	61	25.0
E-P17	Planned School	1.5	107	21.2	82	35.2	61	25.0
E-P19	Planned Residential	1.5	110	22.1	82	35.2	61	25.0
E-P20	Planned Residential	1.5	98	22.4	84	35.7	63	25.4
E-P21	Planned Residential	1.5	106	21.0	82	35.2	61	25.0
E-P23	Planned Residential	1.5	97	22.0	84	35.7	63	25.4
E-P25	Planned Fire Station & Ambulance Depot	1.5	102	25.6	84	35.8	63	25.5
E-P29	Planned Residential	1.5	98	22.2	84	35.7	63	25.4
E-P31	Planned Residential	1.5	94	20.9	85	35.8	64	25.4
E-P32	Planned Residential	1.5	105	21.1	82	35.4	61	25.2
E-P33	Planned Residential	1.5	91	20.9	85	35.8	64	25.4
E-P34	Planned Residential	1.5	95	21.7	85	35.8	64	25.4
E-P40	Planned Residential	1.5	95	21.4	85	35.8	64	25.4
E-P42	Planned Residential	1.5	94	20.4	85	35.8	64	25.4
E-P45	Planned School	1.5	97	21.4	85	35.8	64	25.4
E-P46	Planned Residential	1.5	103	20.2	82	35.4	61	25.2
E-P47	Planned Residential	1.5	93	19.9	85	35.7	64	25.4
Planned ASRs – PDA Area 1								

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m^3) [Year 2042, With Recommended Direct Technical Noise Remedies]					
			NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	200	40	100	50	75	35
W-P2	Planned Residential	1.5 / 10	98	23.5	81	34.8	61	24.8
W-P4	Planned Residential	1.5	96	21.8	83	35.7	62	25.3
W-P6	Planned Residential	1.5	97	22.1	83	35.7	62	25.3
W-P7	Planned Residential	1.5 / 15	99	23.1	83	35.7	62	25.4
W-P9	Planned Residential	1.5 / 15	98	21.9	83	35.7	62	25.3
W-P10	Planned Residential	1.5	99	23.2	81	34.8	61	24.7
W-P11	Planned Residential	1.5	101	24.1	81	34.8	61	24.8
W-P12	Planned Residential	1.5	102	25.3	82	35.1	62	25.0
W-P14	Planned Residential	1.5 / 15	102	24.9	82	35.1	62	25.0
W-P16	Planned Residential	1.5	96	22.8	84	35.7	63	25.4
W-P17	Planned Residential	1.5	101	24.8	82	35.1	62	24.9
W-P20	Planned Social Welfare Facility	1.5	106	26.2	82	35.2	62	25.0
W-P21	Planned Residential	1.5	107	26.9	82	35.2	62	25.0
W-P22	Planned Residential	1.5	106	26.6	82	35.2	62	25.0
W-P25	Planned Residential	1.5	107	26.9	82	35.2	62	25.0
W-P26	Planned Residential	1.5	105	25.1	82	35.1	62	25.0
W-P27	Planned Residential	1.5	98	23.9	84	35.8	63	25.4
W-P29	Planned Residential	1.5	97	22.8	84	35.7	63	25.4
W-P31	Planned Residential	1.5 / 15	96	21.1	83	35.6	62	25.3

4.5.4.7 It is found that the predicted pollutant concentrations at existing ASRs in Year 2042 are generally lower than those in Year 2027, except some existing ASRs located near the planned road network as well as the proposed development.

4.5.4.8 For planned residential uses under this Project, sufficient setback distance of at least 100m between the Yuen Long Highway has already been considered in the layout design. The predicted pollutant concentrations are generally lower compared to the existing ASRs, whereas the predicted annual NO_2 concentrations are ranging from about 20 to 29µg/m^3 only, which are well within the AQO. Planned uses within the buffer between planned residential developments and YLH generally comprise storage site, workshop, and area reserved for public utilities only, and it is predicted that there would be no exceedance of AQO in these landuses. Given the predicted pollutant concentrations within the PDA boundary in Year 2042 are higher than those in Year 2027, contours of 19th highest 1-hour and annual NO_2 concentrations, 10th highest 24-hour and annual RSP/FSP concentrations at 1.5mAG over the PDA

boundary in Year 2042 are plotted in **Figures 4.12k-p**. It is observed that there are no exceedances at all air sensitive uses in Year 2042.

4.5.4.9 On the other hand, the air quality implication at each representative ASRs in Year 2042 due to the recommended direct technical noise remedies (i.e. noise barrier and enclosure, see **Figure 5.7**) has been evaluated, as tabulated in **Table 4.27**. With the implementation of the proposed noise mitigation, reduction of pollutant concentrations is generally predicted, e.g. cumulative annual NO_2 concentrations will be reduced by a maximum of 1.2μg/m^3, and the cumulative air quality impact at all ASRs for without the recommended direct technical noise remedies would also comply with the AQO. No adverse air quality impact during operational phase of the Project is therefore anticipated.

Table 4.27 Air quality implication due to the recommended direct technical noise remedies

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
	Criteria		**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
Existing / Planned ASRs (outside PDA boundary)														
A2	House no. 40, Tai Tao Tsuen	1.5	103	27.1	81	35.0	61	24.9	0	-0.2	0	0.0	0	0.0
A3	House no. 95, Tai Tao Tsuen	1.5	108	28.5	81	35.1	61	25.0	0	-0.1	0	0.0	0	0.0
A4	Sheltered Structure, Fui Sha Wai South Road	1.5 / 5	110	27.9	81	35.1	61	25.0	-1	-0.1	0	0.0	0	0.0
A6	House no. 176A, Fui Sha Wai	1.5 / 5	112	27.8	81	35.1	61	25.0	-2	0.0	0	0.0	0	0.0
A7	Village House, Ping Tong Street South	1.5 / 5 / 10	113	28.4	82	35.3	61	25.1	0	-0.1	0	0.0	0	-0.1
A8	House no. 48, Tong Yan San Tsuen Road	1.5	121	33.4	83	35.5	62	25.3	-1	-0.4	0	0.0	0	0.0
A9	New Territories Assemblies of God Church	1.5	105	26.9	83	35.3	62	25.1	0	-0.2	0	0.0	0	0.0
A10	Block 10, Jasper Court	1.5	115	29.8	83	35.4	62	25.2	-3	-0.8	0	0.0	0	0.0
A11	School	1.5	110	26.8	82	35.2	62	25.1	-1	-0.1	0	0.0	0	0.0
A12	Village House, Lam Yu Road	5	116	27.6	83	35.3	62	25.1	-1	-0.1	0	0.0	0	0.0
A13	Elchk Lutheran Academy	1.5	107	26.3	82	35.2	62	25.0	0	-0.1	0	0.0	0	0.0
A15	Village House, Lam Hei Road	1.5	120	30.3	82	35.4	61	25.2	0	0.0	0	0.0	0	0.0
A16	Village House, Lam Hei Road	1.5	119	30.2	82	35.3	61	25.2	0	-0.1	0	0.0	0	0.0
A17	Sheltered Structure, Lam Hei Road	1.5	120	32.4	82	35.4	61	25.2	0	-0.2	0	0.0	0	0.0
A18	Sheltered Structure no. 66 Kiu Hing Road	1.5	120	31.0	82	35.4	61	25.2	1	-0.1	0	0.0	0	0.0
A19	Sheltered Structure no. 196A, Lam Hau Tsuen	1.5	102	24.6	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	-	-	-	-	-	-
A20	House no. 89A, Lam Hau Tsuen	1.5	117	28.8	83	35.3	62	25.2	-3	-0.7	0	0.0	0	-0.1
A21	House no. 324, Shan Ha Tsuen	1.5	99	22.3	84	35.7	63	25.4	0	0.0	0	0.0	0	0.0
A22	House no. 645, Shan Ha Tsuen	1.5	96	22.0	84	35.7	63	25.4	0	0.0	0	0.0	0	0.0
A23	House no. 193, Kung Um Road	1.5	115	28.2	82	35.2	61	25.1	0	-0.1	0	0.0	0	0.0
A24	House no. 132, Sun Mei Garden	1.5	114	26.6	82	35.4	61	25.2	0	-0.2	0	0.0	0	0.0
A25	House no. 293, Kung Um Road	1.5	107	22.4	82	35.3	61	25.1	-1	0.0	0	0.0	0	0.0
A26	Block 10, Chun Fai Garden	1.5	108	22.4	82	35.3	61	25.1	-1	-0.2	0	0.0	0	0.0
A27	House no. 161, Kung Um Road	1.5	107	21.5	82	35.2	61	25.0	0	0.0	0	0.0	0	0.0
A28	Village House, Kung Um Road	1.5	106	20.4	81	35.2	61	25.0	1	0.0	0	0.0	0	0.0
A29	House no. 366A, Kung Um Road	1.5	105	21.2	82	35.2	61	25.0	0	-0.1	0	0.0	0	0.0
A30	Sheltered Structure, Kung Um Road	1.5	96	20.5	85	35.8	64	25.4	0	0.0	0	0.0	0	0.0
A31	House no. 31B, Kung Um Road	1.5	105	21.1	82	35.4	61	25.2	0	0.0	0	0.0	0	0.0
A32	House no. 241, Kiu Hing Road	1.5	103	21.1	82	35.4	61	25.2	0	-0.1	0	0.0	0	0.0
A33	House no. 67A, Wong Nai Tun Tsuen	1.5	101	19.6	82	35.4	61	25.1	0	0.0	0	0.0	0	0.0
A34	House no. 128, Kung Um Road	1.5	92	19.1	85	35.7	64	25.4	0	0.0	0	0.0	0	0.0
A35	House no. 117, Kung Um Road	1.5	95	19.4	86	36.8	64	26.1	0	0.0	0	0.0	0	0.0
A36	Sheltered Structure no. 375, Tai To Tsuen	1.5	96	25.8	83	35.9	62	25.5	0	0.0	0	0.0	0	0.0
A37	Uptown Tower 1	1.5	106	28.1	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A38	Chinese Mission Seminary	1.5	100	24.8	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A39	Fui Sha Wai Playground	1.5	104	25.8	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A40	Tower 8, Imperial Villas II	1.5	100	25.4	82	35.2	61	24.9	0	0.0	0	0.0	0	0.0
A41	Energy Industrial Centre Block B	1.5	101	26.3	90	39.9	64	26.5	0	-0.1	0	0.0	0	0.0
A42	Block 1, Parkside Villa	5	114	27.5	82	35.2	62	25.1	-1	-0.2	0	0.0	0	0.0
A43	Block 2, Emerald Green	1.5	116	27.8	82	35.2	61	25.0	0	0.0	0	0.0	0	0.0
A44	House no. 49A, Lung Tin Tsuen	1.5	120	30.9	82	35.3	61	25.1	0	-0.2	0	0.0	0	0.0
A45	House no. 139A	1.5	118	30.5	82	35.4	61	25.2	0	-0.2	0	0.0	0	0.0
A46	House no. 101, Sham Chung Tsuen	1.5	118	30.0	82	35.3	62	25.2	0	-0.2	0	0.0	0	-0.1
A47	Meadowlands Block 33	1.5	103	27.9	82	35.9	61	25.5	0	0.0	0	0.0	0	0.0
A48	Po Kok Branch School	1.5	107	28.5	82	35.9	61	25.5	0	0.0	0	0.0	0	0.0
A49	Treasure Court Block 7	1.5	107	27.7	81	34.8	61	24.8	0	0.0	0	0.0	0	0.0
A50	Beauty Court Block 2	1.5	106	27.2	81	34.8	61	24.8	0	-0.1	0	0.0	0	0.0
A51	Park Nara Tower 1	1.5	108	27.6	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A52	Hung Uk Garden	1.5	97	23.4	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A53	Green Lodge House 16	1.5	106	25.9	82	35.2	62	25.0	-1	-0.2	0	0.0	0	0.0
A54	Ping Shan Garden Block 6	1.5	112	28.6	82	35.0	61	24.9	-1	-0.1	0	0.0	0	0.0
A55	Villa Sunshine Block 1	1.5	112	29.3	82	35.0	61	24.9	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A56	Park Royale Block 10	5	113	27.6	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A57	Park Royale Block 2	5	114	28.8	81	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A58	Po Leung Kuk Law's Foundation School	1.5	109	25.7	82	35.1	62	25.0	-1	-0.1	0	0.0	0	0.0
A59	Yuen Long Public Secondary School	1.5	107	25.5	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A60	Villa Art Deco Block 2	1.5	106	26.4	82	35.2	62	25.0	-1	0.0	0	0.0	0	0.0
A61	Gertrude Simon Lutheran College	1.5	120	30.9	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A62	Yuen Long Public Middle School Alumni Association Primary School	1.5	123	32.7	82	35.4	61	25.2	1	0.0	0	0.0	0	0.0
A63	Silver Field Garden Block 17	1.5	120	29.2	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A64	La Grove Block 1	1.5	120	33.0	82	35.4	61	25.2	-1	-0.8	0	0.0	0	0.0
A65	Ma Tin Tsuen House 242	1.5	118	29.2	82	35.3	61	25.1	0	-0.1	0	0.0	0	0.0
A66	Teng Lung Villa	1.5	117	28.1	82	35.2	61	25.1	0	0.0	0	0.0	0	0.0
A67	Yuen Long Baptist Church	1.5	116	30.2	82	35.3	61	25.1	-1	0.0	0	0.0	0	0.0
A68	Fraser Village House 54	1.5	119	32.3	82	35.4	61	25.2	0	-0.1	0	0.0	0	0.0
A69	The Brand	1.5	117	28.4	82	35.2	61	25.1	0	0.0	0	0.0	0	0.0
A70	Sereno Verde Block 2	1.5	111	22.8	81	35.0	61	24.9	0	-0.2	0	-0.1	0	0.0
A71	La Pradera Block 12	1.5	108	21.6	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A72	The Reach Tower 2	1.5	107	22.1	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A73	The Reach Tower 6	1.5	113	25.0	81	35.0	61	25.0	0	-0.2	0	0.0	0	-0.1
A74	Tai Kei Leng House 145	1.5	111	25.3	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A75	Christian & Missionary Alliance Chui Chak Lam Memorial School	1.5	112	24.8	81	35.0	61	24.9	-1	0.0	0	0.0	0	0.0
A76	Grand de Sol Block 15	1.5	109	22.2	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A77	Grand de Sol Block 8	1.5	109	21.9	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A78	Hoover Garden Block 4	1.5	110	23.0	81	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A79	Ha Yau Tin Tsuen House 2	1.5	106	21.0	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A80	Buddhist Wing Yan School	1.5	110	22.1	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A81	Fortune Centre	1.5	109	24.7	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A82	YOHO Town Block 6	1.5	104	21.7	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A83	YOHO Town Block 9	1.5	106	22.2	82	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A84	YOHO Midtown Block 5	15	107	23.2	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A84a	Shopping Mall under YOHO Midtown (Fresh Air Intake)	6	114	28.1	82	35.1	62	25.0	0	-0.1	0	0.0	0	0.0
A84b	Shopping Mall under YOHO Midtown (Fresh Air Intake)	15	107	23.2	82	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A84c	Shopping Mall under YOHO Midtown (Fresh Air Intake)	20	106	22.1	82	34.9	61	24.8	-1	0.0	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A84d	Kindergarten under YOHO Midtown (Fresh Air Intake)	2	111	26.6	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A85	Ho Shun Yee Building Block 2	1.5	109	24.0	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A86	Cheong Wai Building	1.5	112	27.4	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A87	Kwong Ming Ying Loi School	1.5	107	25.7	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A88	Sun Yuen Long Centre Block 5	20	110	22.9	82	34.9	61	24.9	0	0.0	0	0.0	0	0.0
A89	Tung Tau Tsuen House 2	1.5	117	23.5	82	35.0	61	24.9	0	0.0	0	-0.1	0	0.0
A90	Tai Wai Tsuen House 30	1.5	115	23.0	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A91	Small Traders New Village	1.5	117	22.7	82	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A92	Pok Oi Hospital	1.5	118	26.5	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A93	Yeung Uk Tsuen House 10	1.5	98	18.0	82	34.7	61	24.6	0	0.0	0	0.0	0	0.0
A94	Chuk San Tsuen House 17	1.5	101	18.2	82	35.1	61	25.0	-1	0.0	0	0.0	0	0.0
A95	Greenfield Lodge Block 1	1.5	92	16.9	82	35.1	61	24.9	0	0.0	0	0.0	0	0.0
A96	Kong Tau San Tsuen House 5	1.5	106	19.6	81	34.8	61	24.8	0	0.0	0	0.0	0	0.0
A97	Tai Kei Leng House 414	1.5	106	21.7	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A98	Shung Ching San Tseun House 49	1.5	118	28.9	82	35.3	62	25.1	0	-0.1	0	0.0	0	0.0
A99	Silver Garden House 125B	1.5	107	22.5	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A100	Tin Liu Tsuen House 32	1.5	116	27.7	82	35.2	61	25.0	-1	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A101	Pak Sha Tsuen	1.5	103	21.2	82	35.4	61	25.2	-1	-0.1	0	0.0	0	0.0
A102	Wong Nai Tun Tsuen House 47C	1.5	103	19.7	82	35.4	61	25.1	0	0.0	0	0.0	0	0.0
A103	Greenwood Gardens House 397	1.5	101	19.6	82	35.4	61	25.1	0	0.0	0	0.0	0	0.0
A104	Park Signature Block 1	1.5	117	30.0	82	35.3	61	25.1	1	-0.2	0	0.0	0	0.0
A105_FA	Market of Hung Fuk Estate (Fresh Air Intake)	4	110	29.4	81	34.9	61	24.9	0	0.0	0	0.0	0	0.0
A105_D	Market of Hung Fuk Estate (Welfare Facilities)	10	109	27.2	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
A106	Village House along Kung Um Road	1.5	109	23.5	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A107	Village House along Kung Um Road	1.5	113	23.4	82	35.3	61	25.1	-1	-0.2	0	0.0	0	0.0
A108	Shan Ha Tsuen House No. 342	1.5	103	23.2	84	35.7	63	25.4	-2	-0.1	0	0.0	0	0.0
A109	Shan Ha Tsuen House No. 343A	1.5	104	24.1	84	35.8	63	25.4	0	-0.2	0	0.0	0	0.0
A110	Shan Ha Tsuen House No. 345C	1.5	102	23.9	82	35.1	62	24.9	0	0.1	0	0.0	0	0.0
A111	Shan Ha Tsuen House No. 346	1.5	102	24.3	82	35.1	62	24.9	0	0.1	0	0.0	0	0.0
A112	Shan Ha Tsuen House No. 611	1.5	97	22.8	84	35.7	63	25.4	0	0.0	0	0.0	0	0.0
A113	Shan Ha Tsuen House No. 613F	1.5	97	23.3	84	35.8	63	25.4	0	-0.1	0	0.0	0	0.0
A114	Sheung Yau Tin Tsuen	1.5	113	24.1	81	35.0	61	25.0	0	-0.1	0	0.0	0	0.0
A115	Le Regent	1.5	109	27.7	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A116	Planned Long Bin Development	5	115	30.1	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A117	Planned Long Bin Development	5	112	29.0	83	35.3	62	25.1	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A118	Planned Long Bin Development	5	116	30.6	83	35.4	62	25.2	-2	-0.9	0	-0.1	0	0.0
A119	Proposed Hung Shui Kiu Development	5	110	29.3	81	34.9	61	24.9	0	0.0	0	0.0	0	0.0
A120	Yee Fung Garden Block A	1.5	118	29.8	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A121	Ming Wan Court	1.5	121	30.2	82	35.5	62	25.2	0	0.0	0	0.0	0	0.0
A122	Yuen Long Villa House No. 252	1.5	117	28.3	82	35.2	61	25.0	0	-0.1	0	0.0	0	0.0
A123	Sun Fai Court Block C	1.5	115	29.0	82	35.2	61	25.1	0	0.0	0	0.0	0	0.0
A124	Caritas Yuen Long Chan Chun Ha Secondary School	1.5	120	30.4	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
A125	Village House along Ma Tong Road	1.5	118	28.3	82	35.2	61	25.1	0	0.0	0	0.0	0	-0.1
A126	Proposed Hung Shui Kiu Development	5	106	28.3	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A127	Uptown House 30	1.5	110	27.9	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A128	The Woodside Tower 5	1.5	107	27.0	81	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A129	Shung Tak Catholic English College	1.5	107	26.3	81	34.8	61	24.8	0	0.0	0	0.0	0	0.0
A130	Village House along Castle Peak Road	1.5	104	25.7	81	34.9	61	24.9	0	-0.1	0	0.0	0	-0.1
A131	Regent's Park	1.5	111	22.9	82	35.3	61	25.1	0	-0.2	0	0.0	0	0.0
A132	One Hyde Park House 7	1.5	107	22.7	82	35.5	62	25.2	0	-0.1	0	0.0	0	0.0
A133	Pak Sha Tsuen	1.5	102	20.7	82	35.4	61	25.2	0	0.0	0	0.0	0	0.0
A134	Shan Ha Tsuen House No. 330	1.5	97	22.0	84	35.7	63	25.4	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A135	Lam Hau Tsuen House No. 110	1.5	101	24.6	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A136	Evergreen Place Tower 5	1.5	105	25.7	82	35.2	62	25.0	-2	-0.2	0	0.0	0	0.0
A137	Proposed Hung Shui Kiu Development	5	103	25.5	81	34.8	61	24.7	0	0.0	0	0.0	0	0.0
A138	Lok Kui Lau	1.5	118	30.5	82	35.1	61	24.9	0	-0.1	0	0.0	0	0.0
A139	Ping Shan Home for The Aged	1.5	117	29.3	81	35.0	61	24.9	0	-0.1	0	0.0	0	0.0
A140	Sheltered Structure along Tai Shu Ha Road East	1.5	121	29.4	81	35.2	61	25.1	-3	-0.7	0	0.0	0	0.0
A141	Village House along Tai Shu Ha Road West	1.5	117	29.1	81	35.2	61	25.1	0	-0.4	0	0.0	0	0.0
A142	Sheltered Structure near Shap Pat Heung Interchange	1.5	114	26.4	81	35.1	61	25.0	0	-0.3	0	0.0	0	0.0
A143	Kong Tau San Tsuen House No, 61	1.5	106	19.8	81	34.8	61	24.8	0	0.0	0	0.0	0	0.0
A144	Pok Oi Hos. Jockey Club Care and Attention Home	1.5	122	26.3	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A145	Proposed Hung Shui Kiu Development	5	109	26.8	81	34.8	61	24.8	0	0.0	0	0.0	0	0.0
A146	Hong Ping Villa Block 1	1.5	119	30.5	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A147	Tai On Home for Aged	1.5	105	26.8	82	35.5	61	25.1	0	-0.1	0	0.0	0	0.0
A148	Ming Sum Home for the Sen	1.5	109	24.2	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A149	Hung Fuk Estate	1.5	103	26.6	81	34.8	61	24.8	0	-0.1	0	0.0	0	0.0
A150	Grand Yoho Block 1	20	111	22.6	82	34.9	61	24.8	0	0.0	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
A150a	Shopping Mall under Grand Yoho (Fresh Air Intake)	8	121	28.4	82	35.1	62	25.0	0	0.0	0	0.0	0	0.0
A151	Grand Yoho Block 5	20	111	22.4	82	34.9	61	24.8	0	0.0	0	0.0	0	0.0
A152	Grand Yoho Block 9	20 / 80	125	23.9	82	35.0	61	24.9	0	0.0	0	0.0	0	0.0
A152a	Shopping Mall under Grand Yoho (Fresh Air Intake)	10	121	29.0	82	35.2	62	25.1	0	0.0	0	0.0	0	0.0
Existing ASRs (within PDA boundary) – PDA Area 1														
W-A1	House 33, Park Villa	1.5	105	25.2	81	34.9	61	24.8	0	-0.2	0	0.0	0	0.0
W-A2	House 11, Park Villa	1.5	98	22.7	81	34.8	61	24.7	0	0.0	0	0.0	0	0.0
W-A3	Existing developments (under construction)	1.5	102	24.5	81	34.9	61	24.8	0	-0.1	0	0.0	0	0.0
W-A4		1.5	102	26.6	81	34.9	61	24.9	0	-0.1	0	0.0	0	0.0
W-A5	Block 1, Recours La Serre	1.5	101	24.0	81	34.8	61	24.8	0	-0.1	0	0.0	0	0.0
W-A6	Block 2, The Parkhill	1.5	100	25.0	82	35.1	62	24.9	0	-0.1	0	0.0	0	0.0
W-A7	Block 7, Greenville Residence	1.5	101	25.3	82	35.1	62	25.0	0	-0.1	0	0.0	0	0.0
W-A8	Block 6, Windsor Garden	1.5	101	24.9	82	35.1	62	25.0	0	-0.1	0	0.0	0	-0.1
W-A9	Block 1, Marbella Garden	1.5	101	24.9	82	35.1	62	25.0	0	-0.2	0	0.0	0	0.0
W-A11	Kisland Villa Phase 2	1.5	105	25.9	82	35.1	62	25.0	0	-0.2	0	0.0	0	0.0
W-A13	Sha Tseng Tsuen	1.5	107	27.2	82	35.2	62	25.0	0	-0.3	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
W-A14		1.5	109	28.4	83	35.3	62	25.1	0	-0.7	0	0.0	0	0.0
Planned ASRs – PDA Area 2 and Area 3														
E-P1	Planned Residential	1.5	115	27.6	82	35.2	61	25.0	-1	-0.1	0	0.0	0	0.0
E-P2	Planned Residential	1.5	116	28.9	82	35.2	61	25.1	0	-0.1	0	0.0	0	0.0
E-P5	Planned Residential	1.5	108	23.4	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
E-P6	Planned Residential	1.5	100	25.7	84	35.8	63	25.5	0	-0.7	0	0.0	0	0.0
E-P7	Planned Residential	1.5	98	23.7	84	35.8	63	25.4	0	0.0	0	0.0	0	0.0
E-P8	Planned Residential	1.5	108	22.7	82	35.3	61	25.1	-1	0.0	0	0.0	0	0.0
E-P14	Planned Clinic / Social Welfare Facility / Community Hall	1.5	109	23.0	82	35.3	61	25.1	0	0.0	0	0.0	0	0.0
E-P15	Planned School	1.5	107	21.5	82	35.2	61	25.0	0	0.0	0	0.0	0	0.0
E-P17	Planned School	1.5	107	21.3	82	35.2	61	25.0	0	-0.1	0	0.0	0	0.0
E-P19	Planned Residential	1.5	110	22.1	82	35.2	61	25.0	0	0.0	0	0.0	0	0.0
E-P20	Planned Residential	1.5	98	22.4	84	35.7	63	25.4	0	0.0	0	0.0	0	0.0
E-P21	Planned Residential	1.5	106	21.0	82	35.2	61	25.0	0	0.0	0	0.0	0	0.0
E-P23	Planned Residential	1.5	97	22.0	84	35.7	63	25.4	0	0.0	0	0.0	0	0.0
E-P25	Planned Fire Station & Ambulance Depot	1.5	102	25.7	84	35.8	63	25.5	0	-0.1	0	0.0	0	0.0
E-P29	Planned Residential	1.5	98	22.3	84	35.7	63	25.4	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
E-P31	Planned Residential	1.5	95	21.0	85	35.8	64	25.4	-1	-0.1	0	0.0	0	0.0
E-P32	Planned Residential	1.5	105	21.1	82	35.4	61	25.2	0	0.0	0	0.0	0	0.0
E-P33	Planned Residential	1.5	91	21.0	85	35.8	64	25.4	0	-0.1	0	0.0	0	0.0
E-P34	Planned Residential	1.5	97	22.9	85	35.8	64	25.5	-2	-1.2	0	0.0	0	-0.1
E-P40	Planned Residential	1.5	96	21.9	85	35.8	64	25.4	-1	-0.5	0	0.0	0	0.0
E-P42	Planned Residential	1.5	94	20.4	85	35.8	64	25.4	0	0.0	0	0.0	0	0.0
E-P45	Planned School	1.5	97	21.4	85	35.8	64	25.4	0	0.0	0	0.0	0	0.0
E-P46	Planned Residential	1.5	103	20.2	82	35.4	61	25.2	0	0.0	0	0.0	0	0.0
E-P47	Planned Residential	1.5	93	19.9	85	35.7	64	25.4	0	0.0	0	0.0	0	0.0
Planned ASRs – PDA Area 1														
W-P2	Planned Residential	1.5 / 10	98	23.6	81	34.8	61	24.8	0	-0.1	0	0.0	0	0.0
W-P4	Planned Residential	1.5	96	21.9	83	35.7	62	25.3	0	-0.1	0	0.0	0	0.0
W-P6	Planned Residential	1.5	97	22.2	83	35.7	62	25.3	0	-0.1	0	0.0	0	0.0
W-P7	Planned Residential	1.5 / 15	99	23.2	83	35.7	62	25.4	0	-0.1	0	0.0	0	0.0
W-P9	Planned Residential	1.5 / 10	98	22.0	83	35.7	62	25.3	0	-0.1	0	0.0	0	0.0
W-P10	Planned Residential	1.5	99	23.3	81	34.8	61	24.7	0	-0.1	0	0.0	0	0.0
W-P11	Planned Residential	1.5	101	24.2	81	34.8	61	24.8	0	-0.1	0	0.0	0	0.0
W-P12	Planned Residential	1.5	102	25.4	82	35.1	62	25.0	0	-0.1	0	0.0	0	0.0

ASR ID	Location	Worst affected Height above Ground	Pollutant Concentration (µg/m³) [Year 2042, Without Recommended Direct Technical Noise Remedies]						Air Quality Implication due to the Recommended Direct Technical Noise Remedies (µg/m³) [With – Without]					
			NO_2		RSP		FSP		NO_2		RSP		FSP	
			1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual	1-hour (19th highest)	Annual	24-hour (10th highest)	Annual	24-hour (10th highest)	Annual
		Criteria	**200**	**40**	**100**	**50**	**75**	**35**	**-**	**-**	**-**	**-**	**-**	**-**
W-P14	Planned Residential	1.5 / 15	101	25.0	82	35.1	62	25.0	1	-0.1	0	0.0	0	0.0
W-P16	Planned Residential	1.5	96	22.9	84	35.7	63	25.4	0	-0.1	0	0.0	0	0.0
W-P17	Planned Residential	1.5	101	25.0	82	35.1	62	24.9	0	-0.2	0	0.0	0	0.0
W-P20	Planned Social Welfare Facility	1.5	107	26.4	82	35.2	62	25.0	-1	-0.2	0	0.0	0	0.0
W-P21	Planned Residential	1.5	107	28.0	82	35.2	62	25.1	0	-1.1	0	0.0	0	-0.1
W-P22	Planned Residential	1.5	107	27.0	82	35.2	62	25.0	-1	-0.4	0	0.0	0	0.0
W-P25	Planned Residential	1.5	106	26.9	82	35.2	62	25.0	1	0.0	0	0.0	0	0.0
W-P26	Planned Residential	1.5	105	25.3	82	35.1	62	25.0	0	-0.2	0	0.0	0	0.0
W-P27	Planned Residential	1.5	98	24.0	84	35.8	63	25.4	0	-0.1	0	0.0	0	0.0
W-P29	Planned Residential	1.5	97	23.1	84	35.7	63	25.4	0	-0.3	0	0.0	0	0.0
W-P31	Planned Residential	1.5 / 15	96	21.2	83	35.6	62	25.3	0	-0.1	0	0.0	0	0.0

Note:

The air quality implication due to the recommended direction technical noise remedies is determined by subtracting the maximum 19th highest 1-hour and annual NO_2, and 10th highest 24-hour and annual RSP / FSP concentrations predicted under the "with scenario" across various assessment heights by those predicted under "without" scenario.

Figure rounded to nearest integer for 19th highest 1-hour NO2 and 10th highest 24-hour RSP / FSP concentrations, and rounded to 1 decimal place for annual NO_2, RSP and FSP concentrations.

4.5.5 Recommended Mitigation Measures

4.5.5.1 According to the operational air quality assessment results, adverse cumulative air quality impact during operational phase is not anticipated. No mitigation measures are required.

4.5.6 Residual Impacts

4.5.6.1 According to the operational air quality impact assessment results presented in **Section 4.5.4**, no adverse residual air quality impact during operational phase is anticipated.

4.6 Operational Odour Impact Assessment

4.6.1 Identification of Pollution Sources

4.6.1.1 During operational phase of the Project, key odour emission sources within the assessment area include the following:

- Proposed sewage treatment works;
- Treated sewage effluent discharge;
- Proposed refuse collection points and sewage pumping stations; and
- Existing livestock farms.

4.6.1.2 Details of these sources are described in the sections below:

Odour Emission from Proposed Sewage Treatment Works

4.6.1.3 A STW is proposed at the southern end of the PDA Areas 2 and 3, as illustrated in **Figure 4.12**. Potential odour emission is anticipated from the proposed STW and hence has been addressed in the odour impact assessment.

Odour Emission from Treated Sewage Effluent (TSE) Discharge

4.6.1.4 Based on the current design of the STW, a combination of secondary plus and tertiary treatment process is proposed and the reclaimed water from tertiary treatment process will be pumped to the planned service reservoir for future non-potable uses. According to DSD's definition on the reclaimed water [4-1], it is highly treated wastewater which is clear in appearance, odourless and is safe for non-potable uses. Hence, potential odour emission from the discharge and storage of TSE is not anticipated.

Odour Emission from the Proposed Refuse Collection Points and Sewage Pumping Stations

4.6.1.5 Two sites are zoned as "Other Specified Uses (Refuse Collection Point and Sewage Pumping Station)" (OU(RCP&SPS)) under the current RODP. For the proposed RCPs, proper ventilation, deodourising and exhaust system will be provided to minimise the potential odour nuisance on nearby ASRs. Good site practices will be also adopted to enhance the hygiene of the RCPs by frequent washing, proper

[4-1] DSD's website : http://www.dsd.gov.hk/EN/Sewerage/Environmental_Consideration/Reclaimed_Water/ (last accessed on 18th May 2017)

covering of refuse bins, closing of roller shutters and proper maintenance of the ventilation, deodourising and exhaust systems. The Food and Environmental Hygiene Department (FEHD) will also conduct regular checks and surprise inspections to ensure proper operation and hygiene of the proposed RCPs.

4.6.1.6 Two SPSs are proposed under the current RODP and their design capacities are about 4,700m^3 per day and 23,280 m^3 per day respectively. These SPSs are located at about 10m and 90m away from the existing ASRs and planned ASRs respectively. Potential odour emission from wet well, inlet chamber and screen chamber are the possible odourous sources of the proposed SPSs. These sources would be housed inside reinforced concrete structure and the foul air would be ventilated to a deodouriser for treatment before discharge to the environment. Activated carbon type deodouriser with odour removal efficiency of 95% and above is not uncommonly used in DSD's pumping stations (e.g. 95% at Queen's Hill SPS [4-2], 99.5% at Jordan Valley Box Culvert SPS [4-3] and Lin Cheung Road Temporary SPS [4-4]) and thus is considered technically feasible. In order to better protect both existing and planned ASRs located in the vicinity of the proposed SPSs under this Project, a deodouriser with at least 99.5% odour removal efficiency will be adopted to avoid the potential odour emission to the maximum practicable extent. The ventilation system would also maintain a negative pressure within the facilities. In addition, the deodouriser exhaust outlet of the proposed SPSs would be located away from the nearby ASRs as far as practicable.

4.6.1.7 With the implementation of appropriate mitigation measures including installation of deodourising units with at least 99.5% odour removal efficiency, it is anticipated there would be no adverse odour impact from the proposed SPSs at nearby air sensitive uses.

Odour Emission from Existing Livestock Farms

4.6.1.8 There are 3 chicken farms and 3 pig farms currently in operation in close proximity with or within the PDA boundary under the current RODP. Based on the current RODP, it is proposed that only 1 chicken farm (with the rearing capacity of 35,000) located at the southern end of the PDA Areas 2 and 3 will continue its operation at the same location. **Figure 4.13** illustrates their location.

4.6.1.9 One chicken farm and one pig farm located within Phase 2 of the Project will be removed in Year 2023 to cater for the construction of the Project, while the remaining one chicken farm and two pig farms located within Phase 3 of the Project will be removed in Year 2031, which are located at about 1.1km and 2km away from the planned ASRs to be occupied in end-2027 (e.g. ASRs E-P1 and E-P5). In addition, based on the current implementation programme, the proposed STW will commence its operation in Year 2030. Hence, there would be potential cumulative odour impact on these planned ASRs during this interim period (i.e. 2030 – 2031) with 2 chicken farms and 2 pig farms within Phase 3, and the proposed STW in operation.

4.6.1.10 Site inspections have been conducted to the chicken farm and pig farm jointly with EPD in May 2015 to identify the potential odour sources and observe its typical operation. Odour measurements have also been conducted subsequently in July /

[4-2] Environmental Permit for "*Queen's Hill Sewage Pumping Station*" (EP-506/2016)

[4-3] Environmental Permit for "*Control of Water Pollution at Jordan Valley Box Culvert – Sewage Pumping Station (JVBCSPS)*" (EP/365/2009)

[4-4] Environmental Permit for "*Public Housing Development at Lin Cheung Road Site – Temporary Sewage Pumping Station and Associated Sewer Pipes*" (EP-495/2015)

August 2015 to determine the odour concentrations and odour emission rates from various identified emission sources for quantitative odour impact assessment.

4.6.2 Assessment Methodology

Odour Emission from Proposed Sewage Treatment Works

4.6.2.1 Odour emissions are anticipated from odourous facilities in the proposed STW, such as the sedimentation tanks, sludge storage tanks, etc. As advised by the Engineer, the design of the proposed STW is made reference to that under the planned Lok Ma Chau Loop development. Hence, odour emission factors for each odourous facilities are referenced from the approved EIA Study "*Development of Lok Ma Chau Loop*" (AEIAR-176/2013). In order to minimise the potential odour impact on nearby ASRs, it will be designed to enclose all the odourous facilities and provision of odour removal system with an efficiency of at least 99.5% at the ventilation exhaust to control odour emission. **Appendix 4.14** presents the calculation of odour emission from the proposed STW.

Odour Emission from Existing Livestock Farms

4.6.2.2 For the chicken farm and pig farm, on-site odour measurements have been conducted in July / August 2015. As advised by the operators, both the chicken farm and pig farm were operating at their full licenced rearing capacity on the date of odour measurement.

4.6.2.3 For chicken farm, odour samples were collected from the excrement collection channels and temporary excrement storage tanks in typical chicken house for medium- and large-size chickens, and excrement collection bins to determine their odour emission strengths (i.e. OU/s/m^2) by Force-choice Olfactometer in accordance with the European Standard Method (EN13725). However, given there was only limited space in the small chicken house and the excrement was collected and dried on waste papers, the equipment for collecting samples for odour emission strengths could not be deployed. Ambient odour concentrations (OU/m^3) were therefore determined inside and immediately outside the openings of the small chicken houses. Wind speed (m/s) and cross-section area (m^2) at each openings of the small chicken houses were also recorded to determine the air flow rate (m^3/s). The odour emission rate (OU/s) from the small chicken house is then determined for the quantitative odour impact assessment.

4.6.2.4 The pig farm consists of sheltered pig houses, enclosed pig houses and various waste water treatment facilities. Due to site constraint and objection by the operator, the equipment for collecting samples for odour emission strengths could not be deployed in the pig houses and hence only ambient odour concentrations were measured inside the pig houses. Similar to measurement in chicken farm, wind speed (m/s) and cross-section area (m^2) were measured at each opening and mechanical ventilation fans to determine the odour emission rate. Odour samples were collected from the waste collection tank, waste water holding tanks, anaerobic tanks, aeration tanks, sedimentation tank and rubbish bin and the odour emission strengths were determined by the EN13725 method.

4.6.2.5 Odour measurements in chicken farm and pig farms were carried out during summer daytime at 36°C and 34°C respectively. Temperatures during night-time period (i.e. 7pm to 7am of the next day) would generally be lower than daytime. Adopting the odour measurement results for night-time period would therefore overestimate the potential odour impact. According to the approved EIA Study "*Habour Area*

Treatment Scheme (HATS) Stage 2A" (AEIAR-121/2008) (HATS EIA), variation of odour emission due to temperature changes can be determined by the following equation:

$$G = M\ [BOD_5]\ 1.07^{(T-20)}$$

where G = sulphide flux, g/m^2h
$[BOD_5]$ = 5-day biochemical oxygen demand
T = temperature, ^oC
M = coefficient, m/h

4.6.2.6 Temperature data for Year 2010 (based on meteorological data from EPD's PATH-2016, Grids 24_43 and 24_44 for chicken farm and pig farm) has been reviewed and the seasonal temperature profiles during night-time are summarised in **Table 4.28**. With reference to the approved HATS EIA, temperature correction was adopted for temperatures ranging from 29 to 33°C. To reasonably assess the potential odour impact, temperature correction has been applied to night-time period (i.e. 7pm to 7am of the next day) seasonally, and the lowest temperature limit is capped at 29°C in this assessment. A summary of correction factors for the temperature decrease from 36°C and 34°C are summarised in the following table.

Table 4.28 Summary of Temperatures and Correction Factors

Season	Night-time Temperature (^{o}C)			Temperature adopted for Correction (^{o}C)	Temperature Correction Factor (Chicken Farm)	Temperature Correction Factor (Pig Farm)
	Max.	Min.	Average			
Spring (March – May)	27	5	21	29	0.62	0.71
Summer (June – August)	31	20	26	31	0.71	0.82
Autumn (September – November)	31	5	22	31	0.71	0.82
Winter (December – February)	23	2	16	29	0.62	0.71

4.6.2.7 As described in **Section 4.6.1.8 – Section 4.6.1.9**, existing livestock farms will be removed in phases and there will be an interim period (2030-2031) that the planned ASRs in Phase 1 of the Project will be affected by the proposed STW and some existing livestock farms before they are removed. In order to determine the net effect of removing existing livestock farms and introducing the proposed STW on existing ASRs, 3 odour assessment scenarios have been conducted to take into account the potential odour impact during different phases of the Project:

- **Basecase Scenario** (existing condition) : Operations of 3 chicken farms and 3 pig farms
- **Interim Scenario** (2030 – 2031) : Operations of 2 chicken farms, 2 pig farms and the proposed STW
- **Ultimate Scenario** (beyond 2031) : Operations of 1 chicken farm and the proposed STW

4.6.2.8 **Appendix 4.14** presents the odour emission inventory for the existing livestock farms. It should be noted that odour measurements were only conducted to the chicken farm and pig farm, for which they have the highest rearing capacity among all livestock farms of the same type. For the other livestock farms, the odour emissions from the measured chicken farm and pig farm are directly adopted as a conservative approach.

Dispersion Model and Modelling Parameters

4.6.2.9 Odour impact is assessed by the EPD approved dispersion model, AERMOD. Odour sources are assumed as "Point", "Area" or "Volume" sources as appropriate. Grid-specific composite meteorological data extracted from EPD's PATH-2016 model are adopted, including hourly wind speed, wind direction, temperature and mixing height. The stability classes are estimated from PCRAMMET model.

4.6.2.10 As required in the EIAO-TM, the odour criterion is defined as 5 OU units based on an averaging time of 5 seconds. Hence, it is required to convert the predicted odour concentration in 1-hour averaging time from the AERMOD model to 5-second average. Reference is made to the peak-to-mean ratio stated in the "*Approved Methods for Modelling and Assessment of Air Pollutants in New South Wales*" published by the Department of Environment and Conservation, New South Wales, Australia (NSW Approved Method). In accordance with the NSW Approved Method, the factors for converting 1-hour average to 1-second average concentration are adopted directly to convert the 1-hour concentration predicted by the AERMOD model to 5-second concentration as a conservative approach.

4.6.2.11 As stated in the NSW Approved Method, where nearby buildings interfere with the trajectory and growth of the plume, the source is called a wake-affected point source. A point source is wake-affected if stack height is less than or equal to 2.5 times the height of buildings located within a distance of 5L (where L is the lesser of the height or width of the building) from the release point. Based on the current design of the proposed STW, the chimneys of the deodourisation units will be 10m high, while some of the treatment facilities, such as sludge thickening house, sludge dewatering house, etc., located adjacent will be at least 6m high. Therefore, chimney emission sources associated with the proposed STW are considered as wake-affected point sources in the assessment.

4.6.2.12 The conversion factors for different types of source and stability classes are listed in **Table 4.29** below. With reference to the "*Approved Methods for Modelling and Assessment of Air Pollutants in New South Wales*", the conversion factor for "Area" source under E and F class is 2.3. However, the conversion factor of 2.5 is adopted for a more conservative assessment.

Table 4.29 Conversion factors for 1-hour to 5-second averaged odour concentration (Near-field for conservative assessment)

Stability Class	Point Source (Wake-affected)	Area Source	Volume Source
A	2.3	2.5	2.3
B	2.3	2.5	2.3
C	2.3	2.5	2.3
D	2.3	2.5	2.3
E	2.3	2.5 [1]	2.3
F	2.3	2.5 [1]	2.3

Note [1]: Conversion factor of 2.5 is adopted as conservative assessment.

4.6.2.13 The general modelling parameters are summarised in **Table 4.9** for ease of reference. Moreover, a 50x50m grid is used to generate pollution contours in order to investigate the pollutant dispersion.

4.6.3 Prediction and Evaluation of Impacts

4.6.3.1 The predicted 5-second maximum odour concentrations during ultimate phases of the Project and the net improvement during both interim and ultimate phases at representative ASRs located within 500m from the odour sources are presented in **Table 4.30**. Detailed results are presented in **Appendix 4.15**. It can be seen from the table below that the predicted maximum cumulative odour concentrations during the Ultimate phase at the identified planned ASRs under this Project are within the criterion of 5OU/m^3.

Table 4.30 Maximum cumulative odour concentrations

ASR ID	Location	Worst affected Height above Ground (m)	Maximum Odour Concentration (OU/m^3) under Ultimate Scenarios	Net Improvement (OU/m^3)	
		Criteria	**5**	**Interim [1]**	**Ultimate [2]**
Existing ASRs (outside PDA boundary)					
A21	House no. 324, Shan Ha Tsuen	10	0.2	0.2	4.0
A22	House no. 645, Shan Ha Tsuen	10	0.4	2.2	6.5
A23	House no. 193, Kung Um Road	10	0.1	2.2	4.6
A24	House no. 132, Sun Mei Garden	15	0.3	4.1	7.1
A25	House no. 293, Kung Um Road	10	0.2	4.9	7.8
A26	Block 10, Chun Fai Garden	10	0.2	6.8	10.0
A27	House no. 161, Kung Um Road	10	0.2	12.2	15.6
A28	Village House, Kung Um Road	10	0.3	56.4	60.3
A29	House no. 366A, Kung Um Road	10	0.3	14.6	18.7
A30	Village House, Kung Um Road	10	0.4	0.1	6.4
A31	House no. 31B, Kung Um Road	10	0.5	7.3	12.2
A32	House no. 241, Kiu Hing Road	10	0.6	2.8	8.4
A33	House no. 67A, Wong Nai Tun Tsuen	10	1.2	0.0	7.4
A34	House no. 128, Kung Um Road	10	2.0	0.1	14.0

ASR ID	Location	Worst affected Height above Ground (m)	Maximum Odour Concentration (OU/m^3) under Ultimate Scenarios	Net Improvement (OU/m^3)	
		Criteria	5	Interim [1]	Ultimate [2]
A35	House no. 117, Kung Um Road	1.5	6.0	0.0	25.4
A43	Block 2, Emerald Green	20	0.3	1.9	4.1
A44	House no. 49A, Lung Tin Tsuen	10	0.1	1.1	3.1
A45	House no. 139A	10	0.1	2.8	4.9
A46	House no. 101, Sham Chung Tsuen	10	0.1	3.7	5.9
A61	Gertrude Simon Lutheran College	20	0.2	1.4	3.7
A62	Yuen Long Public Middle School Alumni Association Primary School	20	0.2	1.2	3.3
A63	Silver Field Garden Block 17	10	0.1	0.7	2.7
A64	La Grove Block 1	20	0.2	1.6	3.8
A65	Ma Tin Tsuen House 242	10	0.1	1.3	3.2
A66	Teng Lung Villa	10	0.1	1.6	3.6
A67	Yuen Long Baptist Church	10	0.1	1.9	3.8
A69	The Brand	20	0.2	3.5	5.9
A98	Shung Ching San Tseun House 49	10	0.2	4.0	6.4
A99	Silver Garden House 125B	10	0.2	6.5	9.3
A100	Tin Liu Tsuen House 32	10	0.2	3.8	6.1
A101	Pak Sha Tsuen	10	0.4	13.8	18.4
A102	Wong Nai Tun Tsuen House 47C	10	1.2	0.1	7.6
A103	Greenwood Gardens House 397	10	1.0	0.1	8.8
A104	Park Signature Block 1	20	0.2	2.5	4.8
A106	Village House along Kung Um Road	10	0.2	5.2	8.1
A107	Village House along Kung Um Road	10	0.4	24.2	28.5
A108	Shan Ha Tsuen House No. 342	10	0.3	0.9	5.4
A109	Shan Ha Tsuen House No. 343A	10	0.3	0.1	5.4
A112	Shan Ha Tsuen House No. 611	10	0.4	2.0	5.7

ASR ID	Location	Worst affected Height above Ground (m)	Maximum Odour Concentration (OU/m^3) under Ultimate Scenarios	Net Improvement (OU/m^3)	
		Criteria	**5**	**Interim [1]**	**Ultimate [2]**
A113	Shan Ha Tsuen House No. 613F	10	0.4	2.5	6.4
A120	Yee Fung Garden Block A	20	0.2	1.0	3.0
A122	Yuen Long Villa House No. 252	10	0.1	1.1	3.0
A123	Sun Fai Court Block C	10	0.1	0.8	2.7
A124	Caritas Yuen Long Chan Chun Ha Secondary School	20	0.2	1.0	2.9
A125	Village House along Ma Tong Road	10	0.1	1.2	3.1
A131	Regent's Park	10	0.5	18.5	22.9
A132	One Hyde Park House 7	10	0.4	12.1	16.7
A133	Pak Sha Tsuen	10	0.5	5.3	10.3
A134	Shan Ha Tsuen House No. 330	10	0.2	0.0	5.0
Planned ASRs (PDA Area 2 and Area 3)					
E-P1	Planned Residential	20	0.3	-	2.5 [3]
E-P2	Planned Residential	20	0.4	-	-
E-P5	Planned Residential	20	0.3	-	3.3 [3]
E-P6	Planned Residential	20	0.5	-	-
E-P7	Planned Residential	15	0.7	-	-
E-P8	Planned Residential	20	0.4	-	-
E-P14	Planned Market and Sports Centre	20	0.4	-	-
E-P15	Planned School	20	0.4	-	-
E-P17	Planned School	20	0.5	-	-
E-P19	Planned Residential	20	0.6	-	-
E-P20	Planned Residential	20	0.6	-	-
E-P21	Planned Residential	20	0.6	-	-
E-P23	Planned Residential	20	0.4	-	-
E-P25	Planned Fire Station & Ambulance Depot	20	0.7	-	-
E-P29	Planned Residential	20	0.8	-	-
E-P31	Planned Residential	20	1.1	-	-
E-P32	Planned Residential	15	0.9	-	-
E-P33	Planned Residential	15	0.8	-	-

ASR ID	Location	Worst affected Height above Ground (m)	Maximum Odour Concentration (OU/m^3) under Ultimate Scenarios	Net Improvement (OU/m^3)	
		Criteria	**5**	**Interim [1]**	**Ultimate [2]**
E-P34	Planned Residential	20	0.8	-	-
E-P40	Planned Residential	15	1.3	-	-
E-P42	Planned Residential	20	1.1	-	-
E-P45	Planned School	15	1.7	-	-
E-P46	Planned Residential	15	1.0	-	-
E-P47	Planned Residential	15	1.4	-	-
W-P16	Planned Residential	15	0.3	-	-
W-P27	Planned Residential	15	0.2	-	-
W-P29	Planned Residential	15	0.3	-	-

Note:
Value in bold means exceedance of its respective criteria.
[1] Net improvement refers to the decrease in maximum odour concentration due to implementation of this Project from existing conditions to the interim period (Year 2030 – 2031)
[2] Net improvement refers to the decrease in maximum odour concentration due to implementation of this Project from existing conditions to the full commission of the proposed YLS development (Ultimate Scenario)
[3] Since these planned ASRs will be occupied in end-Year 2027, the net improvement refers to the decrease in odour concentration from the interim period to the full commission of the proposed YLS development

4.6.3.2 Upon full commissioning of the Project, 2 chicken farms and 3 pig farms will be removed. This will substantially reduce the total odour emissions in the area and improve the odour conditions in future from existing situation. It would reduce the maximum cumulative odour concentrations at all existing ASRs by a range of 2.7 OU/m^3 to 60.3 OU/m^3. The predicted maximum cumulative odour concentrations would comply with the criterion of 5OU/m^3 at all ASRs except the existing ASR A35 (House no. 117, Kung Um Road), which is located close to the existing chicken farm. It is however anticipated that without this project in place, this ASR is currently subject to an even worse cumulative odour impact as it is exposed to other nearby existing livestock farms. From **Table 4.30** above, it is shown that the implementation of Project will greatly benefit the ASR A35 by decreasing the maximum cumulative odour concentration of about 25.4OU/m^3 from its current condition (from 31.4OU/m^3 to 6.0OU/m^3). Improvement in cumulative odour concentrations is also predicted for all year round (**Appendix 4.16**). The potential odour nuisance perceived by the existing ASR A35 would still occur under southerly to westerly winds. According to the hourly meteorological data from EPD's PATH-2016 model for area in the vicinity of the chicken farm (i.e. Grids 24_43 and 24_44), southerly to westerly wind will only occur less than 20% time of the year. And it is predicted that only 0.068% of the time in a year (6 out of 8760 hours) with the odour concentration at ASR A35 will exceed the odour criterion, albeit a net improvement in cumulative odour concentrations by removing the 5 livestock farms in ultimate phase. In other words, the cumulative odour concentration at ASR A35 would comply with the criterion in most of the time. During this 0.068% of time with odour exceedance, the proposed STW contributes only 0.01 OU/m^3 to 0.04 OU/m^3 at this ASR.

4.6.3.3 Contours of 5-second odour concentrations at 1.5mAG and 10mAG for Ultimate Scenario are presented in **Figures 4.14a – Figure 4.14b** to illustrate the cumulative

odour impact on ground level sensitive uses and worst-hit level respectively. During the formulation of the RODP, considerations on the odour issues have already been given to the planned residential and school developments so that sufficient setback (at least 150m) from the odour sources are allowed to avoid the potential odour impact. According to the assessment results, no exceedance of odour concentrations at the planned residential and school developments is predicted. However, exceedance is anticipated at some of the planned landuses within the buffer area, including the planned district open space (DO3.1) and government land (G3.3 and G3.5). Locations of these landuses are illustrated in **Figure 3.1c**.

4.6.3.4 For the district open space (DO3.1), it is recommended that no sensitive uses should be introduced in the areas with exceedance of odour concentrations. On the other hand, based on the current RODP, the planned government lands (G3.3 and G3.5) are intended for government depot, which will generally be unmanned, and hence it will not be subject to odour impact.

4.6.3.5 Some existing ASRs, including A35 (House no. 117, Kung Um Road), are located within the odour exceedance area which is close to the existing chicken farm. The extent of exceedance covers approximately 0.4km from the southern end of the PDA. The community likely affected includes few temporary structures and few scattered village houses located near the southern end of the PDA areas. It is however anticipated that without this project in place, a much larger extent of exceedance is expected as it is exposed to other nearby existing livestock farms. As discussed above, with implementation of this Project, five of the existing livestock farms will be removed and there will likely be a net improvement in the odour condition on the existing ASRs in the area during its operational phase in future years.

4.6.3.6 The interim phase of the development will remove one pig farm and one chicken farm. During this interim period (i.e. Year 2030 – 2031), there is also a general improvement in odour conditions at all existing ASRs from existing situation by a maximum of about 56.4OU/m^3.

4.6.3.7 Some existing ASRs will still exceed the odour criterion for a short duration of time (from 0.01% to 3.16% of time in a year, **Appendix 4.16**) during interim phase, which are mainly contributed by the existing livestock farms. The contribution from the proposed STW during the odour exceedance period is very small, with a maximum of 0.05OU/m^3 only. Despite the predicted odour exceedance at these ASRs, the cumulative odour impacts during the exceedance period will be improved compared to that of existing condition due to removal of one pig farm and one chicken farm. Overall, improvement in cumulative odour concentrations is predicted at all these ASRs for more than 95% time of a year, with most of them with net improvement for even 100% time of a year.

4.6.3.8 Although there will be a slight increase in the predicted cumulative odour concentrations under the downwind condition of STW (for maximum 4.7% time of a year) at some of these ASRs, all would comply with the odour criterion of 5OU/m^3 during this period and thus no adverse odour impacts are anticipated. Detailed analysis is given in **Appendix 4.16**.

4.6.3.9 During interim phase, the predicted maximum cumulative odour concentrations at the planned ASRs in Phase 1 (i.e. E-P1 and E-P5) range from 2.8 to 3.6 OU/m^3, which comply with the criterion of 5 OU/m^3. Contour of 5-second odour concentrations during interim period at 1.5m and 20mAG are given in **Figure 4.14c – Figure 4.14d** to illustrate the cumulative odour impact on ground level and the worst affected level of the planned ASR E-P1 and E-P5. It is observed that there are

no exceedance of cumulative odour concentrations at all the planned ASRs in Phase 1.

4.6.4 Recommended Mitigation Measures

4.6.4.1 The following at-source mitigation measures should be implemented to control odour emission from the proposed STW and SPS:

- Potential odour sources should be enclosed;
- Negative pressure should be maintained within the facilities;
- Installation of deodouriser with an odour removal efficiency of at least 99.5% to control odour emission via ventilation exhaust;
- Exhaust of the deodouriser should be oriented away from sensitive receivers and vertically upwards to avoid direct facing to any sensitive receivers; and
- Maintenance of deodouriser should be regularly conducted to ensure good condition.

4.6.4.2 For the planned air sensitive uses within the odour exceedance area, the following measures should be adopted:

- No sensitive uses should be introduced in the exceedance area within the planned DO3.1.

4.6.5 Residual Impacts

4.6.5.1 With the implementation of the recommended mitigation measures at the proposed STW and SPS as discussed in **Section 4.6.4** above, the predicted cumulative odour concentrations at all planned ASRs within the PDA boundary would comply with the relevant criterion during both interim and ultimate phases of the development, except for the planned District Open Space DO3.1 to the east of the proposed STW and at south of the PDA where no sensitive uses should be introduced within the exceedance area. With all these mitigation measures and landuse requirements in place, adverse residual odour impact on the planned developments is not anticipated.

4.6.5.2 Some existing ASRs will exceed the odour criterion for a short duration of time during both interim and ultimate phases of the Project. Nonetheless, as detailed in **Section 4.6.3**, the Project will bring about a net improvement in cumulative odour concentrations at all existing ASRs for most of the time over a year. Despite the predicted odour exceedance at some existing ASRs, the cumulative odour impacts during the exceedance period will be improved compared to that of existing condition without removal of the five livestock farms. Under downwind condition, although the proposed STW would bring about a slight increase in the predicted cumulative odour concentrations at these ASRs, they would all comply with the odour criterion for both interim and ultimate phases. It is therefore concluded that there is no adverse residual odour impact arising from the Project.

4.7 Conclusions and Recommendations

4.7.1 Construction Phase

4.7.1.1 Potential construction dust impact would be generated from site clearance, ground excavation, site formation, etc. during the construction phase. Quantitative construction dust impact assessment have been conducted. Results have concluded that there will not be any adverse residual air quality impact during construction phase given frequent watering on all works area once per hour during working hours.

4.7.2 Operational Phase

4.7.2.1 Quantitative operational air quality assessment has been conducted, taking into account the vehicular emission impact associated with the Project and existing road networks, and industrial emission in the vicinity of the Project. It is concluded that the predicted cumulative air quality impacts on all air sensitive receivers would comply with the AQOs during the operational phase of the Project.

4.7.3 Odour Issues during Operational Phase

4.7.3.1 Quantitative odour impact assessment has been conducted, taking into account the proposed STW and existing livestock farm to be retained under the current RODP. It is concluded that, during full commissioning of the Project, the predicted cumulative odour concentrations on all planned air sensitive receivers would comply with the criteria during the operational phase of the Project, except for the planned District Open Space DO3.1 where no sensitive uses should be introduced within the exceedance area. Nevertheless, exceedance is predicted at some existing ASRs.

4.7.3.2 Given the fact that, without the Project in place, existing sensitive receivers are subject to an even worse odour impact as they are exposed to other nearby livestock farms. Moreover, a total of five existing livestock farms within the Project site will be removed under this Project which would reduce the total odour emission in the area and improve the odour conditions in future from existing situation.

4.7.3.3 During the interim period, there is also a general improvement in odour conditions, in terms of maximum odour concentration and/or percentage time of the year, at all of the existing ASRs. Predicted odour concentrations at the planned ASRs during the interim period, i.e. Phase 1 would also comply with the criterion.

4.8 References

Environmental Protection Department, A Study to Review Hong Kong's Air Quality Objectives 2009 (http://www.epd.gov.hk/epd/english/environmentinhk/air/studyrpts/aqor_report.html)

Environmental Protection Department (2015), 2015 Hong Kong Emission Inventory Report (http://www.epd.gov.hk/epd/sites/default/files/epd/2015_EI_Report_Eng_v2.pdf)

Environmental Protection Department (2016), Air Quality in Hong Kong 2016 – Statistical Summary (http://www.aqhi.gov.hk/api_history/english/report/files/2016%20Statistical%20Summary_Final_en.pdf)

Environmental Protection Department (2015), Air Quality in Hong Kong 2015 (http://www.aqhi.gov.hk/api_history/english/report/files/AQR2015e_final.pdf)

Sun Y., Wang L.L.., Wang Y.S. (2010) "In situ measurements of SO_2, NOx, NOy, and O_3 in Beijing, China during August 2008" Science of the Total Environment 409 (2011), P933-940

Sun Y., Wang L.L.., Wang Y.S. (2010) "In situ measurements of NO, NO_2, NOy, and O_3 in Dinghushan (112°E, 23°N), China during autumn 2008", Atmospheric Environment 44 (2010), P2079-2088